

CARSPAN for Windows



User's manual

Corresponding to version 0.0.1.36

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Preface

CARSPAN for Windows largely has the same functions as the previous, DOS-version of CARSPAN. Now, however, the possibilities of Windows are used to facilitate the users with more options for data viewing, interactive artifact correction and with more flexibility in data handling. Most of the computational procedures work satisfactory, as well as most of the artifact correction routines. The current version, however, is certainly not the final version, and has to be seen as an improved beta-version. Flexibility not only has advantages, but also has a backside. In particular the session structure in combination with going forward and backward through the analysis procedure is error-prone and is therefore not full proof at the present.

The computational algorithms in principle are the same as in the previous DOS-version, which were based on cardiovascular research initiated by Bert Mulder. Together with Arie van Roon and Hans Veldman, years ago I was in the situation to develop the spectral analysis routines, which were implemented in the DOS environment by David Schweitzer. The present Windows version has been developed by Sicco Strampel and Bas Kortmann, while we were very grateful that Jaap Bos could continue this development into the current state.

Many, many early users were involved in testing and improving previous incomplete versions, either planned or unplanned. We are very grateful to them. In particular, I like to mention: Ydwine, Anje, Bert, Jaap, Diana, Marnel, Chris, Heleanne, Dick, Arjan, Margreet, Elly, Hetty, Harriëtte, Andrea, Monika, Yvonne and Anna.

Further development of CARSPAN will continue along two lines. At first, the present version will be improved by adapting the present version at several, merely minor points to make it more convenient for the user. Moreover, some additions will be made with respect to a number of computational routines and some tools that may improve data handling at the input- and output side of the program. The second line requires much more time and includes rewriting the session structure as well as redesigning the interface.

The current version of the manual is rather complete but certainly not yet finished. In this respect I want to thank Hedwig Hofstetter for her tedious work in writing the current users guide and her help in re-writing large parts of the manual. I hope further completion of the manual will go along with further progress in developing next versions of CARSPAN for Windows.

Ben Mulder,
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1. Introduction

1.1 A view on CARSPAN

CARSPAN for Windows version 1.0 is an environment for processing cardiovascular time series, such as heart rate and systolic blood pressure, in an easy way. Its main function is to compute spectral measures of cardiovascular variables. It provides tools for artifact detection and correction of time series, other pre-processing options, and graphical display functions of time series as well as of spectra. CARSPAN was developed for use in studies on mental work. However, it may be useful in other fields such as stress research, industrial psychology, physiology, and medicine as well.

With respect to the data to be used it is supposed that the user has two types of data files: an R-peak event file in which the heart rate data are stored and a sampled data file with blood pressure and/or respiration data (in case such signals have to be analyzed). However, some tools will become available to convert data from other formats.

In psychophysiology, heart rate variability (HRV) is used as an index of mental workload. In several studies it is found that HRV decreases as a function of invested effort in the task to be performed. Relations have been studied between heart rate, blood pressure, respiration and the fluctuations in these variables during task performance and in mentally demanding situations.

In blood pressure, similar fluctuations as those in heart rate are found. It can be shown that the baroreflex, an important loop for controlling short term blood pressure, plays a dominant role in the occurrence of these similarities. The loop gain of this reflex is dependent of the mental state of the subject and of invested effort in a mental loading task. Spectral analysis of these concurring fluctuations in heart rate and blood pressure is a useful tool in detecting such changes in baroreflex sensitivity (BRS).

Nowadays, blood pressure indices can be assessed easily on a beat-to-beat basis, for example by means of the FIN.A.PRESTM finger blood pressure measuring device. By using a simple averaging procedure samples of the respiratory signal can be obtained on a beat-to-beat basis as well.

CARSPAN is a software package for computation of spectral measures of all kinds of variables that are available on a beat-to-beat basis. Not only spectral density functions, but also transfer and coherence functions can be computed. A power density spectrum gives the distribution of the signal variability as a function of frequency. The coherence is a measure of similarity of two time series (based on a linear relation). The transfer function gives an estimate for the gain (modulus) and the phase difference between two time series as a function of frequency. Spectral power, weighted coherence as well as mean modulus and phase are computed in specified frequency bands, which are user defined or dependent on the respiratory frequency in the analysis period.

Also a simple but effective type of time-frequency analysis is included in the program: short segment spectral analysis. The resulting functions, so-called spectral profiles, include essentially the same spectral outcomes but now as a function of time. Mean values, spectral power, mean modulus, phase and weighted coherence in specified frequency bands are computed from small, partly overlapping analysis segments which enables the study of the effects of time varying task demands and the use of non-stationary data segments. This technique is closely related to digital band filtering of a time series, wavelet analysis or

complex demodulation (Van Steenis, 2002).

Several steps are necessary in the processing of cardiovascular time series, such as deriving time series on a beat-to-beat basis from continuously measured signals, artifact detection and correction. All these steps are included in CARSPAN and can be executed from the environment. Additionally, a number of tools are connected to the program for computation of blood pressure scale factors, for data conversion of other types of (sampled) data files to the necessary binary format as well as for software R-peak detection. Some of these tools are already available; others will be implemented in the next versions of CARSPAN. The basis of the environment consists of a menu structure from which all parameter settings and specifications can be given. On the one hand this concerns specification and selection of data files, data segmentation, channel definitions and labels, and scale factors. On the other hand it concerns functions which can be applied: heart rate artifact detection and correction, construction of time series of other types of cardiovascular and respiration data on a beat-to-beat basis as well as detection and correction of these data. Next to automatic procedures, data can be monitored and corrected manually as well. Finally, functions are available for data analysis and presentation of results (plotting of time series and computed spectral functions). All specifications can be saved, re-loaded and changed later on, which enables an easy, flexible and consistent use of all options on large sets of data files. The program runs on MS-Windows XP systems.

1.2 Structure of this manual

The aim of the CARSPAN program is explained in chapter 2. The last part of the chapter is an application example of CARSPAN.

A detailed description of the theory, choices made and used algorithms can be found in chapter 3. Pre-processing (like artifact correction) and analysis are discussed here. Applications in other than cardiovascular research are briefly mentioned. The mentioned example is worked out further here. A list of references is at the end of this chapter.

Chapter 4 helps you to get started with the user environment. Part of the same example as used in chapter 2 is discussed to make the user familiar with the menus and options of CARSPAN.

All features of the program are discussed in the user's guide, chapter 5. This part explains the details of the functions and menu items.

Chapter 6 will give the technical requirements of the systems CARSPAN runs on. How to install CARSPAN on these systems will be described in chapter 7.

In chapter 8 an elaborated example is described further. Most common processing is shown and some pre-processing possibilities are discussed.

1.3 Differences and similarities with the CARSPAN DOS-version 2

Users of the older DOS-version of CARSPAN may be surprised by the large similarities in available functions in this windows version, but hopefully also by the largely enhanced flexibility of the program. In our view the program has gained enormously with respect to signal monitoring functions, which makes it much more suitable and faster to use on large data sets. One of the additional functions is that raw ECG, blood pressure, respiration or other

signals can be checked in the relevant data segments in order to see whether artifact correction is adequate. In order to fulfill these and other analysis functions it was decided to integrate all routines in one windows program, instead of having an interface program connecting a series of separate sub-programs, as was the case in the DOS-version of CARSPAN.

One of the nice features of the present program is that when derived time series are (partly) corrected it is still possible to go back to the original heart rate data without losing all previous work. This can be seen as a normal function in such kind of analysis, but it requires secure administration of required data-adaptation. As a matter of fact, this functionality is achieved by keeping track of a list of corrections to be performed, while the corrections itself only take place in the final phase of the process. Unfortunately, flexibility has a backside: the chance that instabilities occur in this phase of the correction process is much larger than when not giving these possibilities. This means that the user needs adequate insight in the on-going data handling processes to prevent such instabilities.

In respect to previous DOS-versions the windows version has more data-output options. Spectral functions as well as spectral profile data can be outputted in Excel format, while the profile output itself can be considered as a new extended time series dataset that can be read in again as a standard input data file; this latter function gives nice possibilities for data monitoring and relating spectral profiles to task related events and markers.

2. Aims of the CARSPAN program

The present chapter gives an overview of the aims and functions of the CARSPAN program. First some background information is given about the research field from which the program has originated and about the type of signals, variables, spectral functions and pre-processing techniques in the described approach. A detailed description is given of the aims that were set before developing the package. At the end of the chapter an example is given, which will be worked out more extensively in the next chapters.

2.1 Backgrounds

Laboratory studies on heart rate variability (HRV) during mental task performance have shown that these fluctuations decrease compared to resting situations (Mulder, G., 1980; Mulder, L., 1988). Some researchers argue (Porges, 1982) that in particular fluctuations in heart rate related to respiratory patterns are affected by these mental processes. Others (Mulder, G., 1980) show that in particular fluctuations related to vasomotor tone are affected. Research has shown that the techniques are very useful in studies on mental workload. HRV is seen as an index of invested effort in the task to be performed. By using specific, well defined test tasks in the beginning and at the end of a working day insight can be obtained in the effects of the work to be performed and in the changes in the physiological state which go along with the performance of heavy mental work during several hours.

The different types of fluctuations in heart rate can be distinguished by using spectral analysis on the series of interbeat intervals derived from the R-peaks in the ECG. The variability can be determined for several frequency bands. At least three frequency bands can be distinguished:

- 0.02 - 0.06 Hz, low frequency band; temperature changes and adaptation to the task demands are mainly reflected in this band;
- 0.07 - 0.14 Hz, mid frequency band; oscillatory fluctuations in blood pressure related to vaso-motor tone regulation are reflected in this band. In particular this band appeared to be very sensitive to changes in mental task load;
- 0.15 - 0.40 Hz, high frequency band; respiratory activity and task induced effects (e.g. effects of repetitive stimulus presentation) are most predominant in this band.

Heart rate variability is both a result of changes in blood pressure which are mediated by the central nervous system and of some mechanisms in the central nervous system itself. Respiratory related changes in heart rate are at least partly caused by central mechanisms. To gain a better insight into the mechanisms that contribute to HRV, and in the differential sensitivity of the mentioned frequency bands to mental load, it is useful to study its relation with blood pressure regulation and with respiratory activity. In order to gain insight into the relations between mental task load and the fluctuations in heart rate, blood pressure, breathing pattern as well as their mutual dependency all these signals have to be measured. For the same reason it is necessary to develop procedures to compute spectral densities for all relevant cardiovascular signals and to compute coherences and transfer functions between the obtained time series.

Several authors have adopted their own procedures for deriving spectral measures of HRV

(Berntson et al., 1997; see Rompelman, Coenen & Kitney, 1977 for an overview). Some of them compute the variability in interbeat interval series (IBI), others the variations in instantaneous frequency (heart rate). Most of them need to interpolate the obtained time series first before they can use an existing Fourier transform algorithm designed for equidistant time series. Rompelman (1985) has chosen for a procedure in which it is not necessary to perform this interpolation step. Unfortunately, the method cannot be used directly to compute spectral functions of other cardiovascular time series. For this reason we set out to improve his procedure and to develop a program that can do the necessary computations for all types of cardiovascular time series and respiration (Mulder, L., 1988).

In blood pressure similar fluctuations as in heart rate are found. It can be shown that the baroreflex, an important loop for controlling short term blood pressure, plays a dominant role in the occurrence of these similarities (Karemaker, 1987, Wesseling & Settels, 1985). It is shown that the loop gain of this reflex is dependent on the mental state of the subject (e.g. awake or asleep, Sleight, 1980) and of invested effort in a mental loading task (Robbe, Mulder L., Rüddel, Veldman & Mulder G., 1987). Mulder L. (1985) indicated already that spectral analysis of these concurring fluctuations in heart rate and blood pressure would be helpful in detecting such changes in baroreflex sensitivity (BRS).

2.1.1 Spectral functions

Several approaches can be followed concerning the computation of the spectral functions of heart rate and blood pressure (Rompelman, Coenen & Kitney, 1977, Mulder L., 1988), depending on the chosen models for the origin of the fluctuations in heart rate and the selected arithmetic methods to process the data. Some authors choose to use heart rate (Rompelman, 1977), while others prefer R-R interval time (Porges, 1982). It can be shown that it really makes a difference which choice is made when effects of mental task performance on heart rate variability are studied. Furthermore, relevant differences are found when either absolute or normalized scores (variation coefficient, modulation index) are used. For this reason we have chosen an approach in which all these options can be made available and in which a simple coupling between heart rate and blood pressure time series can be obtained (Mulder L., 1988). A beat-to-beat approach is chosen without having the necessity to interpolate between beats for reasons of equidistance between succeeding sampling values. There are two main consequences of this choice: the first is that the standard fast Fourier transform (FFT) cannot be used. This is not really a problem because the discrete Fourier transform (DFT) can be adopted in a simple and direct way, both for heart rate (Rompelman, 1985) and for blood pressure and other derived time series (Mulder L., 1988). The second consequence is related to the handling of signals which do not have that beat-to-beat character, e.g. respiration. In the approach chosen it will be necessary to derive a sample value of such signals for each heart beat in order to facilitate the computation of spectral relations (coherence, transfer functions) between such signals and variations in heart rate and blood pressure time series. It can be shown that as long as only linear relations are studied between all the time series no information is lost.

The spectral analysis program contains computational functions for power spectral density functions to estimate the spectral distribution of variability, as well as coherence and transfer functions (modulus and phase) between time series. Raw spectral values as well as total variance, weighted coherence, and/or mean modulus and phase in pre-selected frequency areas can be computed. The selected frequency bands are either user defined, or computed in

accordance with the main respiratory frequency in the selected measuring period.

2.1.2 Spectral profiles

One of the major restrictions of using spectral techniques in (semi-)real life situations, such as car driving, process control and air traffic control, is that task demands are changing rapidly. This means that when analysis segments of for instance 5 minutes are selected, these may include both standard or easy workload periods as well as high task load conditions. When such periods are mixed, clear task effects probably will not be found. When very short task segments are selected, for instance 30 seconds, and when these are related to specific task events, this will greatly increase the chance that relevant connections are found between task load and HRV or HR effects. However, now two problems still have to be solved. The first is making an adequate selection of the (small) time segments related to specific task related events during the work. The second problem has to do with the reliability of the computed spectral values; assuming normal statistical distributions the number of degrees of freedom will be linearly related to the segment length. However, this type of problem can be overcome by (weighted) averaging over several segments.

Another restriction of standard spectral analysis is related to the issue of non-stationarity of the time series to be analyzed. A time series is supposed to be stationary when the characteristics of the signal are the same in the whole analysis period. This includes not only the mean and the standard deviation of each small time segment, but also the spectral distribution of the signal. The issue is not only a theoretical one, but it causes practical problems as well. Interpretation of the outcomes of statistical tests for this purpose (Weber, Molenaar & Van der Molen, 1991) are hindered by dependency on the chosen segmentation of the time series to be analyzed.

As a matter of fact two main aspects can be distinguished in this issue. The first aspect which can be seen as the most relevant is related to slow fluctuations in cardiovascular time series due to changing task demands. For example, in the task of the subject there are periods that ask for more attention or more effort to be invested in order to insure a correct task performance. The physiological system adapts to such a new situation: heart rate and blood pressure increase or decrease to a new level and the researcher wants to know what happens with variability measures during such an adaptation period. Clear examples can be found in air traffic and driving research, such as take-off and landing of air planes or car driving in and outside of urban cities. However, the same type of problems is present in a lot of other working situations.

The second aspect is related to the assumptions which are formulated with respect to spectral analysis (Chatfield, 1975). It is assumed that the signal can be extended periodically outside the measuring window without losing its random character, which means without introducing new rhythms that are related to the length of the selected analysis segment. It should be kept in mind, however, that the spectral computations itself are not hindered nor are the results corrupted. In fact a precise description of the signal is obtained in the frequency domain, while the interpretation of these results may cause some difficulties because of unwanted trend patterns. The most important thing is that the signal under analysis has to be a good representation of the process to be studied. In this respect the assumptions for spectral analysis are not really different from those for computation of means and standard deviations during statistical analysis.

Both task related issues as well as effects of non-stationarity problems can be successfully attacked by an approach which is called here the 'spectral profile' technique. As a matter of

fact the same spectral computations are carried out as used for the spectral functions, but now on short periods of time (e.g. 30 or 60 seconds) which are called segments. By shifting such segments over the time series to be analyzed and by introducing a certain overlap between succeeding segments (e.g. 50 up to 95 % overlap) a series of spectral values is obtained indicating, for instance, a varying spectral power in a selected frequency band.

The same functions and options are available as in 'normal' spectral analysis and all values (spectral power, weighted coherence, mean modulus and phase) are computed in a number of pre-selected and user-defined frequency bands. The same holds for mean values of the time series in those time segments. The described technique offers the possibility to study cardiovascular responses during changing task demands as well as looking in more detail to the effects of non-stationarities in the data on the obtained spectral measures. In this respect it is a nice feature that the segment length and the overlap factor can be adapted in a wide range.

The spectral profile technique is not really different from the well known approach of band pass filtering followed by computation of the variance (power) in short time segments (see Mulder, G., 1980 and Mulder, L., 1985 for examples). However, the here proposed technique has the advantage that the applications easily can be extended to the study of baroreflex sensitivity as a function of time because of the availability of coherence, modulus and phase functions. Furthermore, results of this technique should give the same information as the complex demodulation technique and wavelet analysis (Bloomfield, 1976; van Steenis, 2002). It has the advantage, however, that the same computational concepts can be used as during normal spectral analysis, which makes it easier to compare results between the two approaches. As a matter of fact one can make a smooth transition from one approach to the other.

2.1.3 Pre-processing

In order to have the possibility for computing spectral functions of heart rate and blood pressure variations and of respiratory patterns different pre-processing is required for each of the signals. From the ECG first the R-peaks have to be detected to be able to compute changes in interval time, which form the basis for the spectral computations. Within the CARSPAN environment it is supposed that the R-peaks from the ECG are already detected in hardware (or in another software program). However, a tool for software R-peak detection is available which is connected to CARSPAN as a tool. From the sampled (continuous measured) blood pressure waveform maxima, minima and mean values within each heart beat have to be computed in order to derive time series of systolic, diastolic and mean blood pressure values. Pre-processing for respiration is quite different because it is a continuous signal which is not dependent of the heart beat. In order to be able to relate changes in heart rate and blood pressure to those in respiration either the beat-to-beat variables have to be interpolated in order to reach a set of equidistant time series, or in each beat a instantaneous (at the moment of the R-peak in the ECG) or an average respiration sample value has to be computed. Mulder L. (1988) argues that this latter beat-to-beat approach has advantages because the other cardiovascular variables to be studied are intrinsically of this beat-to-beat type. It can be shown that no information is lost by this approach as long as only linear relations between the beat-to-beat variables and respiration are studied.

Finding a beat-to-beat representation (i.e. constructing a time series) for each of the variables under study is not the only pre-processing step necessary before spectral analysis can be performed. Artifacts in these time series have a serious effect on the computed spectral

results. It is clear that in particular detection of these artifacts may be specific for each of the time series. For instance, artifacts in the R-R interval series may be caused by errors in the R-peak detection process (either misses or false alarms) or originate from physiological sources (extra-systoles). Repeatedly occurring 'recalibrations' in the continuously measured finger pressure by the FIN.A.PRESTM instrument could be the reason of missing values in the systolic blood pressure time series (e.g. 2 or 3 consecutive missing beats in a sequence of 70 beats). However, after detection of these errors similar procedures for artifact correction can be used for different types of cardiovascular time series.

The procedures mentioned above are only a part of the routines necessary in a software program for cardiovascular spectral analysis. It is clear that most of these processing steps are so specific that it is not possible to use a standard spectral analysis package using the fast Fourier transform (FFT) without the need for a large number of separate programs for all pre-processing functions. Therefore a windows program has been developed that integrates all specification, pre-processing, analysis and data-viewing functions, in contrast to the approach followed in the earlier Dos environment that coupled separate programs by means of an interface program.

2.2 The design of a processing environment

For several reasons it was useful to develop a program like CARSPAN. Some are summarized here:

- Almost all standard signal analysis packages need an equidistant time series for the computation of spectra. Interpolation or filtering procedures to obtain such time series are very specific for cardiovascular signals and are different depending on whether instantaneous heart rate or interbeat interval are used (Rompelman et al., 1977).
- A number of cardiovascular variables can be expressed in absolute scales (for instance interval time in ms and blood pressure in mmHg) or in variations relative to the mean value in the chosen measurement period (variation coefficient, modulation index). In several packages, not specifically suited for cardiovascular analysis, it is not so easy to express the results in absolute units; often arbitrary units are used. This makes it difficult to exchange results between research groups.
- It is useful to compute the power in some pre-selected frequency bands (see section 2.1). With the standard packages this is, in general, only possible by calculation of the power in such bands afterwards. A standardized output containing not only spectral values but band values as well is required. This output should be suitable for statistical analysis and/or for MS-Excel.
- In psychological studies (and many others) it is often necessary to use the same program parameters for a number of experimental conditions and subjects (some kind of batch processing). These parameters and information about conditions and subjects should be available in the output. These kinds of options are usually not provided by the standard packages. Moreover, the same set of specifications has to be available for analysis of a new set of data; so data specification, processing specification and data analysis have to be separated. Additionally, such requirements, related to batch processing, should not hinder fast exploratory analysis, as for instance for educational purposes. Therefore, a fast and really interactive procedure must be available as well.

2.2.1 Aims

With respect to the specifications of the program the following starting points were chosen:

Integration of processing steps

In the whole processing cycle a large number of steps are included, either different or identical, for the variables to be handled. One of the main goals of the analysis environment is to integrate all these steps in one program in order to facilitate the use of such procedures for both inexperienced and experienced users. These processing steps include, among others: pre-processing, artifact detection and correction, spectral analysis and graphical functions to display both input time series and obtained results.

Specification and processing separated

In many applications it is useful to perform the same operations on several data files. This holds both for signal pre-processing as for spectral analysis in the final phase.

A clear reason for using the same processing parameters on the whole data-set is that this is the only way to assure that the results of the spectral analysis can be used correctly for statistical analysis afterwards. This means that it is important to specify all the processing steps precisely, saving the derived parameters in a specification file that can be used on a complete directory containing all data files to be analyzed. Another relevant step is a check whether all the data files to be processed are available.

In the described way a kind of batch processing procedure can be realized which combines the advantages of batch processing and working interactively with the analysis system. In the specification phase the user can work fast and interactively with the system without the need to wait for results of previous processing stages.

Documentation of processing steps

A procedure will be realized in which not only the processing parameters are stored in files for repeated use, but from which also a good readable documentation can be obtained of the performed processing steps, including the specified parameters.

Invisible internal command and parameter structure

Another aim of the program is to keep the internal structure of command and parameter files invisible to the user. The user has to specify all the functions to be performed in terms of his data-set and the analysis techniques to be used. He should not be bothered by the internal organization of these functions within the analysis environment.

Extendibility

Despite the requirement to make an environment in which internal command and file structures are kept far from the user it is important to build an open structure in which it is not too difficult to add internal and external functions to the environment. An example of such an extension might be the addition of external computed time series such as stroke volume values and/or systolic time intervals to an existing file with R-R interval, blood pressure and respiratory values. Of course, not the addition itself is a problem but to keep the administration intact without bothering the user with problems concerning file structures etc.

2.2.2 Functions

In the basic form of the current version of CARSPAN several functions and parameter settings can be specified. Time series can be build and processed for heart rate and R-R interval time separately (see Mulder,L., 1988), for systolic, diastolic and mean blood pressure as well as for a respiratory sample per heart beat. Other functions can be added, or the obtained time series can be extended using external programs.

In the user environment the following functions are included:

File menu

This option gives the possibility to open or save sessions and to export computed event files.

Specification

The specification menu includes data set specification, data specification, as well as specification of the spectral analysis options to be performed.

During data set specification the user can choose the data directory to be handled and select a series of files within this directory. The availability of the files is checked and the user can determine whether only R-peak event files are handled (then the analysis is restricted to heart rate and/or interbeat interval (IBI) spectra) or sampled data files as well, enabling the possibility for including blood pressure and respiratory data.

In the data specification menu data segmentation (blocks, periods), event codes for R-peaks and other user defined codes can be specified,. At this level the coupling is made between sampled data files (blood pressure, respiration) and event files in which, among others, the R-peak events are included. This latter function is called 'time series creation'. At this level calibration values (scale, zero) can be inputted manually. Finally, at this level it can be specified whether there are already user defined (pre-computed) time series available in the event file.

All the processing options available can be entered in the analysis specification menus. This part consists of two options: a. the normal spectral functions such as power spectral density, coherence and transfer functions of the defined time series and its relations and b. the spectral profile option. In this latter technique, short segment spectral analysis, the same spectral functions are computed, but now over short analysis periods (segments) which are shifted in time. The resulting functions - spectral results as a function over time - are called 'spectral profiles'. More details concerning this technique are given in the next paragraphs. If all these specifications are entered and saved, the next menu '(pre-) processing' becomes relevant. All specifications can be saved and re-loaded via the file menu in order to be used in other analysis sessions or in combination with other data files.

Processing

This menu includes monitoring of heart rate data (R-peaks series), checking data segmentation, artifact detection and correction of the R-R interval series (constituting the time base of the data), deriving beat-to-beat values for blood pressure and respiration from the sampled data files, monitoring these data and, at the end of this processing stage, performing artifact detection and correction on these series.

Analysis

After data pre-processing the spectral analysis itself, both for the spectral functions and the

spectral profiles follows as a final stage.

Viewing data files and results

Graphical functions are included to preview both raw sampled data files and spectral results.

Tools

From the tools menu some separate programs can be reached for file conversion, software R-peak triggering, blood pressure calibration (computation of zero and scale factors) etc. At the present only a limited number of tools is available; in the next versions of CARSPAN this list will be extended.

2.3 Scope of applications

Initially, spectral computations as well as pre-processing procedures within CARSPAN have been developed for studies on mental work load using cardiovascular measures. However, now the scope is certainly wider and it can be concluded that the program is highly suitable for use in other research fields, such as stress research, industrial psychology, physiology, cardiology and other fields of medicine.

In the set-up of the program the starting point was the availability of R-peaks on a continuous time axis, continuously measured blood pressure and respiration. However, there are options available to include other types of cardiovascular data. When a researcher has the possibility to obtain beat-to-beat values of e.g. systolic time intervals, T-wave amplitude, or plethysmographic amplitudes, such values can be added to the data file as well. Additionally, the 'pre-computed time series specification option' in the data specification menu allows for importing other types of data that are available on a beat-to-beat basis.

Respiratory data are handled by estimating beat-to-beat samples from the respiratory signal, either by computing a value around the time point of the R-peak or by computing the mean respiration value of the time period between two R-peaks. This same approach can be followed for other slow varying signals, such as temperature, movement activity or even EMG-activity and EEG-alpha-activity. For skin conductance or skin potential probably a method has to be found which takes into account the response characteristics of these signals. Using such derived time series of other than cardiovascular signals the possibility exists to estimate relations between autonomic activity, motor activity, activity of the central nervous system and changes in these relations.

Another field of interest that can be connected to the previous suggestions is the relation of drivers' performance (steering angle, position on the road, velocity etc.) to cardiovascular indices. There are some studies in which this method has already been followed using CARSPAN (Brookhuis, de Waard, and L. Mulder, 1994).

Of course, the spectral computational method used in CARSPAN is not only suitable in cardiovascular research. In all types of data-sets in which only irregularly spaced samples are available a technique as used in CARSPAN is to be recommended. In some cases CARSPAN is directly suitable. It has to be recognized, however, that some specific choices in the program have been made which may be not all relevant in other types of research.

In some cases it even might have advantages to use the CARSPAN program for equidistantly sampled data. A nice example is a study in traffic research described by Brookhuis et al. (1993) in which car-following is studied. From coherence and phase

functions time delay characteristics between 'leader' and 'follower' car have been estimated.

2.4 CARSPAN example

In this users guide an example will be worked out which is related to a mental load study. In this section the research questions to be answered in the study will be described, while in the following sections a more complete description will be given of the relevant program options that are necessary to solve the stated problem.

This example concerns data from a study on the effects of mental task load (5 minutes), after a five minutes baseline registration, on heart rate, blood pressure and its variability. For this purpose subjects performed a memory search and counting task.

Actually, the example file is part of a longer lasting registration in which additional rest and task situations were included (L. Mulder et al., 1992b). Based on the results of previous experiments it is to be expected that HRV in all frequency bands is decreased during task performance as compared to rest, while at the same time variability in all bands of SBP is decreased. Additionally, a decrease of baroreflex sensitivity is expected.

Usually mental load induces an increase in blood pressure and heart rate and a reduction in HRV when compared to a preceding baseline-rest; effects in HRV are most clearly seen in the mid frequency region (about 0.10 Hz). Such effects have been found in several short lasting laboratory experiments (Mulder G., 1980; Mulder L., 1988; Mulder & Mulder, 1987). Other studies (Kamphuis & Frowein, 1985; Veldman, 1992) show that HR decreases and HRV increases as a function of time without a clear indication of a continuing change in invested effort in the task to be performed. At the same time it can be shown that the initial increase in blood pressure gradually diminishes. This is interpreted as an activated baroreflex, which corresponds in many cases with increased baroreflex sensitivity (Mulder, van Roon, Veldman, Laumann, & Burov, 2003).

It is supposed that starting with a mental task induces a new cardiovascular state, including a higher blood pressure and a different pattern in sympathetic/para-sympathetic activation. Blood pressure is seen as the variable to be controlled in short term blood pressure regulation. There are a number of effector systems (HR, heart contractility, total peripheral resistance, venous volume) which are activated to reach this goal, from which HR is the fastest one. After a few minutes other mechanisms are becoming effective: for instance peripheral resistance is increased by vasoconstriction.

The continuing decrease of blood pressure can simply be explained by the fact that despite the different sympathetic/para-sympathetic activation pattern during mental task performance the set-point of the blood pressure control system has not really changed. Therefore, it has to be expected that the baroreflex as well as other control systems, more involved in long term blood pressure regulation, are becoming activated in order to bring blood pressure back to normal levels.

What information has to be obtained from the cardiovascular system and which analyses have to be performed to study these phenomena? Although it would be very interesting to determine total peripheral resistance it is almost impossible to do so with non-invasive methods on a beat-by-beat basis. Therefore, we restrict ourselves to continuous registration of heart rate and blood pressure. Blood pressure is measured continuously at the finger, using the FIN.A.PRESTM method (Settels & Wesseling, 1985). In each beat systolic (SBP), diastolic

(DBP) and mean blood pressure (MBP) are determined. Measures of heart rate can be derived from the interbeat interval times (IBI's). With respect to the computation of these measures it has to be mentioned that the transition from HR to IBI indices (and vice versa) is a non-linear transform. This has strong implications for spectral density measures: it makes a lot of difference whether heart rate variability measures are computed from IBI's or from HR (heart rate), see Mulder (1988).

To overcome this problem in the program mean heart rate values are computed from the number of beats in the measuring period, while spectral variability measures are handled and expressed in heart rate terms, according to the IPFM-model (see Rompelman, 1985). In order to avoid large differences in spectral measures based on IBI rather than HR all spectral measures are normalized to the mean values in the concerning measuring period. By doing the same for blood pressure indices all cardiovascular spectral measures can be viewed of as 'variations around the mean value', i.e. modulation depth (L. Mulder et al., 1992b).

In many measuring systems only uncalibrated respiration is available in which the mean value is close to zero. Normalization will then result in incorrect indices and/or errors in the program. So for respiration absolute spectral values are computed.

Another important variable which can be estimated with spectral techniques is baroreflex sensitivity (BRS, Robbe et al, 1987). In literature BRS is defined as the ratio between changes in IBI and changes in SBP with the units ms/mmHg. For this reason, the gain function (modulus) between SBP and IBI is computed in our example.

Variability patterns in cardiovascular measures can be estimated by means of spectral analysis. HRV in the higher frequency range (Porges et al., 1982) is seen as an index of para-sympathetic activation, under the restriction that respiration rate and depth have not changed too much. Indices for sympathetic activation can be derived either from low frequency changes in blood pressure (Wesseling & Settels, 1985; Akselrod et al., 1985), from the blood pressure level (Van Roon, L. Mulder, Althaus, & G. Mulder, 2004) or from the ratio of HRV in the mid frequency band compared to HRV in the high frequency area (Pagani et al., 1986).

To summarize, for a complete analysis, the following variables have to be derived:

- mean values of IBI, SBP, DBP, MBP;
- spectral power values in low, mid and high frequency band of HR and SBP (in terms of modulation indices);
- coherence and modulus values between IBI and SBP changes in mid and high frequency band;
- spectral power values of respiration in the high and mid frequency band as well as modulus values between respiration and HR changes in the same bands.

In order to see how the transitions between rest and task periods and vice versa will take place as well as to see how cardiovascular indices can change as a function of time spectral profiles have to be computed. These will be restricted in this example to:

- mean values of IBI and SBP;
- spectral power in mid frequency band of HR and SBP;
- modulus in mid frequency band between SBP and IBI.

Before the computations can be done a series of pre-processing steps has to be conducted. These will be described in chapter 3.5 and chapter 8 in which this example will be continued.

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3. Signal processing

In the present chapter the theoretical background and the used algorithms within the CARSPAN package are described. Attention is given to the necessary data-acquisition and calibration procedures to obtain cardiovascular signals from which time series can be derived on a beat-to-beat basis. One of the important steps to obtain a time series on which spectral analysis can be applied is artifact detection and correction. Some theoretical background is given why spectral measures are based on heart rate instead of interbeat interval times and it is described in which way the spectral computation of other types of cardiovascular and non-cardiovascular time series fits nicely well in this concept. The more theoretical part is concluded with some definitions of spectral indices.

The part which describes the algorithms used within the package follows about the same partitioning as used in the theoretical part. All procedures and criteria used for artifact detection and correction as well as the algorithms which are used for computing spectral functions and profiles are described extensively. A separate section is dedicated to the physical units of the calculated measures. The chapter is concluded with a section on applications outside the cardiovascular research.

3.1 Theoretical basis

3.1.1 Data-acquisition

Cardiovascular time series on a beat-to-beat basis have to be derived from a number of signals which can be assessed from the cardiovascular system. The manner in which some of these indices can be obtained from the raw signals will now be discussed in more detail.

- a. Timing information i.e. occurrence time of each QRS-complex is the most relevant aspect of the ECG signal when short term blood pressure regulation has to be studied. This is because interbeat interval time (or heart rate) is one of the main effectors in the short term blood pressure control system. Electrical activity in the heart within a contraction period starts in the SA-node of the right atrium. Conductance of this excitation along the atria and the ventricles results in the P-wave and the QRS-complex, respectively. For practical reasons the R-wave is most frequently taken as the reference point of the complex (see for a more complete description; Rompelman, 1987). From successive R-wave occurrence times interbeat intervals (IBI's) can be derived. Rompelman (1980) indicated that it is more useful to consider the system as a frequency modulator than as an interval generator. This means, that there are more advantages in using heart rate rather than IBI to describe HRV. In addition, Rompelman (1985) argued that spectral density functions from HRV can be obtained directly from the R-wave occurrence times. It is therefore not always necessary to compute IBI or instantaneous heart rate.
- b. Relevant information from the continuously measured arterial blood pressure signal within each heart period includes systolic, diastolic as well as mean pressure, where systolic pressure is the maximum and diastolic pressure the preceding minimum signal value in the considered interval. The mean blood pressure is defined as the mean signal value between two diastolic phases of the heart. Additionally, the pulse pressure can be computed as the difference between systolic and diastolic pressure values. In psychological studies it is usually not possible to obtain invasive blood pressure indices. Non-invasive finger

pressure measurements according to the principle of the unloaded vascular wall (Penaz, 1973) by means of the FIN.A.PRESTM device (Settels & Wesseling, 1985) are therefore a welcome alternative. This device also delivers a continuous signal from which systolic, diastolic and mean values can be derived in exactly the same manner as with arterial blood pressure recordings.

- c. Other cardiovascular indices such as stroke volume, cardiac output, systolic time intervals etc. can in principle be handled in the same way as systolic and diastolic blood pressure values. Additionally, an option for 'pre-computed time series' is available enabling the use of extra time series that cannot be derived with CARSPAN itself. Signals and derived time series have to satisfy certain prescribed criteria in order to be able to handle the specific character of each of these variables and to correct for artifacts in an adequate way. Processing of respiration will be discussed in another section.

Detection of the R-wave in the ECG

The first aspect in achieving an accurate and artifact-free automatic acquisition of R-wave occurrence time is an optimal electrode positioning: this implies that the R-wave amplitude should be high compared to the amplitudes of the other peaks in the ECG and movement artifacts should be held to a minimum. An adequate lead is a pre-cordial bipolar one with electrodes on position V6 (according to Wilson, see Keidel, 1973) and the common electrode at the sternum.

Usually, not only R-wave amplitude but also ECG baseline varies as a function of time which may mainly be caused by respiratory activity. This possible problem affects the design of an automatic R-wave detector. A normal Schmitt-trigger is certainly not sufficient for R-wave detection with an accuracy required for spectral analysis, namely better than 4 ms. Respiratory associated R-wave amplitude variation of about 50% may cause detection errors up to 15 ms, what can be related to the duration of the Q-R interval (about 30 ms). In addition, two types of artifacts may be present under these circumstances: double triggering on both the R-wave and the T-wave (the adjusted level is too low) as well as missing R-waves (the adjusted level is too high). These errors can be prevented by using an adequate R-wave detector (Van Alsté, 1984). In the next version of CARSPAN a tool will become available for software R-wave detection in the ECG.

Calibration of signals

Using cardiovascular signals, calibration is necessary for blood pressure (arterial and venous, as well as for finger pressure), and for respiration when an attempt is made to calculate respiration depth from a combination of chest and abdominal leads. For blood pressure registration an asymmetrical block signal (duty cycle for instance 60%) with a repetition rate of 1 Hz and amplitudes between 0 and 1 V (corresponding with 0 and 100 mmHg respectively) is usually adopted. From these values scale and zero values can be derived; in the next version of CARSPAN a tool will become available determining scale and zero values from these calibration levels. A similar procedure is suitable for other signals.

Sampling and event collection

In psychophysiological research subjects usually obtain information from a display or from other sources and have to respond to it. Time point of presentation and of response as well as signal type (category) and type of response constitute the information to be obtained during data collection. This type of information is indicated as a collection of events. R-wave occurrence times can also be considered as events. At the same time continuous signals (blood pressure, respiration or EEG channels) have to be digitized. These two processes have to be synchronized; otherwise timing errors will occur (Gerritsma, 1982). In the literature, the sampling of continuous signals usually receives much more attention than does event collection, while a combination of both being only seldom used. Mays & McElligott (1976) describe such a combination approach and report a data-acquisition and analysis procedure in which neural spike trains are related to physiological and behavioural data.

In the laboratory of experimental psychology of the University of Groningen a similar approach is chosen and the following procedure is adopted:

- there is one computer clock which controls both processes of event collection and sampling of continuous signals;
- event channel codes and occurrence times are recorded and collected in event files; these events may consist of stimulus and response codes,

R-wave occurrence times from the ECG, inspiration and expiration triggers, start and stop codes for experimental conditions (either introduced by hand or by a computer program), events indicating the occurrence of specific artifacts etc. The required time resolution is 1 ms. Care has to be taken that no events are lost if more than one occurs at the same time (buffering is necessary). Part of the file format which is used is given in figure 3.1. More details can be found in appendix A;

- for the sake of simplicity in data handling, the same sampling rate is chosen for all channels. For pressure signals and respiration this rate is usually 100 or 200 Hz, which is

```
.....
0 303.747 R-peak event code
0 304.583
0 305.388
0 306.142
0 306.852
0 307.589
0 308.392
0 309.225
0 310.028
0 310.788
11 311.514 Begin-of-block code
0 311.734
0 312.265
0 313.096
21 313.900 Begin-of-period code
0 314.649
0 315.369
0 316.083
.....
0 607.844
0 608.562
0 609.320
0 610.101
0 610.858
0 611.594
22 611.740 End-of-period code
0 612.340
0 613.120
0 613.882
0 614.596
0 615.268
0 615.937
.....
```

Figure 3.1: Parts of an event file. Each line contains the event code and event time.

high with respect to frequency contents, but adequate for determining maxima and minima and other derived values.

The resolution required for the A/D converter is at least 10 bits for the blood pressure signals if differences of about 0.5 mmHg have to be detected.

3.1.2 Cardiovascular time series

Beat-to-beat approach

Characteristic values for the different cardiovascular signals have to be derived in each interval. This gives a substantial data reduction. For blood pressure, for example, systolic, diastolic and mean pressure values have to be determined between two R-waves in the ECG. This is done by taking the maximum, the preceding minimum and the mean of the sample values in the selected R-R interval. Data are stored in a 'time stamped data' file which has an ASCII format. In such a file these derived values are combined with the collected events in such a way that the derived values from a certain R-R interval are selected as sample values to be assigned at the time point of the second R-wave of the considered interval. The same procedure as described for arterial pressure can be used for other derived time series.

Respiration

A specific procedure has to be followed for respiration and other slow varying signals if (linear) relationships between these signals and cardiovascular time series are to be studied. The procedure to be followed can now be considered as signal re-sampling at a lower rate: namely, at the time points of each R-wave. Because of the low re-sampling rate this may only be performed on slow varying signals. The procedure that is used includes the following steps:

- determining the new 'sample value' from the sampled data within each set of two consecutive R-peaks. Such a value could be the mean value of a number of samples around the R-peak ('peak value'; this value is called 'sampled volume' in the program) or it could be the mean of the samples in the preceding interval ('beat average'; this value is called 'mean volume' in the program). Of course, other procedures could also be adequate.
- assign the obtained original sample value to the time point of the second R-peak in an R-R set.

This can be done at each R-peak and the obtained sample values together form a new time series representing the time course of the respiration signal. If required, the same procedure can be used for temperature measures, skin conductance, or other slow varying signals.

Some problems may occur with respiration when breathing rate is in the area of half the mean heart rate, which would be excessive high. The proposed method will then distort the respiratory signal. At lower breathing rates, however, usually the main frequency in the respiratory signal remains, while higher harmonics are (partly) removed. However, when only the dependency between heart rate and respiration is studied there is no problem: the spectrum of heart rate as well as systolic blood pressure does not contain any relevant information above half the mean heart rate. Therefore, it would not be meaningful to study the linear (!) relationship between respiration and heart rate or blood pressure for frequencies higher than this intrinsic Nyquist frequency.

Constructing the time series

The series of consecutive beat-to-beat values is called a cardiovascular time series. The time base of this series is formed by the series of R-peak time points. All information which is relevant for HRV spectral analysis is included in this specific time series, namely the time points of the R-peaks constitute the varying heart rate, represented by the time differences between succeeding R-peaks (or its inverse). Examples of a data file in which these time series are stored are given in figure 3.1 and Appendix B.2.

However, before spectral analysis can be applied we have to be sure that no artifacts are present in the derived time series. Additionally, data segments that have to be analysed have to be distinguished from those that are not relevant for further processing. So, three additional steps have to be performed in data pre-processing: data segmentation, artifact detection and correction. It is important that first all these steps are finished with respect to R-peak handling before the next step is taken: creating additional time series with values for e.g. systolic blood pressure, respiratory amplitudes etc. The series of R-peaks constitute the time axis that forms the basis for all other beat-related time series.

After this step, other time series can be created, while, as a next step, error detection and correction has to be performed in about the same way as for the R-peak event series. These data will then be written to a so-called extended event file; see Appendix B.3 for an example.

All information that is relevant for spectral analysis of all time series is then available: e.g. varying systolic blood pressure as a function of time, the respiratory pattern represented by the non-equidistant sampled (at time points of R-peaks) original respiration signal, etc.

3.1.3 Pre-processing

Data segmentation

Data files can be segmented in 'blocks' and 'periods'. Periods are the analysis segments on which the computation of the spectral functions is carried out. Blocks may contain more than one period. It is expected that within blocks there are no time gaps longer than normal interbeat interval times. Files may contain more than one block; time gaps between blocks within a file are allowed. In order to control data segmentation, the length of periods can be specified in terms of time units (seconds). Another option is to insert start and stop codes in the data file for both blocks and periods which can be used optionally to specify the length of the data segments. The availability of both procedures guarantees a flexible and simple data segmentation in all kinds of laboratory and field experiments.

The procedure for the computation of spectral profiles, i.e., spectral power in a specified frequency band as a function of time, makes use of this block structure also. For this procedure the division into periods is no longer relevant. Spectral values in the specified frequency bands are now computed from small time segments called 'windows'. Usually, consecutive analysis segments in this profile procedure are partly overlapping, which is indicated by the 'step size'. The ratio between 'window' and 'step size' gives the amount of overlap between consecutive analysis segments.

Artifact detection and correction

R-peak event series

Artifacts can be divided into two categories:

- I. Measurement or technical artifacts; due to failures in the registration system (disconnected ECG electrode) or noise (e.g. body movements);

II. Physiological irregularities; the heart rhythm is irregular (arrhythmia), caused by extra systolic beats or respiratory phenomena (sighing).

It is clear that artifacts of category I have to be corrected because they cause additional variability that is not related to the phenomena to be measured. Extra beats have to be removed and missing beats should be reconstructed, if possible and in case not many beats are affected. But first of all the artifacts must be detected. Whether artifacts of category II have to be corrected (or can be corrected) is dependent on several factors, including the origin of the irregularities and the number of artifact beats per unit of time.

Measurement artifacts in a series of R-wave occurrence times may consist of either missing R-waves or additional events within one R-R interval due, for instance, to T-wave triggering. Additionally, deviating patterns in IBI sequences may be caused by rhythm disturbances, which may be called 'physiological artifacts': disturbances in the atrial-ventricular conductivity or in the generation mechanism of the QRS-complexes in the ECG. All these artifacts may have an appreciable effect on HRV, particularly when intrinsic signal variability is low. The following example may illustrate the seriousness of the problem. Suppose an artifact free series of 101 IBI values is present with a mean of 'MIBI' and a variation coefficient which is at the upper end of the normal range, say 10%. The variance corresponding to this 10% variation coefficient is equal to:

$$s^2 = 0.1^2 \times MIBI^2 = 0.01 \times MIBI^2$$

Next, one event at a random point in time is removed from the series, resulting in a prolonged IBI, with expected value of $2 \times MIBI$, at the occurrence time of the R-wave event following the inserted artifact. The additional variance caused by the prolonged IBI equals:

$$s_{add}^2 = \frac{1}{100} (2 \times MIBI - MIBI)^2 = 0.01 \times MIBI^2$$

This means that one prolonged interval contributes as much variability as the 100 remaining intervals of the original series! This example shows the relevance of detecting existing artifacts and of correcting them whenever possible (Mulder, 1992a).

Two types of problem may arise during the detection and correction of artifacts: the first is related to the question of whether a seemingly deviating IBI value really is an artifact. The second problem occurs when decisions have to be made about whether or not it is permissible to correct deviating IBI's when the cause of these disturbances has a physiological background. In the second case such characteristic patterns may occur repeatedly within the registrations of a particular subject. If this occurs it can be advisable to remove the data of such subjects from the final analysis.

Artifact detection and correction remains a difficult problem. At present our opinion is that complete automatic detection, classification and correction is almost impossible for all types of artifact and all types of data. Therefore, we have chosen for the approach to restrict the effort in automatic detection and classification and to ask the user of the program to visually check the outcomes of the artifact correction procedure and to correct manually where necessary.

The present detection procedure has two components: 1. checking whether a specific IBI deviates more than a number of standard deviations (SD) from a running mean; 2. checking whether the difference between two consecutive IBI's is larger than a number of SD's (e.g. 5) from the running mean of consecutive IBI-differences. Deviations are seen as artifacts when one of these two criteria has met. More specifically the procedure works as follows: at first the mean IBI value and its standard deviation are computed in a time window preceding the

present interval; the windows duration can be specified (e.g. 60 seconds). The data window is shifted throughout the data file, while mean and SD values are updated at each correct beat. According to this criterion, IBI values are seen as artifacts if the deviation of the current window mean is larger than a certain number of standard deviations to be specified (e.g. maximal deviation is 5 SD). The second component in the procedure works in about the same way, but now it is checked whether the successive difference value at a certain beat is larger than a number of 'SD of successive differences' (e.g. 5). The 'SD of successive differences' is computed from the same time window that is used for the running mean and SD of IBI-values. Several types of classification of artifacts are possible (Mulder, 1992), however, experience has learned that the way of correction is not strongly dependent on the type of artifact; therefore classification is restricted to a couple of classes. The correction itself is based on a linear interpolation technique. A new IBI value (based on R-peak times) is calculated for an artifact using an interpolation between two preceding and two succeeding correct values. Additionally, care will be taken that the overall time axis will not be changed outside the corrected window. Finally, a small amount of noise is added in order to prevent excessive reduction in variability due to the correction itself. The algorithms used for detection, classification and correction are described in section 3.3.1.

Other time series

Every other time series, such as systolic or diastolic blood pressure, may require artifact correction. However, the effects are usually less disastrous than for heart rate. The deviations found are seldom of the same order of magnitude as the mean value of the series itself which differs from the IBI series. However, if also in these time series artifacts with very large deviations from the mean are present exactly the same effects should be expected as in the IBI example.

For the detection of artifacts in these types of signal the same main rules as described for IBI fluctuations are followed. For this purpose, a general artifact detection and correction option is available. A value in a time series is an artifact if it is out of range. The range is determined by the current running mean plus or minus a number of SD's in the same way as explained for the IBI's. The procedure for successive differences is not implemented for other time series. With respect of correction, also the same procedure is used as for IBI's: a new value is calculated for an artifact using an interpolation between two preceding and two succeeding correct values. For other time series there is no noise addition implemented

In order to improve artifact handling it might be useful to obtain as much information as possible from other independent sources in order to examine a possible cause of artifacts; this may characterize the types of deviation to be found. For example, artifacts in finger pressure measurements can be due to the automatic 'physiocal' procedure of the device which is applied every 60 or 70 beats (Settels & Wesseling, 1985). On the other hand, artifacts in respiration might be due to body movements.

Because the respiration signal is not of the beat-to-beat type, artifact detection and correction with respect to respiration frequency is not possible in CARSPAN in the same way as it is for cardiovascular data. However, it is possible to change several parameters in RespTool to optimize the recognition of respiratory cycles. See Appendix ... for further details.

In chapter 2.2.3 and 2.3.5 of Mulder, 1988 more information on this topic can be found. The used algorithms for the derivation of systolic, mean and diastolic pressure, plethysmogram, and respiratory time series are in section 3.3.2.

3.1.4 Spectral processing

Heart rate spectra

There are many ways to compute spectral functions of heart rate (Berntson et al., 1997; Rompelman, Coenen and Kitney, 1977). In CARSPAN an approach is followed which is based on the fact that the only information which is available in the heart rate signal is related to the time points of arrival of the series of R-peaks. The waveform of the QRS-complex in the ECG itself is not of any relevance for determining heart rate. The second assumption that can be made is that fluctuations in heart rate have to be studied (i.e., heart rate variability). This implies that only the low frequency part of the spectrum, which means below half the mean heart rate, has to be studied. Hyndman & Mohn (1975) have shown that such a slow varying heart rate signal can be obtained by low pass filtering of the series of unity pulses corresponding with the time points of the R-peaks in the registration. By choosing an ideal low-pass filter (rectangular transfer function in the frequency domain, $\sin x/x$ impulse response function) all information with respect to heart rate variability is regained theoretically in the filter's output. Rompelman (1975) has shown that it is not necessary to do this filter operation in the time domain. Exactly the same spectra are obtained by directly calculating the Fourier spectrum of the original series of unity pulses, corresponding to the R-peak event-times. For heart rate variability we have chosen exactly the same approach as Rompelman did.

It is important to know that the resulting spectra are representing heart rate (HR) changes and not changes in inter beat interval (IBI). A theoretical reason for this approach is related to the adoption of the Integral Pulse Frequency Model (IPFM) as a model for the generation of heart beats (Hyndman & Mohn, 1975). A practical advantage of the chosen technique is that it is not necessary to perform interpolation between successive heart beats in order to obtain an equidistant time series: the spectral computations can be based directly on non-equidistant time series. A disadvantage with respect of computational speed is that the Fast Fourier Transform (FFT) cannot be used. However, also the restrictions of the FFT technique are out of order; in principle spectra can be computed of each specified length of time segment. Practically, the computational speed is high enough for all types of application met up to now.

Other time series

Fortunately, about the same approach can be followed for other time series than heart rate. The procedure works as follows. Suppose, at each heart beat a value is obtained of another signal, e.g. systolic blood pressure. The obtained series of (non-equidistant) sample values of systolic blood pressure can be Fourier transformed by applying the Direct Fourier Transform (DFT) on this series. It can be proven that the results of such a transformation are not different from an equidistant Fourier transform as long as the contributions at each beat are weighted with the length of the preceding interval time (Gaster & Roberts, 1977). Details concerning this technique are described in section 3.3.; more information can be found in Mulder, L. (1988, chapter 3.2.).

IBI-series

An additional feature of the described technique is that spectra of IBI can be obtained also. By defining the series of IBI's as an additional time series, in exactly the same way as with systolic blood pressure values, the same spectral procedure can be followed for the IBI-series. This enables even the possibility for determining the relation between changes in heart rate and IBI.

Spectral Functions

The spectral functions which are computed are power spectral density, coherence, modulus (gain) and phase functions. Derived spectral measures are related to spectral power, weighted coherence, mean modulus and phase in specified frequency bands. See section 3.1.5 for more details.

Spectral profiles

Spectral profiles are functions giving the time course of specified spectral measures, for instance the power in the mid frequency band of heart rate as a function of time. The computational procedures are exactly the same as those used in normal spectral analysis. However, now each of the spectral measures is computed on a small time segment (window) of for instance 30 or 50 seconds. A series of such spectral measures, constituting a new time series is obtained by selecting a new analysis segment which is partly overlapping with the previous one. The technique can be applied not just for spectral power measures, but for weighted coherence, mean modulus and phase indices as well. The described technique is in particular useful in those situations in which non-stationary patterns of cardiovascular state are expected (Mulder, Rusthoven, Kuperus, De Rivecourt, & De Waard, 2007). More details about the implementation of this technique and choices concerning window size and the amount of overlap are given in the next sections.

Normalised spectral power measures

De Boer, Karemaker & Strackee (1985) and Van Dellen et al. (1985) have shown that absolute spectral variability measures of HR and IBI are strongly dependent of mean IBI. De Boer et al. (1985) calculated that the ratio of HR spectral variability values and IBI spectral density values at a certain frequency is inversely proportional to $(\text{mean IBI})^4$. Van Dellen et al. (1985) illustrated this same effect with experimental data. It was shown that the dependency of the mentioned ratio and mean IBI disappeared when normalized data were used. Normalized data here means: variations relative to the mean value (modulation index, variation coefficient). In the concept of the IPFM, the modulation index refers to the depth of frequency modulation. According to this model it is more reasonable to use normalized values rather than absolute values in either IBI or HR.

3.1.5 Spectral Indices

Definitions of frequency bands

Spectral variability measures are defined as the total spectral power in a considered frequency band, i.e. the integral of the spectral density function in the relevant frequency area. The frequency resolution chosen between consecutive points in the spectrum is equal to 0.01 Hz (see section 3.3.4). Therefore boundaries of the spectral bands are indicated in multiples of 0.01 Hz. The following bands are usually applied in studies on mental load (Mulder, L., 1988):

- TOTAL spectrum, between 0.02 - 0.50 Hz. Spectral computations are in any case restricted to one half of the mean heart rate (inherent Nyquist frequency). The lowest frequency point (0.01 Hz) is not included in order to reduce possible trend effects.
- LOW frequency band, between 0.02 - 0.06 Hz, in which variations due to temperature regulation and adaptations to the task demands are reflected.
- MID frequency band, between 0.07 - 0.14 Hz, in which variations due to regulation of vasomotor tone are reflected. Fluctuations in this band are connected to resonance phenomena (Wesseling & Settels, 1985) or even oscillation of the short term blood pressure control system (Hyndman, Kitney & Sayers, 1971).
- HIGH frequency band, between 0.15 - 0.40 Hz (or 0.15 - 0.50 Hz). Respiratory activity and effects of task induced signal repetition are mainly reflected in this region; as to whether these effects are really found in this area depends on the main breathing frequency and/or signal repetition rate.

For other applications, some further specific frequency bands are defined:

- LF (Low) frequency band, between 0.04 and 0.14 Hz, according to Pagani and colleagues reflecting sympathetic activity of the autonomous nervous system (Pagani et al, 1986; Berntson et al., 1997).
- HF (High) frequency band, between 0.05 and 0.40 Hz, according to Pagani and colleagues reflecting parasympathetic activity of the autonomous nervous system (Pagani et al, 1986; Berntson et al., 1997).
- VLF (Very Low frequency band), between 0.004 and 0.03 Hz.
- RESP band: a small frequency band just around the main respiration rate ($F_{resp} \pm 0.03$ Hz). Of course, since the respiratory band depends on the respiration rate of the subject, it will be different for each subject and measuring period. Such a small band can only be used if the respiratory signal is measured and if the main breathing frequency can be adequately determined. Results indicate that it is certainly useful to distinguish the respiratory frequency area from the whole frequency region above 0.15 Hz, i.e., to sharpen the relevant area and to remove non-relevant variance. This option is particularly useful if the band can be adapted dynamically to the subjects' respiratory pattern.
- TASK band: a small band just around the event rate of the task. In general, this band is only relevant in measuring conditions in which subjects have to perform a machine-paced task. In many cognitive tasks this frequency is higher than 0.15 Hz (interstimulus time smaller than 7 seconds).

Weighted coherence

In some studies the index 'weighted coherence' is used as a measure for dependency between two cardiovascular time series. The coherence function indicates the amount of (linear) coupling between two signals as a function of frequency. The function approaches 1.0 if signals are strongly (linear) related and diminishes to zero if there is no linear relationship at all in that frequency range (L. Mulder, 1985). Function values can be compared with variance explained (squared correlation coefficient) in a linear regression equation. Porges et al. (1981) presented the 'weighted coherence' as a summarized coherence measure in the considered frequency band, indicating the amount of variance explained in signal x by signal y . The precise definition is given in section 3.3.4. This measure is used several times (Robbe et al, 1987) for estimating the dependency between fluctuations in systolic blood pressure and IBI. It appears (L. Mulder, 1985) that this coherence is only large in the MID and HIGH frequency band.

Baroreflex sensitivity

Baroreflex sensitivity (BRS) reflects the ratio between changes in IBI and systolic blood pressure. This ratio can be determined in several ways (Smyth, Sleight & Pickering, 1969, Eckberg, 1977). In CARSPAN a spectral approach is chosen. According to this method, BRS is determined from spontaneous fluctuations in systolic blood pressure and IBI. Only those spectral areas (MID and HIGH frequency band) are selected in which there is a strong relationship between these time series. Also other, more specific, bands in these frequency regions, such as the RESP and TASK band, can be used for this purpose. The procedure used within CARSPAN works as follows: the modulus (gain) function is computed from the Fourier transformed IBI and systolic blood pressure series. Then the mean modulus is computed in the considered frequency band while only those frequency points are included which have a squared coherence higher than a specified value (e.g. 0.30).

3.2 Choices made during the program development

Before the implementation of the algorithms to be used actually can start some choices have to be made that were not discussed in the previous sections and that will not be handled in the next sections either. The following choices are relevant in order to understand the described algorithms:

- All input and output files are ASCII files, except for the sampled data file which has a structured binary format.
- Prior to spectral computation, all time series are multiplied with a 5% tapering window to prevent strong boundary effects at the beginning or end of each data segment. In this way leakage effects (Otness & Enochson, 1972) are restricted.
- In CARSPAN no filter or other de-trending algorithm is used to remove trends from data. Years of experience with linear trend compensation, higher order detrending and digital high pass filtering has brought us to the conclusion that the only good way is high pass filtering. However, there is almost no difference between high pass filtering and leaving out the very low frequencies (up to 0.01 Hz or 0.02 Hz). Because of the $1/f$ nature of cardiovascular spectra (Kobayashi, 1982) it is not to be advised to use higher order de-trending, while linear de-trending is too much dependent on the length of the analysis

segment in relation to the time duration of the ‘trend’ that occurs. However, it should be kept in mind that there are exceptional situations in which trends determine the whole lower part of the spectrum. This can only be seen by visual inspection of the data. Influences on the mid frequency region are, however, really very seldom.

- No smoothing of the spectra is carried out on the spectra before computing the spectral band values, while a moving average window over three frequency points (0.03 Hz bandwidth) is applied before plotting the spectral functions. In the literature several methods are described to increase the reliability (degrees of freedom) of the obtained spectral estimates (Otness & Enochson, 1972) and to get smoothed plots of spectra. In CARSPAN it is decided not to implement such smoothing routines on the spectral bands because it does not improve the estimated spectral band values. The only factor which increases the reliability of each spectral (output) point is the integration/interpolation routine which is used to get a frequency resolution of 0.01 Hz. As an example: the number of degrees of freedom per spectral point is increased in this way with a factor 3 when the analysis period length is increased from 100 to 300 seconds. If the user wants to have more smoothed spectra it is advised to use a smoothing algorithm on the (interpolated) spectral output data. Another exception on this approach is, of course, the necessary frequency averaging over three points (i.e. a bandwidth of 0.03 Hz) which is used for coherence, modulus and phase computation (see section 3.3.4).
- No statistical analyses are performed by CARSPAN. The output files are suitable for most packages for statistical analysis. An additional tool is becoming available in the next CARSPAN version to select specific variables from the output records to make it more suitable for further statistical analysis and/or further processing with the baroreflex model (Van Roon, Mulder, Althaus & Mulder, 2004).

3.3 Used algorithms in CARSPAN

In this chapter all algorithms used in CARSPAN are discussed. Again, a distinction is made between the R-peak events series and the other time series. The algorithms are in many cases very similar, but never exactly the same. Firstly, the signal pre-processing (sections 3.3.1 and 3.3.2) is discussed. The user can follow the processing step by step and can check the calculations made in the CARSPAN computing environment. Secondly, the analysis algorithms are presented in the same way (sections 3.3.3-3.3.5). Finally, the units of the computed variables are discussed in section 3.3.6.

3.3.1 Pre-processing of R-peak events

The pre-processing of the R-peak events is a procedure for artifact detection and correction. The detection of artifacts is a simple test whether an interbeat interval IBI_i and the difference between 2 succeeding beats is within certain limits. If the decision is made that the interval is too large or too small, a classification of this beat is made. Every class requires a specific type of correction.

In this section the implemented decision rules are presented.

Definitions

Current Interbeat Interval length	IBI_i
Previous Interbeat Interval length	IBI_{i-1}
Window length for running mean calculation	T_w
Running IBI mean	$\overline{IBI}_r$
Running IBI standard deviation	sd_{IBI_r}
Running IBI Successive Difference standard deviation	sd_{SuDr}
Refractory period	T_{refr}
Maximum acceptable interpolation period	T_{max}
Maximum deviation in no. of standard deviations	N_{IBI}
Maximum deviation in no. of successive differences	N_{SuD}

Running IBI Successive Difference is determined by calculating the average of the difference between every two consecutive IBI's in the preceding window.

Detection algorithm

$$\text{Normal beat if: } -N_{IBI} \cdot sd_{IBI_r} < IBI_i - \overline{IBI}_r < N_{IBI} \cdot sd_{IBI_r} \text{ and} \quad (3.1)$$

$$-sd_{SuDr} \cdot N_{SuD} < (IBI_i - IBI_{i-1}) < sd_{SuDr} \cdot N_{SuD} \quad (3.2)$$

$$\text{Short beat if: } IBI_i \leq \overline{IBI}_r - N_{IBI} \cdot sd_{IBI_r} \text{ or } (IBI_i - IBI_{i-1}) \leq -sd_{SuDr} \cdot N_{SuD} \quad (3.3, 3.4)$$

$$\text{Long beat if: } IBI_i \geq \overline{IBI}_r + N_{IBI} \cdot sd_{IBI_r} \text{ or } (IBI_i - IBI_{i-1}) \geq sd_{SuDr} \cdot N_{SuD} \quad (3.5, 3.6)$$

Classification and correction

The classification of the type of artifact (a kind of artifact 'diagnoses') is necessary to decide whether a correction of the artifact is allowed. Each class requires its own kind of artifact correction (this can even be no correction at all). When a beat is considered to be normal (according to the detection algorithm) no classification is made. For abnormal intervals, the procedure is split into two cases: interval too short and interval too long. For each case a classification scheme is given. In these schemes abbreviations are used which are explained in figure 3.2. In this figure three R-peaks are shown, at t_{i-1} , t_i , and t_{i+1} . The interval between t_i and t_{i-1} has length IBI_i and the interval between t_{i+1} and t_i has length IBI_{i+1} . For the classification not only interval i plays an important role (that is the interval which is too long or too short) but also the next interval $i+1$.

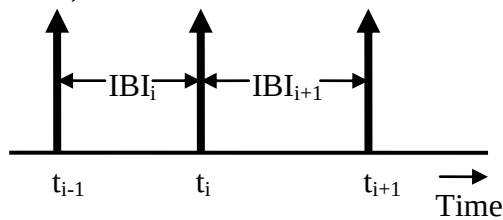


Figure 3.2: Definitions for artifact classification and correction schemes.

Actually, there are three steps in the detection and correction process:

1. In the first step detection of either too short or too long intervals or detections of too large positive or negative changes in successive beats are performed, according to the given definitions.
2. In the second step certain IBI patterns are searched for. These include: a. ‘Special artifact’: an abrupt, impulse like vagal inhibition/activation patterns; in the IBI series sudden vagal activation patterns can, for instance, be characterized by a sharp increase in IBI (e.g. a doubled interval length), followed by an exponential decrease to the preceding level in about 3 to 4 beats. b. A short/long IBI pattern e.g. an (S)VES. c. A short-normal-short IBI combination.
3. The third step includes a classification of the detected pattern, followed by a correction where possible and allowed. The following classification/correction scheme is used in the program.

<u>Classification</u>	<u>Action</u>
- IBI_i = too short:	
If $IBI_i < T_{refr}$:	Remove R-peak
If $IBI_i > T_{refr}$:	
If number_fitting intervals = 1:	Remove R-peak
If number_fitting intervals = 2:	Remove R-peaks
	Interpolate
Otherwise: unknown artifact:	No correction
- IBI_i = too long:	
If $IBI_i > T_{max}$:	No correction
Otherwise: unknown artifact:	Interpolate
- IBI_i = vagal activation:	Warning: ‘special artifact’; no correction
- IBI_i = short-long:	Warning: "possibly (S)VES"
	Remove R-Peak, interpolate
- IBI_i = short-normal-short:	Warning: "short-normal-short"
	Interpolate

In the final correction part R-peaks can be removed or intervals can be interpolated (insertion of additional R-peaks), however, the whole time axis will never be shifted. This is important because otherwise phase distortion with other signals might occur.

Interpolation: the value of the IBI is estimated by linear interpolation, using two preceding and two succeeding correct IBI's. The number of beats that fit best in the gap is found by minimizing the squared difference e_m between the gap length and m average beats:

$$e_m = \left(m \cdot \overline{IBI_r} - IBI_i \right)^2 \quad e_m > e_{\min} \text{ for } m \neq \min \quad (3.7)$$

A linear interpolation will reduce the variance in the signal. To prevent a large reduction in variance, a small amount of noise is added to the interpolated values. The noise that is used for this correction is normally distributed with zero mean and standard deviation equal to a

quarter of the segment standard deviation. However, the total length of the sum of the new intervals is always equal to the original IBI_i .

Parameters

The parameters used in the pre-processing of R-peaks in the program are default set in the CarspanDefaultSettings.ini file; the values in the next table can be useful to start with.

Parameter	Value	Unit	Description
T_w	60.0	s	time window
N_{Cor}	10.0		maximum number of successive beats to be corrected
T_{refr}	0.3	s	refractory period (cannot be changed)
N_{IBI}	5		number of standard deviations
Min SD	5	%	if N_{IBI} is lower than this value, Min SD is used for setting detection boundaries
Max SD	15	%	if N_{IBI} is larger than this value, Max SD is used for setting detection boundaries
N_{SuD}	5		number of standard deviations for successive differences
Min SuD	5	%	if N_{sud} is lower than this value, Min SuD is used for setting detection boundaries
Max SuD	25	%	if N_{sud} is larger than this value, Max SuD is used for setting detection boundaries

All values except T_{refr} can be specified by the user. This refractory period is equal to the inactive state of our hardware R-peak trigger. This means that the highest heart rate that can be measured is less then 180 beats/minute, which is sufficient for most applications.

3.3.2 Pre-processing of other time series

The pre-processing of the other time series is a two step process. In the first step, the series of interest are derived from an equidistant sampled signal. The algorithms for two signals are discussed in this section: blood pressure and respiration. A special case of another time series is the interbeat interval series; it is discussed as a third signal.

The artifact detection and correction is the second process and can be the same for all other time series, including a time series that has not been derived by means of CARSPAN (A user-defined time series). This artifact handling is discussed at the end of this section. The interbeat interval series will never need correction, because it has been corrected already in its original form as a series of R-peak events (see section 3.3.1.)

Derivation of other time series

For all these kind of variables, the values are derived between two R-peaks. The value determined between these peaks is assigned as the ‘beat value’ at the second R-peak of the interval. The reason for this choice is that blood pressure values for example are only available after the beat is finished.

WARNING: This is an important notice when phase relations are studied and a comparison is made with studies in literature. If the first peak is chosen in such a study, the phase function $\phi(f)$ will be different due to a difference in method. The difference is in principle unknown

(different shift for each beat), but can be estimated by expressing the mean interbeat interval (in seconds) in radians:

$$\phi(f) = 2\pi \cdot \overline{IBI} \cdot f \quad (3.8)$$

Usually, recalculation of the sample values is necessary to get right dimension. A scale factor (*scale*) and zero value (*zero*) can be assigned to each time series. The original time series $x_{org}(t)$ is converted to new time series $x(t)$ according to:

$$x(t) = x_{org}(t) \cdot scale + zero$$

Blood pressure

The blood pressure signal can provide the following variables:

- Systolic blood pressure SBP [mmHg]
- Diastolic blood pressure DBP [mmHg]
- Mean blood pressure MBP [mmHg]
- Pulse blood pressure PBP [mmHg]

The SBP is detected as being the maximum of the signal between two R-peaks. The DBP is found by detecting the pressure minimum preceding the SBP maximum. The MBP is calculated as the mean of pressure values between the two R-peaks. The PBP is calculated as the difference between SBP and DBP. In case of a partly missing blood pressure signal (like the physio-cals with the FIN.A.PRES™ finger pressure device) zero values are returned (indicating missing values).

Respiration

Several types of time series can be derived from the sampled respiration signal; these include:

- Re-sampled amplitude values
- Respiration cycle time related values, such as respiration frequency, respiration cycle time, duration of inspiration and expiration.

The respiration signal is re-sampled at the R-peaks, to obtain beat-to-beat values. There are two options available to determine a sample value:

- beat average; the average of the respiration samples between two R-peaks is calculated and assigned as beat value; this is indicated in the program as 'mean volume.'
- peak average; a number of samples preceding and succeeding the R-peak are averaged and assigned as the beat value; this is indicated in the program as 'sampled volume.' The user can specify how many samples are used for this peak average. In case of noisy signals the number of samples to be averaged can be chosen larger; for a first indication: the effective bandwidth of a corresponding low pass filter is about $2/T_d$, in which T_d is the time duration of the interval included in the average.

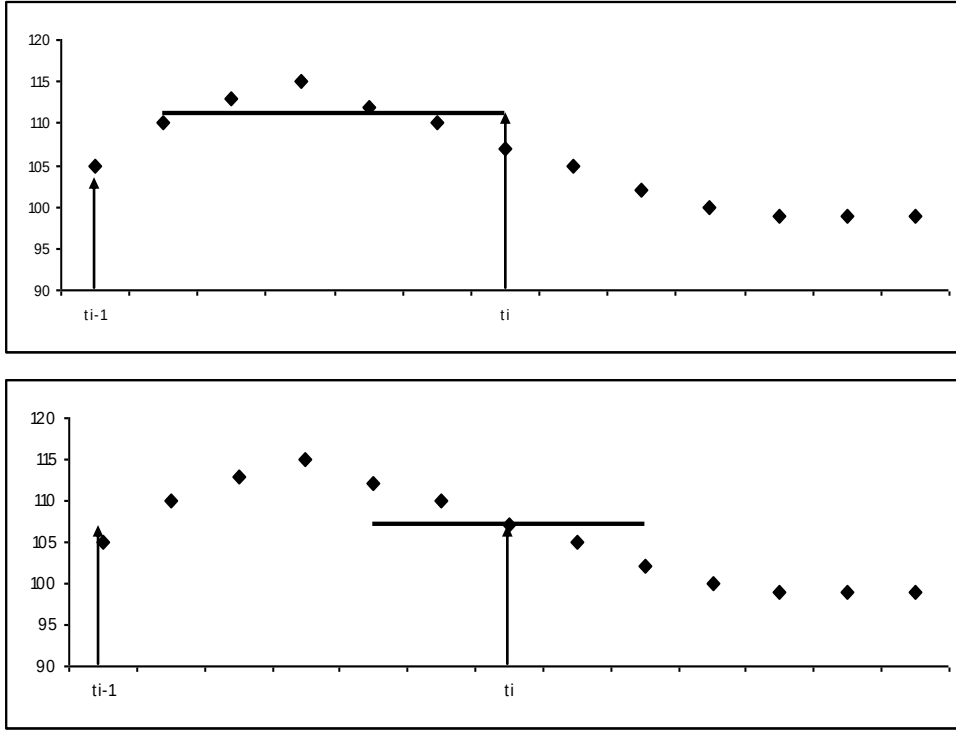


Figure 3.3: Re-sampling of the respiration signal. The new value is assigned to t_i

In figure 3.3 is shown which section of the signal is averaged for these two methods. It is clear that the phase shift between respiration and other time series is different for the methods. The phase difference between the methods can be approximated with the following function:

$$\phi(f) = \pi \cdot \overline{IBI} \cdot f \quad (3.9)$$

This is the phase shift of a time delay of half the mean interbeat interval, expressed in radians.

Respiration cycle times

It is informative to have at any time point an indication of the respiration frequency or related values. In the current version of CARSPAN the respiratory frequency is determined by estimating cycle duration. This can be achieved by means of the program RespTool, which differentiates the respiratory cycli in a period of interest.

The basis is formed by the maximum values of the respiratory signal and its corresponding time points. Because the respiration cycle is always longer than an R-R interval, at each beat (R-peak time point) the value of the cycle related parameter is kept the same as the previous value, unless a new maximum value of the respiration signal is detected in the preceding R-R interval. If such a maximum is found, for the corresponding R-R interval an interpolation is performed between the preceding and the successive value. RespTool creates a new time series under the name of Resp Period, showing the duration (in seconds) of the respiratory cycli. For further details and a users guide regarding RespTool, see Appendix ...

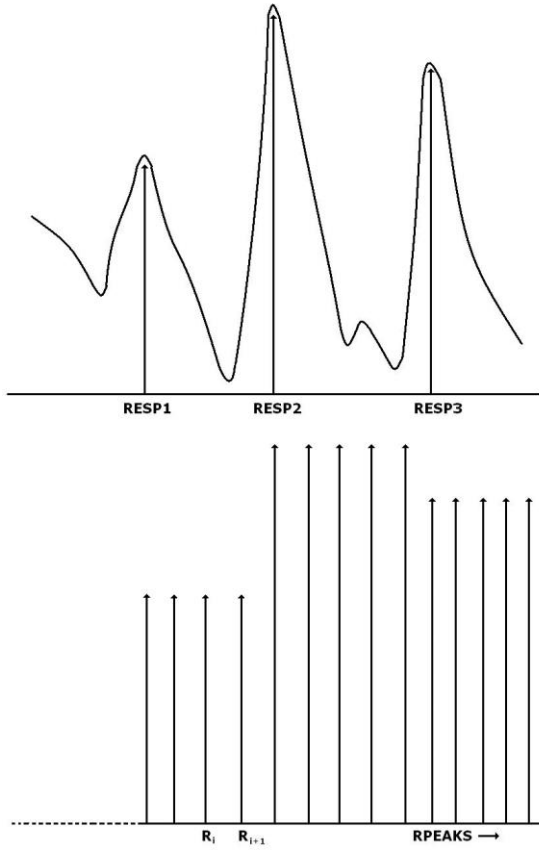


Figure 3.4: Respiration signal (top) and derived respiratory amplitudes at R-peak time points (bottom) are shown. For each heart beat the corresponding respiration amplitude is calculated.

Note: the described algorithms for respiration are not yet all available but will be implemented in one of the next versions of CARSPAN.

Interbeat interval

In section 3.1.4 it is made clear that the spectrum of the R-peak event series is a spectrum of heart rate. In some cases it is required to calculate the spectrum of interbeat interval (IBI series). An example is the estimation of the baroreflex sensitivity, which is defined as the relation between changes in systolic blood pressure and interbeat interval values. Therefore an IBI series is derived from the R-peak events with a simple algorithm. The value of IBI_i is assigned to the R-peak event at the end of the interval (see figure 3.2).

Other time series

Other types of time series can be added by the user as precomputed time series. Examples are plethysmogram values, values of a second blood pressure signal, and stroke volume values. When you add an external signal, be careful and aware of the warning concerning the phase shift, mentioned above. Use zero values as missing values, which are interpolated by the next pre-processing step, the artifact correction.

Correction of time series

The second processing step is artifact detection and correction of the newly created time series (such as systolic blood pressure or a precomputed time series.) The time series $x(t_i)$ is checked for values that are missing (zero values) or that are out of range. The method for the

range-check is based on running statistics. From a small part of the signal the mean and standard deviation is calculated. The next value is compared with the running mean. If the value differs more than a number of standard deviations from the mean, it is considered to be an artifact because the value is out of range.

If it is within the range (not an artifact) the new statistics are calculated of the time series including this value. Otherwise, the next value is checked. This is repeated until two succeeding values are found in range.

The values which are out of range are linear interpolated, using two values in range before and two values after the artifacts. Note: one of the differences with artifact detection in Rpeak series is that the check on successive differences is not performed during time series correction.

During the derivation of time series, missing values can be returned. These missing values (coded as zero) are processed in exactly the same way as detected 'out of range' values.

Definitions

Current value	$x(t_i)$
Signal length for running statistics calculation	T_w
Running mean of $x(t)$	$\bar{x}_r$
Running standard deviation	$sd_{x,r}$
Maximum deviation in no. of standard deviations	N_x

Detection algorithm

Missing values: 0 (zero)

In range if: $-N_x \cdot sd_{x,r} \leq x(t_i) - \bar{x}_r \leq N_x \cdot sd_{x,r}$ (3.10)

Out of range if: $x(t_i) < \bar{x}_r - N_x \cdot sd_{x,r}$ or $x(t_i) > \bar{x}_r + N_x \cdot sd_{x,r}$ (3.11)

Correction

When a number of succeeding artifacts (missing and/or out of range; not more than the maximum number as specified in the CarspanDefaultSettings.ini file) are preceded and succeeded by two correct values, linear interpolation is used to calculate new values. If there are no preceding or succeeding values (at the beginning or end of an analysis segment) the running mean is assigned as new value(s).

In those cases that there are too many succeeding artifacts, no correction is made. The user has to inspect the data carefully, and probably divide the analysis segment into two segments, excluding the part with too many artifacts.

Parameters

They are specified in the CarspanDefaultSettings.ini file; useful starting values are listed in the table.

Parameter	Value	Unit	Description
T_w	60.0	s	time window for running statistics
N_{Cor}	10		maximum number of successive values to be corrected
N_x	5		number of standard deviations
Min SD	5	%	if N_x is lower than this value, Min SD is used for setting detection boundaries
Max SD	15	%	if N_x is larger than this value, Max SD is used for setting detection boundaries

These values can easily be changed by the user. Although five times the standard deviation seems a wide range, you must remember that the range is based on the running mean and standard deviation. In some parts the standard deviation can be excessively (accidentally) small what may result in a very small range when a small detection boundary is used.

3.3.3 Definitions and measures of time series

The time-axis is defined by means of appearance of R-peak events. A mathematical description of the R-peak event series is:

$$x(t) = \sum_{i=1}^N \delta(t - t_i) \quad (3.12)$$

Where: i peak number
 N number of R-peaks
 t_i time of occurrence of R-peak event i
 $\delta(t-t_i)$ unit Dirac pulse at time point t_i

This is the mathematical representation of series of unit Dirac pulses assigned to every R-peak (see section 3.1.4).

In case of (other) time series than R-peak events, the definition is extended. These time series $x(t)$ can be estimated by:

$$x(t) = \sum_{i=1}^N \Delta t_i x_i \delta(t - t_i) \quad (3.13)$$

where: $\Delta t_i = t_i - t_{i-1}$
 x_i = sample value for the relevant time series in beat i

These signal descriptions will be used in the following section on the algorithms used by CARSPAN. Two time domain measures are derived for each segment; the mean and variation coefficient. For the R-peak event series these measures are defined as:

$$\overline{HR} = \frac{1}{\frac{1}{N} \sum_{i=1}^N (t_i - t_{i-1})} \quad (3.14a)$$

$$\overline{IBI} = \frac{1}{N} \sum_{i=1}^N (t_i - t_{i-1}) \quad (3.14b)$$

$$sd_{IBI}^2 = \frac{1}{N-1} \sum_{i=1}^N [(t_i - t_{i-1}) - \overline{IBI}]^2 \quad (3.14c)$$

$$v_{C_{IBI}} = \frac{sd_{IBI}}{\overline{IBI}} \cdot 100\% \quad (3.14d)$$

For the other time series, these measures are defined as:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (3.15a)$$

$$sd_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (3.15b)$$

$$v_{C_x} = \frac{sd_x}{\bar{x}} \cdot 100\% \quad (3.15c)$$

The standard deviation is only used for the calculation of the variation coefficient vc and is not an output variable. The variation coefficient loses its meaning when the mean of the other time series is close to zero. 'Close' is defined as an amplitude of the signal larger than its mean value, meaning that the standard deviation becomes larger than the mean ($vc > 100\%$).

The units of the variables are discussed in section 3.3.6.

3.3.4 Spectral density functions

Using the definitions of time series from the previous section 3.3.3, spectral analysis is described for these signals. It is not possible to use the fast Fourier transform for these signals, because of the non-equidistant character of the series. Therefore the discrete Fourier transform is used and described here.

More details concerning this analysis can be found in Mulder (1988). A useful book on spectral analysis in general is written by Bendat and Piersol (1971, 1986).

Power density spectra

R-peak events

The basis of analysis of the time series is the direct Fourier transform. Assuming periodicity outside the analysis segment T, the Fourier transform of x(t), $X_x(f)$, in the interval [0,T] can be computed by:

$$X_x(f) = \int_0^T x(t) e^{-2\pi f t} dt \quad (3.16)$$

where $j^2 = -1$ and $f = \text{frequency}$.

The transform of x(t) becomes (interchanging the order of the summation and integral and substitution of x(t) according to formula (3.17) in $X_x(f)$):

$$X_x(f) = \sum_{i=1}^N e^{-2\pi f t_i} \quad (3.17)$$

When such a spectrum from a series of events is computed in the segment [0,T], a line spectrum appears at discrete frequencies f_k with a frequency resolution Δf (the difference between f_k and f_{k-1}) of $1/T$:

$$X_x(f_k) = \sum_{i=1}^N e^{-2\pi j f_k t_i} \quad (3.18)$$

The spectral density function $S_{xx}(f_k)$ can be calculated by:

$$S_{xx}(f_k) = \frac{2}{T} X_x(f_k) X_x^*(f_k) = \frac{2}{T} \{ [\sum_{i=1}^N \cos(2\pi f_k t_i)]^2 + [\sum_{i=1}^N \sin(2\pi f_k t_i)]^2 \} \quad (3.19)$$

where X_x^* is the complex conjugate of X_x . The spectral density function represents the distribution of the power in the time series over frequencies. The power in a small frequency band is related to the amplitude of the sine wave at that frequency. Therefore, the spectral density function is sometimes called the power density spectrum.

If the power for a frequency is very small (compared to the other frequencies) this frequency is not present in the signal. If the power for a frequency is very high compared to the others, the signal is dominated by this frequency and will look almost like a sine wave with this specific frequency.

An example is shown in figure 3.5. It is a spectrum of a signal which contains four sine wave with frequencies 0.08, 0.16, 0.24, and 0.42 Hz. The largest wave is found at 0.16 Hz, the smallest at 0.42 Hz.

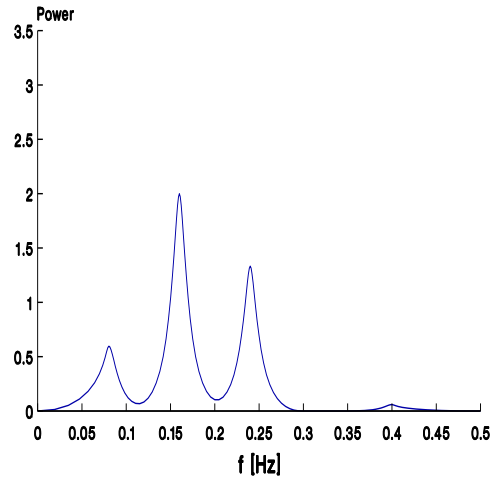


Figure 3.5: An example of a power density spectrum from a signal which is a combination of four sine waves.

The spectral values can be normalized at the mean. The variations are then expressed as a proportion of the mean of the signal $x(t)$. The spectra are then an index of modulation depth of the signal at every frequency (L. Mulder, 1992). The corresponding units are ‘squared modulation index’. These normalized spectra are calculated by:

$$S'_{xx}(f_k) = \frac{S_{xx}(f_k)}{\bar{x}^2} \quad (3.20)$$

In this case, with $x(t)$ the R-peak event series, the mean ($\bar{x}$) of $x(t)$ is equal to the reciprocal value of mean interbeat interval. An advantage of the normalization is that the spectrum becomes dimensionless. In that case spectra of heart rate and blood pressure can be very well compared (see section 3.3.6 for dimensions).

An important advantage of normalized values is that spectral power indices derived from R-wave event series (heart rate) and from IBI series are about equal. This is certainly not the case for non-normalized values (De Boer et al., 1985; Van Dellen et al., 1985). For details see Mulder, 1988.

Other time series

In the same way as for the R-peak event series, a spectrum can be calculated of (other) time series, transforming the signal of equation (3.14) (for details: see Mulder (1988)):

$$\tilde{S}_{xx}(f_k) = \frac{2}{T} X_x(f_k) X_x^*(f_k) = \frac{2}{T} \{ [\sum_{i=1}^N \Delta t_i x_i \cos(2\pi f_k t_i)]^2 + [\sum_{i=1}^N \Delta t_i x_i \sin(2\pi f_k t_i)]^2 \} \quad (3.21)$$

The same normalization as for R-peak events is possible:

$$\tilde{S}_{xx}(f_k)' = \frac{\tilde{S}_{xx}(f_k)}{\bar{x}^2} \quad (3.22)$$

However, for signals with their mean value close to zero, the normalization is not useful. Therefore, in those cases that the mean is smaller than the standard deviation ($vc > 100\%$, equation (3.16)), this measure is considered to be not meaningful.

Transfer function

The relation between two signals (R-peak events and other time series, or between other two time series) can be expressed by the transfer function and coherence function. Let $x(t)$ be the input signal of a system H and $y(t)$ be the output of the system (figure 3.6). The system's functioning (or the relation between in- and output) can be described by a transfer function. This function is a complex function and can be represented by its modulus and phase.

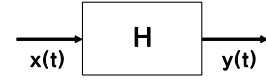


Figure 3.6: A system with input signal x and output signal y can be described by its transfer function H .

The modulus (gain) represents the multiplication factor for the amplitude of the input signal to get the amplitude of the output signal. This factor can be different at each frequency, so the modulus is a function of frequency. The phase (sometimes called phase shift) is a measure for the time delay of the output signal in relation to the input. A negative phase means that a change in the input signal is found after some time in the output signal. The length of the delay is expressed relative to the period length of the given frequency.

A transfer function can only be used to describe linear systems. The coherence function is a measure for the linearity of the transfer (for a perfect linear relation the coherence is equal to 1.0). The transfer function can therefore only be interpreted if the coherence is high (e.g. above 0.5) in the frequency range of interest.

In order to get an idea of the meaning of these functions they can be compared to the parameters in a regression analysis. The slope of a regression line is comparable with the modulus; the correlation r^2 (explained variance) is comparable to the coherence. However, as can be expected from frequency analysis, these measures are now calculated for each frequency.

Given the line spectra of $x(t)$ and $y(t)$, the transfer function can be calculated, using the cross spectrum of x and y :

$$S_{xy}(f_k) = \frac{2}{T} X_x^*(f_k) X_y(f_k) \quad (3.23)$$

The transfer function between x and y can be calculated by:

$$H_{xy}(f_k) = \frac{S_{xy}(f_k)}{S_{xx}(f_k)} \quad (3.24)$$

This complex function can be represented by the modulus and phase:

Modulus:

$$|H_{xy}(f_k)| = \sqrt{\text{Re}^2\{H_{xy}(f_k)\} + \text{Im}^2\{H_{xy}(f_k)\}} \quad (3.25)$$

Phase:

$$\text{Arg}\{H_{xy}(f_k)\} = \text{atan}\left(\frac{\text{Im}\{H_{xy}(f_k)\}}{\text{Re}\{H_{xy}(f_k)\}}\right) \quad (3.26)$$

The coherence function γ_{xy}^2 is calculated by:

$$\gamma_{xy}^2(f_k) = \frac{|S_{xy}(f_k)|^2}{S_{xx}(f_k)S_{yy}(f_k)} \quad (3.27)$$

Coherence, modulus and phase values are always averaged over three frequency points to get a more reliable estimate.

Note: Because of estimation inaccuracies, related to the small averaging bandwidth, it proves to be possible that the estimated coherence becomes larger than 1.0. This strange phenomenon may occur in short measuring periods in which time series consist of distinct frequency peaks with a very small bandwidth.

Band values

In section 3.1.5 was made clear that it can be useful to calculate the power in some frequency bands. It was mentioned in the previous section that the modulus and phase functions are useful in those frequency areas at which there is a high coherence, which is an other reason to restrict the analysis to specified bands.

In the CARSPAN spectral analysis environment, a number of frequency bands can be predefined by setting its boundaries in the CarspanDefaultSettings.ini file. For each frequency band $f_l \dots f_h$ the power in the spectrum of segment i , $B_{x,i}$, in the spectrum is calculated by integrating the power density spectrum over the band. Since a spectrum is a function of discrete frequencies f_k only, the integration is the summation of the spectral density values multiplied by the frequency resolution:

$$B_{x,i}(f_l, f_h) = \sum_{f_k=f_l}^{f_h} S_{xx}(f_k) \Delta f \quad (3.28)$$

It is not necessary to distinguish between spectra from R-peak events and other time series for the band calculations. In the same way, the band values of normalized spectra (spectra with modulation index, equation (3.21) and (3.23)) can be calculated:

$$B'_{x,i}(f_l, f_h) = \sum_{f_k=f_l}^{f_h} S'_{xx}(f_k) \Delta f \quad (3.29)$$

In order to get an idea of the magnitude of these band values for power density spectra the following example may be relevant. Suppose an R-wave event series with a variation in IBI duration of 10% (coefficient of variation). Squared modulation index equals then 0.01; however, in CARSPAN power values are multiplied with 10^6 in order to obtain a nicer range of output values. So, a coefficient of variation of 10% corresponds to a total power of 10.000. In the mid frequency band, representing about one quarter of the total power, values of about 2500 may then be expected. If logarithmic values are used: the natural log (Ln) of 10000 equals: 9.2 and of 2500 equals: 7.8.

The values of modulus and phase of the transfer function are averaged over the frequency band. If the specified frequency band contains n frequency points, the band value for these variables is calculated by:

$$M_{xy,i}(f_l, f_h) = \frac{1}{n} \sum_{f_k=f_l}^{f_h} |H_{xy}(f_k)| \quad (3.30)$$

and

$$P_{xy,i}(f_l, f_h) = \frac{1}{n} \sum_{f_k=f_l}^{f_h} \text{Arg}\{H_{xy}(f_k)\} \quad (3.31)$$

The present CARSPAN version distinguishes between wrapped and unwrapped phase. As equation (3.31) shows, the phase is computed from the inverse tan function, resulting in values between $-\pi$ and $+\pi$. However, in some cases this function shows large phase jumps at a specific frequency that occur in cases that the time differences between input and output time series exceed the value of half a period at that specific frequency. When averaging phase values in a certain frequency band it may occur that a phase jump within this band gives a large bias to the computed mean phase value. Therefore, in the present version of CARSPAN a phase unwrapping algorithm is added to the original phase computations (cf Matlab 7). This means that both the wrapped and the unwrapped phase can be computed and compared when necessary.

It is possible to specify a minimum coherence level for the band values of modulus and phase. Frequency points with coherence lower than the specified level are excluded from the average of equation (3.30) and (3.31). Therefore, the number of points n included in the computation can be variable and is part of the output. Practically, a minimum coherence value of 0.30 proves to be a good boundary. This value seems to be a reasonable compromise between too unreliable modulus or phase estimates and too little spectral points in the computation.

To improve the estimation of the coherence, the weighted coherence is used to compute a coherence measure in a specified frequency band (Porges et al., 1980):

$$CW_{xy,i}(f_l, f_h) = \frac{\sum_{f_k=f_l}^{f_h} \gamma_{xy}^2(f_k) S_{xx}(f_k)}{\sum_{f_k=f_l}^{f_h} S_{xx}(f_k)} \quad (3.32)$$

Frequency resolution and highest frequency

The frequency resolution is always 0.01 Hz. Without any interpolation this would correspond to a segment size of 100 seconds. By applying linear interpolation, in principle it is possible to compute spectra of time segments of any length, although the routine is meant for segments longer than 100 seconds. The reliability of the spectrum increases with the number of frequency points averaged (Bendat and Piersol, 1971); therefore a longer measurement segment will not increase the frequency resolution, but the reliability of the spectral measures.

The highest frequency $f_{\max}$ in the spectra is determined by the mean interbeat interval, according to:

$$f_{\max} = \frac{1}{2 \cdot IBI} \quad (3.33)$$

3.3.5 Profiles

The spectral profiles functions are calculated by means of the algorithms of section 3.3.4. However, the interpolation to a fixed frequency of 0.01 Hz is not applied.

The profiles are calculated using a WINDOW that is shifted in time by a STEP. For each STEP the specified band values are computed, applying the algorithms of 3.3.4 on the analysis segment WINDOW. In this way a number of new time series are generated, which describe the time course of for instance the spectral power in a specified band; this pattern is called a (spectral) profile. This type of analysis can be applied on all types of spectral function (power, coherence, modulus and phase), but also on the mean value of a time series (running mean) or on the variation coefficient of a time series.

The STEP is always smaller than WINDOW, so there is an overlap between segments. The definition of the profile function is conform for the other time series and can be derived from equation (3.13) by substitution of the parameters $x_i = B_{x,i}(f_l, f_h)$ and $\Delta t_i = \text{STEP}$. The new time series is:

$$\bar{x}(t) = \sum_{i=1}^N \text{STEP} \cdot B_{x,i}(f_l, f_h) \delta(t - t_i), t_i = t_0 + \text{WINDOW}/2 + (i - 1) \cdot \text{STEP} \quad (3.34)$$

where t_0 is the real start time of the segment.

Notice that the sample interval, STEP, is now a constant which results in an equidistant time series of band values. The band value $B_{x,i}$ can be any band value (modulation index values $B'_{x,i}$, modulus $M_{xy,i}$ or phase average $P_{xy,i}$, and weighted coherence $CW_{xy,i}$ as defined in equation (3.30), (3.31) and (3.32). Profiles can be computed as well for the mean values of the time series (e.g. mean interbeat interval, mean systolic blood pressure) and corresponding variation coefficients.

The number of time points in the output file (the number of WINDOW analyses) N is determined by:

$$N = \text{int}\left(\frac{T - \text{WINDOW}}{\text{STEP}}\right) + 1 \quad (3.35)$$

where T is the segment length and the function $\text{int}(x)$ truncates the real argument x to the greatest integer less than or equal to x .

The STEP size should be chosen in such a way that the changes of interest in the profile function will not be finished within one STEP (the same criterion as for sample interval time: at least two samples per highest signal frequency). However, the minimum STEP is 1 second. The WINDOW size is related to the frequency band of interest and the reliability. If the specified frequency band ranges from $f_l..f_h$, then the lowest frequency f_l in the specified band must be larger than $1/\text{WINDOW}$. The number of frequency points in the band $f_l..f_h$ is equal to:

$$n = \text{int}(f_h \cdot \text{WINDOW}) - \text{int}(f_l \cdot \text{WINDOW}) \quad (3.36)$$

For reasons of reliability it is advised to choose the window length at least three times the minimum length. The step size can be any multifold of 1 second; a small step size will increase fluency of the profile, while the dependency between succeeding profile-samples is increased as well.

It is possible to specify a minimum coherence level for the band values of modulus and phase as well. Frequency points with coherence lower than the specified level are excluded

from the average of equation (3.30) and (3.31). Therefore, the number of points included in the spectral band, for modulus and phase profiles, is not equal for all time windows

The computation time increases rapidly with WINDOW and decreases with STEP. The computation time per block is proportional to:

$$\text{Computation time} \sim \frac{\text{WINDOW}^2}{\text{STEP}} \quad (3.37)$$

The smallest segmentation for this analysis is a block, the period segmentation is replaced by the WINDOW segmentation and therefore period segmentation is not available for the users.

3.3.6 Units of the calculated measures

The units of the measures derived from the R-peak events series are summarized in table 3.1. Note, again, the spectrum of this time series is a heart rate spectrum, while the time series mean is heart rate (HR). A unit has been defined for the normalized spectra. The division by the squared mean results in dimensionless measure (defined in eq. (3.6) and (3.24). But it is a very specific one and called the squared Modulation Index or MI^2 . It is a measure of relative changes, like the variation coefficient. A scaling is performed within CARSPAN, the spectral values are multiplied by 10^6 . Using the SI-unit convention, the unit is squared milli-modulation index, or mMI^2 . It is not a physical unit; the unit type is comparable with the phase unit radian. Additional to the HR indices also the IBI-measures are given in table 3.1. The unit of other time series can be derived from the unit of the variable that is derived as a time series. We will use a blood pressure time series for an example. The unit for blood pressure is mmHg. The units of the measures for such a time series are shown in table 3.2. The relations between time series are expressed in modulus, phase, and coherence functions.

Measure	Unit	Eq.
HR	Hz	(3.14)
sd_{HR}	Hz	(3.14)
vc_{IHR}	%	(3.14)
S_{xx}	mHz^2/Hz	(3.4)
S'_{xx}	mMI^2/Hz	(3.20)
$B_{x,I}$	mHz^2	(3.28)
$B'_{x,I}$	mMI^2	(3.29)
IBI	ms	(3.14)
sd_{IBI}	ms	(3.14)
vc_{IBI}	%	(3.14)
S_{xx}	ms^2/Hz	(3.4)
S'_{xx}	mMI^2/Hz	(3.20)
$B_{x,i}$	ms^2	(3.28)
$B'_{x,I}$	mMI^2	(3.29)

Table 3.1: Units of R-peak event series measures as well as those for IBI, including a reference to the equation of definition.

The units for coherence and phase are independent from the chosen time series. The modulus unit is equal to the unit of time series y per unit of time series x .

In table 3.3 the units are shown for the example of the transfer function between blood pressure (x) and interbeat interval (y , derived from the R-peak event series). In that case the modulus has the unit ms/mmHg.

The normalization of the power spectra (eq. (3.20) has no consequences for the units of the transfer function; transfer functions are computed preceding the normalization process.

3.4 Applications outside cardiovascular research

It is possible to analyze any event series with intervals in the range of about 500-1500 ms. These events can be for instance the responses of a subject performing a task (in a psychology laboratory or in an operator room etc). Of course, it must be useful to make a spectrum of these events.

A totally different application is the following. Suppose there is an object moving along a certain track. The length of the covered distance by the object is measured every second and these distances are stored in a file. Now, the observer is interested in the speed variations, not only in time but also along the track. The parallel between this example and R-peak events is that the R-peak time is now the covered distance and the interbeat interval is the speed (because it was measured every second in meters, the speed is in m/s). The amplitude of the speed variations along the track and the length of the variations can be calculated using CARSPAN. A spectrum of the speed signal on the horizontal axis is now the reciprocal value of the length of the speed variations. The vertical axis is the squared amplitude of the speed variations with a certain length. For instance, our object can accelerate 2 m/s and decelerate 2 m/s every 10 m (independent from the time, it depends on its speed in the past what the time is!).

Another example might be the following (Brookhuis et al., 1994). A car driver receives the task to follow a preceding car on a fixed distance on a highway. Distance between the two cars is measured using a distance radar. The transfer between velocity variations of the leading car and the follower is velocity (i.e. frequency) dependent. The CARSPAN program can be applied by either using equidistant or non-equidistant sample points. Of course, other frequency bands have to be chosen than relevant for cardiovascular spectral analysis.

Measure	Unit	Eq.
$\bar{x}$	mmHg	(3.15)
sd_{BP}	mmHg	(3.15)
vc_{BP}	%	(3.15)
S_{xx}	mmHg ² /Hz	(3.21)
S'_{xx}	mMI ² /Hz	(3.22)
$B_{x,i}$	mmHg ²	(3.28)
$B'_{x,i}$	mMI ²	(3.29)

Table 3.2: Units of other time series measures using blood pressure as an example (mmHg), including a reference to the equation of definition.

Measure	Unit	Eq.
$ H_{xy} $	ms/mmHg	(3.25)
$\text{Arg}\{H_{xy}\}$	rad	(3.26)
γ_{xy}^2	(none)	(3.27)
$M_{xy,i}$	ms/mmHg	(3.30)
$P_{xy,i}$	rad	(3.31)
$CW_{xy,i}$	(none)	(3.32)

Table 3.3: Units of the transfer function between two time series, using blood pressure and interbeat interval as an example, including a reference to the equation of definition.

3.5 Example (continued)

In this section the example of section 2.5 will be worked out in respect of describing the necessary steps in data-segmentation to be performed before the actual processing can take place, in respect of time series to be derived, the type of artifact processing which is needed and finally the analysis options to be chosen.

Data segmentation

Subjects perform a mental task of about 5-minutes duration after a baseline-rest of about 5 minutes. In this way two analysis periods are obtained. Begin and end of the useful data segments are indicated by a start and stop code (event code 21 and 22 respectively) in the data files. (See appendix B.2).

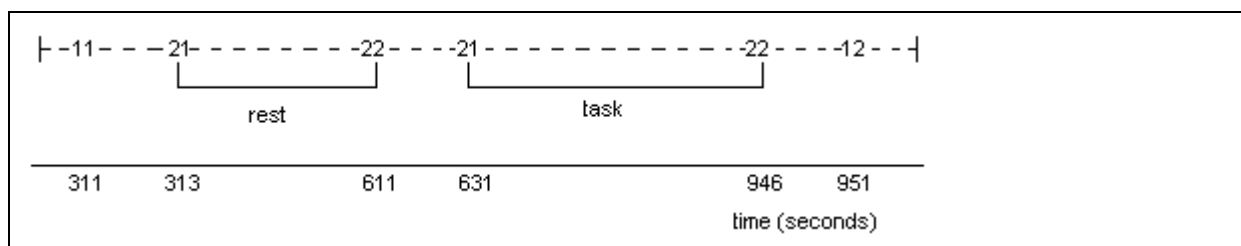


Fig 3.7 Data segmentation: The experimental session is divided in two parts: a rest and a mental task. Codes 21 and 22 are seen as start and stop respectively of analysis periods. Codes 11 and 12 are start and stop codes for a block, respectively. The lower line shows the time span.

The analysis segments between the codes 21 and 22 will be called 'periods', while these two periods together will be called 'block'. In this example there is 1 block in the data file. For this purpose two additional codes (11 and 12, included in figure 3.7) are inserted as begin and end of the blocks. This procedure is used in order to enable the computation of spectral profiles across the borders of the rest-task transitions. Another possible choice could have been to distinguish two separate blocks: the baseline-rest and the task performance segment, respectively. However, in such a case the mentioned transitions could not be included in the analysis. All data pre-processing options work with block boundaries; only the spectral functions option works with 'periods'.

In the remainder of this example the spectral analysis and the description of the frequency bands will be restricted to the 5 minutes period of the baseline condition and a task period of 5 minutes.

Working procedure and specification

After specifying the files to be analyzed, the next step is specification of all processing options. This starts with data segmentation, specification of block and period codes, followed by selection of additional time series (IBI, systolic, diastolic, mean blood pressure, and respiration time series). Then the analysis functions are specified, including type of data-segments, spectral bands, computation of mean values, use of modulation index (values relative to the mean), use of logarithmic power values, power spectra of time series to be computed and finally coherence and transfer functions (modulus and phase) between pairs of time series. In order to compute spectral profiles (spectral band values as a function of time) window length, step size, spectral band to be used and the relevant time series (and relations between time series) have to be specified. Before going on with data pre-processing, the derivation of time series needs additional attention.

Derivation of time series

A number of time series have to be derived in order to perform the selected computations. These are:

- Heart rate

Heart rate spectra as well as mean values of IBI and its coefficient of variation are derived from the R-peak event series (R-peak event-code and the accessory event-time), see Appendix B.2.

- IBI series

In order to compute BRS-values (modulus between changes in SBP and IBI) an additional channel for IBI has to be specified which is computed from the R-peak event series. In this way both variability measures of HR and IBI can be computed as well.

- SBP, DBP and MBP

Systolic, diastolic and mean blood pressure have to be derived within each beat from the equidistant sampled (rate = 100 Hz) finger blood pressure signal.

- Respiration volume

Within each R-R interval the mean value of the equidistant sampled respiration signal (rate = 100 Hz, same data file as finger pressure signal) is computed and assigned at a new series expressing the respiratory pattern.

- Respiration frequency

Durations of the respiration cycles are estimated in RespTool, so that for every R-peak event code a duration in seconds is given. RespTool adds a column to the event file containing the above-mentioned time series.

Event-check and artifact processing

At first, it will be checked whether the necessary codes for begin an end of blocks and periods are indeed available and whether there are no hindering additional begin/end codes. Correction can be done using the pre-processing and check-blocks options in CARSPAN.

Several types of artifacts may occur. These will be processed in two steps: in the first, automatic detection and correction is applied, while visual checking and manual correction takes place in the second step.

Artifacts in the R-peak event series may consist of either too long or too short intervals. Long intervals will automatically be corrected; short intervals will be detected, however, correction possibilities are limited. When a correctly working (hardware or software) R-wave detector is used too short intervals occur very seldom, unless the detected artifacts have a physiological origin. In the latter case it has to be considered whether it is justified to correct more than a limited number of intervals in a measuring period.

After R-peak correction, time series processing takes place. This means that first IBI, systolic, diastolic and mean blood values are computed, followed by artifact correction. This latter proceeds in the same way as with heart rate: first automatic correction, followed by visual checking and manual correction where necessary.

Most of the artifacts in finger blood pressure are related to the re-justification procedure (physiocal) of the measuring device which takes place about every 70 beats and which lasts for about 3 beats. Correction is performed by means of linear interpolation between two preceding and two succeeding correct beats.

'Artifacts' in the respiratory volume time series are mostly related to chest movements (depending on the measuring method) and sighing and coughing. When such types of artifact

occur it has to be taken into account that such respiratory patterns are not really artifacts and that these induce cardiovascular effects also. Usually, correction of respiratory time series is not necessary. Given the fact that respiratory frequency is computed in the program RespTool, a correct differentiation of the respiration cycles is depended upon choosing the right values for the parameters (see Appendix ...).

Analysis

The analysis part contains two options: spectral functions and spectral profiles. The first option specifies the computation of spectral density, coherence and transfer function from specified data segments, while in the second option the same spectral computations are performed on (partly) overlapping data-windows which results in spectral profiles (e.g. spectral power in a certain frequency band as a function of time).

Spectral functions

Power density spectra, coherence, modulus and phase spectra are the spectral functions to be computed. The default frequency bands are included in a settings-ini file; these will be used in the present example. For the power spectra and derived band measures of HR and SBP the modulation depth option (MI) will be used; for respiration this option is not adequate.

After the spectral computations numerical and graphical information is available about these spectral functions. The graphical facilities show for each computed power spectrum per period (analysis segment) the original time series and the corresponding power density spectrum. For the relation between time series, next to the input and output time series and the corresponding power density spectra, coherence, modulus and phase functions are shown. The numerical output consists of a summary and a numerical output file; both can be shown in the CARSPAN environment. The numerical output file is also available in an Excel format. Both the summary and the numerical output file contain all the specifications, the block and period segmentation, the mean values and the band values. Additionally, the numerical file contains the raw spectral values.

Spectral profiles

Spectral profile analysis will be restricted to mean values of IBI and SBP, spectral power in the mid frequency band of HR and SBP and the modulus in the mid frequency band between SBP and IBI. Also in this analysis the modulation depth option will be used.

In chapter 8 the example will be worked out further.

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4. How to get started

4.1 Introduction

The purpose of the present chapter is to get started with a simple example and a restricted analysis. The example file only contains heart rate data. From this time segment a power density spectra will be computed; only a few analysis options will be specified. Before this can be done the program has to be installed correctly (see chapter 7) and data files have to be available in the specified formats in a data directory.

In the program different phases can be identified for dataset, data- and analysis functions specification, for pre-processing, for the actual analysis and for viewing/plotting the results. These phases can be recognized in the menu structure of CARSPAN.

The program works with a session structure, as well as with procedures to save and load this session information. Corrected event files can be exported.

4.2 Using the first example

In order to get in touch with data analysis as soon as possible, a simple analysis procedure will be started. Only a limited number of pre-processing and analysis options will be used in this first approach of working with CARSPAN.

What is necessary before getting started?

- Is the program correctly installed? See chapter 7.
- Is the CarspanDefaultSettings.ini file available in the start-up directory of CARSPAN?
- Is the data file Example1.evt to be analyzed available in a data directory?

```
.....
0  303.747 R-peak event code
0  304.583
0  305.388
0  306.142
0  306.852
0  307.589
0  308.392
0  309.225
0  310.028
0  310.788
11 311.514 Begin-of-block code
0  311.734
0  312.265
0  313.096
21 313.900 Begin-of-period code
0  314.649
```

```

0 315.369
0 316.083
.....
0 607.844
0 608.562
0 609.320
0 610.101
0 610.858
0 611.594
22 611.740 End-of-period code
0 612.340
0 613.120
0 613.882
0 614.596
0 615.268
0 615.937

```

Figure 4.1: Parts of an example event file. Each line contains the event code and event time. (The comments at the end of some lines are NOT part of the file.)

Structure of the data file

The ASCII data file to be processed contains a header with specified event-information, the events that are recorded and the time point at which they occur (see figure 4.1). The first column in the file represents the event code; the second column represents the time points of these events (in ms) on a continuous time axis. Code 0 corresponds to an R-peak; code 21 is the beginning of an analysis segment (period); code 22 is the end of such a period (see chapter 3.1). The file contains 2 analysis periods, both with the same start/stop code combination (21 and 22). In this example, two of these periods together constitute a 'block'; beginning and end of a block are indicated by the codes 11 and 12, respectively. In the example that is used here only the first period will be considered.

Processing the example file

The simple example that will be worked out here contains the computation of power spectral functions of heart rate. Before this can be done, a check is performed whether the necessary event codes are present in the data file for defining begin and end of the data segment (period) to be analyzed. Additionally, a procedure is executed to detect and correct errors (artifacts) in the heart rate data.

Now, at first the data file and the accessory event codes have to be specified. Then the pre-processing options are given while finally the options are specified which are used in the spectral computation itself.

Start CARSPAN:

Launch the program by double-clicking the CARSPAN icon on the desktop, or through Start > Programs > CARSPAN.

File: New session;

A file menu is opened to specify the configuration file (.csf); browse to the data directory; select sub-directory Sessions and type: 'Start_CARSPAN' as session name;

Specification: Specify data set;

The browser tree is opened (same as starting window of CARSPAN); browse to the data directory where the example file Example1.evt is available; if more event files are available in this directory, delete them from specification list (select, delete); the other files are no longer part of the dataset to be analysed; of course they are not removed from the data directory.

Specification: Specify data;

Fill out in the form: Rpeak: 0; BOB (begin-of-block): 11; EOB (end-of-block): 12; BOP (begin-of-period): 21; EOP (end-of-period): 22.

Now the relevant data with respect of event-codes is specified.

File: Save session (information thus far specified is stored in the session file)

Specification: Specify Analysis Functions:

Segments: select 'periods marked by period code'

Frequency bands: press double-arrow-left (frequency bands are selected, according to preset values in the CarspanDefaultSettings.ini file)

Choose tab: Power functions (at left side)

Select input: Rpeaks (at right side)

Set the parameters for Modulation index, use logarithmic values, Mean, Variation coefficient, all frequency bands (all are set by default)

Press .

Now the relevant data with respect of the spectral computation is specified.

File: Save session (information thus far specified is stored in the session file)

Pre-processing; process R-peaks;

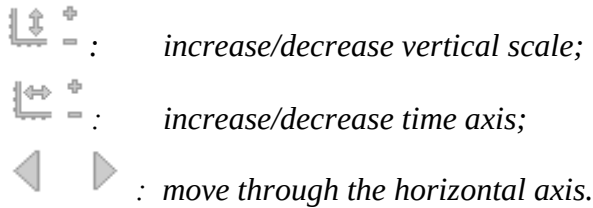
A new graphical window is opened with two graphs, both representing the Rpeak series; the Y-axis represents the IBI-durations in seconds; the X-axis the time axis in seconds.

Check blocks; check current data pair; check; no block-specification errors found

This means that all begin and end codes of the blocks and periods are available in the right order. The block structure can be displayed by decreasing the time axis of the upper graph in such a way that the whole file becomes visible, including the block and period codes.

Box 4.1 Basic navigation

Each of the graphs has navigation options in the action bar (see also Appendix A):



If these options are not visible for the lower window: press the down arrow at the left side of the graph. The graph above will always represent the original Rpeak data; the window below contains corrected data after applying data correction. Navigation occurs separately for each window; this means that time axes of different windows may become desynchronized. This can be restored by pressing the right mouse button and shifting over one of the graphs. In this case the middle of the time windows will be synchronized, while the time scale of each window will not change. By shifting the mouse with the right mouse button pressed over the graph, this graph is shifted over the screen, enabling a more continuous navigation than by means of the left/right arrows. Another probably useful function is that part of the original Rpeak time series can be selected in the upper screen and displayed in the screen below. Try this by: pressing the left mouse button in the upper screen, keep depressed, select a piece of data. More navigation functions available are discussed in chapter 5.

Autocorrect Rpeaks;

Select: Correct current data pair; select RpeaksCor; OK. Forget about more advanced settings.

The small window on the left side contains 1 artifact; indicated is type of artifact (complex) and begin- and end-times. The lowest line shows the total number of artifacts as well as the total artifact time in the heart rate channel. In the graph below the artifact is visible and the proposed corrections are displayed in red.

*** Interpretation:**

One bad interval was found, and in the correcting phase the program decided to correct this artifact.

Box 4.2 Manual correction

For reasons of understanding how it works, try manual correction:

Select a small window around the corrupted beat by positioning the mouse at the left side of the artifact; keep the left mouse-button pressed and move the mouse over the artifact to the right. Press the interpolation button () and see what happens. This manual correction is visible in the left window and can be un-done by deselecting the artifact in this window.

Now the artifact correction is finished; as a matter of fact the corrections are available in a list and will be used during further processing; however, the original data has not changed and no corrected data file has been saved up to now.

File; save session

All specifications and data corrections are stored in the session file

File; Export event files

Choose a new file extension: EV1 (automatically proposed); select export all event files and select new event file format; Export.

The new file Example.EV1 now contains the corrected Rpeak data.

Analysis; Calculate spectral Functions

The spectra are calculated and an output window is opened, containing a tree structure with graphical and numerical output.

With the plot option both the time signals and the spectral power distribution (power density spectrum) will be displayed per selected segments (analysis period). The numerical function shows the numerical output in two different formats: a summary (man readable) format and a numerical (listed) output format, suitable for further statistical processing. Both types of output are also written to output files; these can be found in a separate Output-directory connected to the input data directory. The summary output only contains mean heart rate (Rpeaks) and the power in each of the selected frequency periods per segment. The numerical output contains all spectral data points, additional to the band values. See Appendix C for examples of summary and numerical output files.

In case the user wants to have a look at the original data or at the specification by moving back to the pre-processing or specification window, he/she can come back to the spectral results by selecting: View > Spectral results. Moving back to the Analysis > Calculate spectral functions option would force the program to re-calculate the spectra.

Finally, going back to the specification menu and changing parameter settings is possible. It depends on the type of changes made, however, whether it is necessary to go next to the pre-processing stage or directly to the analysis menu. The safest way is to choose the Pre-processing > Analysis path or to try in which cases first an update of block-

and period codes and necessary artifact corrections have to be made before renewed spectral computations.

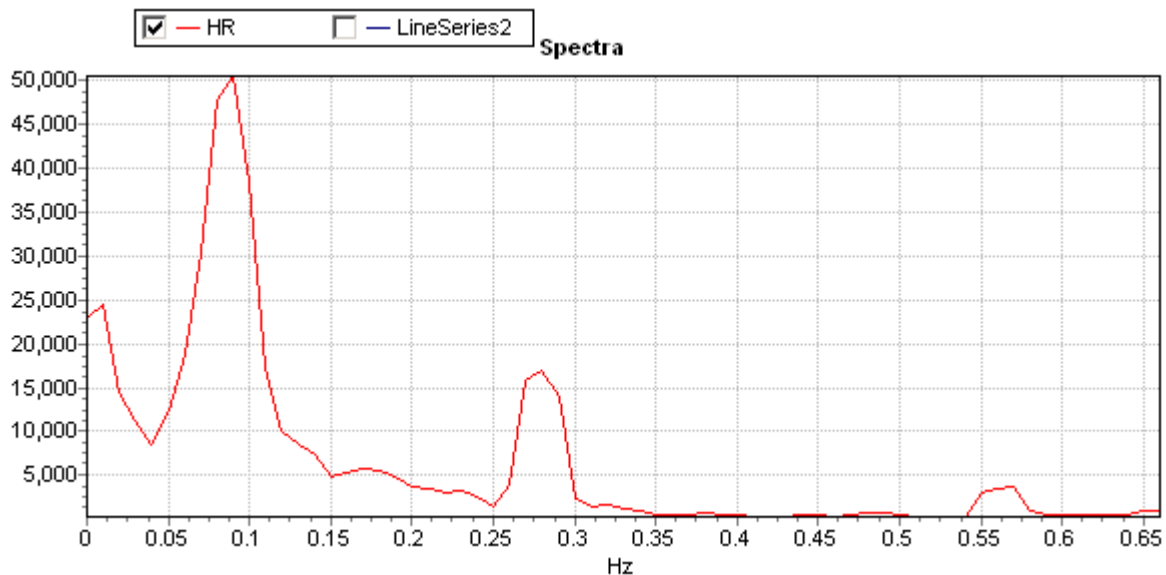


Figure 4.3: Power density spectrum of heart rate in the period analysed in this example.

The numerical output contains the data in a format which is well suited for plotting purposes. Each spectral computation delivers a data record, which contains experiment and subject number, mean value of the time series (for heart rate HR values are given in Hz, corresponds to beats/second), band values and finally, each of the frequency points of the computed spectral function (fixed resolution: 0.01 Hz). In this example it can be seen that the spectral power is divided over three spectral regions: the lower frequencies (up to 0.04 Hz), the mid frequency band with a peak around 0.09 Hz and a high frequency peak around 0.28 Hz, representing respiratory related variations in heart rate. Also the double frequency of the respiratory peak (second harmonic) is visible, at 0.56 Hz.

Save session; all specifications in this session are saved.

4.3 How to continue?

There are several ways to go on with the CARSPAN program for analysis of your data, depending on the already available (theoretical) knowledge of cardiovascular signal processing, the availability of data in a format which is necessary for the CARSPAN program, and on the type of time series to be analyzed (e.g. just heart rate or a combination of several cardiovascular time series).

In any case it is advised to read first the chapters 1 and 2 of the manual, which give some of the background ideas of the program, including the summary of available functions and spectral routines and the aims of the program. When the user wants to know more about the theoretical basis of the spectral routines that are used in the program or about the algorithms it is advised to read chapter 3 first.

Furthermore, it has to be mentioned that a complete example is worked out in the

manual throughout a number of chapters (2.5, 3.5 and 8). This example can be read separately from the remainder of the text in the chapters 2 and 3.

Also, the example data files and completed example session are all distributed with the CARSPAN package; it is possible to look at the processing options and the files that result from this.

Finally, a User Guide is found in chapter 5.

5. User Guide

In this chapter, all features of the program will be discussed. A detailed description of the menu items is provided.

Launch the program by double-clicking the CARSPAN icon on the desktop, or through Start > Programs > CARSPAN.

5.1 Files

CARSPAN input and output files

CARSPAN uses several input, output and data specification (settings) files, which are either located in the same directory as the program itself (CarspanDefaultSettings.ini file) or in the data area that can be chosen freely. It is advised to make a working directory in which sub-directories can be chosen for input data files, session files and corrected event files. The program stores the spectral results in a sub-directory (Output) of the input data file directory.

- CarspanDefaultSettings.ini.

This file has to be located in the same directory as the CARSPAN program itself; it contains a number of initial settings of the program, such as band definitions and correction parameters. Each time a new session is opened this information is renewed from the **.INI** file.

- Input files:

- Event files, containing the time points of the Rpeak series, in combination with markers giving segmentation information; in most cases the extension **.EVT** is chosen; the program searches for extensions starting with the letter E.
- Sampled data files, further called 'data files', containing the sampled values of the ECG, blood pressure, respiration etc. These files have the extension **.NFF**; it is required that these files have exactly the same name as the corresponding event files. The combination of an .EVT and .NFF file is called a data pair. If only spectra of heart rate have to be computed, the .nff files are not required.
- Extended event files; these have the same basic structure as the event files, but now additional columns are added giving the beat-to-beat cardiovascular data (e.g. IBI, systolic and diastolic blood pressure). In most cases these files are the result of previous CARSPAN sessions. The extension of these files starts with the letter E and can be chosen further freely.

- Session files and exported event files:

- All CARSPAN sessions start with defining (or opening) a session file, containing information about the dataset used, the data and analysis definitions as well as all data corrections. These files have the extension: **.CSF**; it is advised to make a separate sub-directory '**Sessions**' in the data directory for storing the session files.
- Corrected event files (and extended event files). These files are the result of the 'Export event file' option in the file menu; these files have the same structure as the original event files, while in the extended event files additional columns are added for derived time series. These files can be exported to the data input directory or to a new sub-directory.

- Output files, in sub-directory '**Output**' of the input data file directory:
 - Summary output file (extension **Summary.OUT**), containing spectral result information from the 'Analysis Spectral Functions' option in a man-readable format.
 - Numerical output files in both Excel format (**Numerical.XLS**) and in ASCII format (**Numerical.OUT**).
 - Profiles output files from the 'Analysis Spectral Profiles' option (extension: **.PRO**). These files have the same format as the extended event files and contain the results of the time-frequency analysis for the selected spectral bands, channels and analysis types. Also these files can be approached by Excel.

See Appendix B for examples of all file types.

Box 5.1 Working with sessions and files

The aim of the program is that data and analysis specifications that are defined for one data set can be used for another one by simple reading a new data set. However, this procedure does not work correctly in all cases. This is related to the fact that the session file contains both a list of files to be handled as well as the lists of corrections and added time series per file. This makes it rather complex and sensitive to errors during switching between data sets. Therefore, it is advised to make an initial session definition, with only 1 data pair containing all data definitions and analysis specifications, without any corrections, and save this session file as a data specification file to be used for each succeeding new data set that needs the same specifications.

Although in principle an unlimited number of files can be handled in a session, it is probably better to restrict the number to about 10 files in order to prevent that session files contain both corrected and uncorrected files which makes it rather inconvenient and error prone. It is advised to divide a large data set in subdirectories that can be handled in 1 session. If a session is not finished completely at the end of a working day (or at another moment in time) it can be saved as it is and the user can go on with data analysis the next time by opening again the saved session file.

In case event files are exported to a new directory while in a succeeding phase, additional time series have to be derived from the sampled data files, .NFF files have to be copied manually to this directory. Therefore, it may be better only to export to a new directory in case of extended event files when further use of .NFF files is not necessary anymore.

5.2. Menu structure

The menu structure of CARSPAN contains the following items:

File ▾ **Specification** ▾ **Pre-Processing** ▾ **Analysis** ▾ **Tools** ▾ **View** ▾ **Help** ▾

File: New session, Open a session, Save a session, Export Event Files,

	Exit CARSPAN
Specification:	Specify Data Set, Specify Data, Specify Analysis Functions, Specify Analysis Structure
Pre-Processing:	Check Blocks, Pre-Processing RPeaks, AutoCorrect RPeaks, PreProcessing TimeSeries, AutoCorrect TimeSeries
Analysis:	Calculate Spectral Functions, Calculate Spectral Profiles
Tools:	Additional programs
View:	View Spectral Results, View Raw Data
Help:	About CARSPAN

5.3. Menu items

5.3.1 File

After starting CARSPAN, the first thing to be done is either opening an existing session file or creating a new session. You can create a new data session or open an existing data session in the **File** menu. If you want to create a new data session file, go to **File > New Session**, select the data directory and choose a file name. The file has the extension .CSF. The directory and session identification is visible in the upper bar. If you want to open an existing session file, go to **File > Open Session**. The (extended) event files in a dataset can be **exported** for using in another CARSPAN session or for further processing in other programs.

Box 5.2 Buttons in the File menu



Create a new session (Ctrl + N)



Open an existing session (Ctrl + O)



Save a session to file (Ctrl + S)



Export event file (Ctrl + E)



Exit CARSPAN (Ctrl + X)

5.3.2. Specification

Specify Data Set

By selecting **Specification > Specify Data Set**, the required data directory can be selected in the browse tree. After selecting a directory, event files (.EVT) and data files (NFF) are visible. The combination of an .EVT and .NFF file is called a data pair. A check will appear in front of the data pair(s) when two requirements are fulfilled: 1) the name of the event file and the sampled data file are the same, and 2) the structure of the sampled data file is the

same as the so-called reference file. In principle, the reference file is the first data pair in the file list; however, another data pair can be selected as reference by selecting this pair and pressing the right mouse button.

Data pairs can be removed from the file list by using the standard windows conventions of selection and deletion (CTRL-DEL, or SHIFT-DEL), or by using the icons in the upper bar (see Box 5.3). The files are no longer part of the dataset to be analysed, but are of course not removed from the data directory.

Two additional options are available:

- *Select file extension:*

A selection can be made on the basis of the file extension of the event files in the current directory.

Note: it may be necessary to first renew the file list (see box 5.3) or even to select another data directory before all available file extensions become visible.

- *Use data files:*

By deselecting this item the sampled data files are not used in the analysis.

Use Data files: ☒

Box 5.3 Buttons in the Dataset Specification menu



Renew the file list by reading all data files in the directory once again



Remove data pair(s) from the file list



Place selection on top



Move selection one file down



Move selection one file up



Place the selection at the bottom

Specify Data

Now, the data can be further specified by choosing **Specification > Specify Data**. Several parameters have to be filled in the form:

- **Standard event codes**

Values can be filled in for:

Rpeak: event code for RPeaks

BOB: event code for the beginning of a block

EOB: event code for the end of a block

Standard eventcodes:

RPeak	
BOB	
EOB	
BOP	
EOP	

BOP: event code for the beginning of a period

EOP: event code for the end of a period.

(See box 5.4)

- **User defined event codes**

Here you can define codes that are included as markers in the event file. These user defined codes have the function of recognizing specific time points in the data; they have no other special function in CARSPAN. The event codes have to be available in the event file or can be inserted during data pre-processing; the name functions as a label in further processing (See box 5.4).

Box 5.4 Event codes

R-peak code

The R-Peak code is necessary for almost all processing. If this code is not specified the Rpeaks and time series can not be processed.

Block and period codes

A block consists of one or more periods. The import for block and period codes is optional, but if these codes are not defined they can not be used for viewing or processing. Artifact correction, for example, only works between block codes (BOB, EOB); if these are not available the whole file is seen as a block. The period codes (BOP, EOP) are necessary for calculating analysis functions.

Block codes can be specified in the standard event codes menu using the event markers in the event file. There are, however, several ways to construct periods. In the standard event codes menu, for example, you can use the event markers that are identifiable in the event file to specify the period codes (in the same way as with specifying block codes). Another option would be to manually add the markers in the graphs, which will be explained in the pre-processing menu (section 5.3.3). Period codes can also be created in the 'specify analysis functions' menu; with the option 'period marked by Time' in the analysis segments section blocks will automatically be divided by CARSPAN in a number of periods according to the specified period length (see section 'Specification > Specify analysis functions').

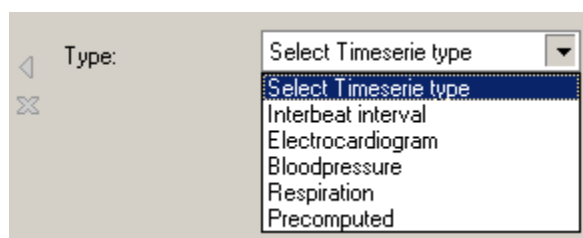
Note that in this case other codes are used as specified in the 'standard event codes' menu.

User defined codes

The option user defined event codes enables the possibility to recognize specific event types in the data, such as start of stimulus presentation, subject responses or specific behavioral patterns (i.e., subject starts speaking, subject is sitting down). If such a code is defined, during pre-processing a vertical line and a small text block is inserted with the specified label.

- **Create Time series**

This means that additional data channels are defined which are added as additional columns to the event files. The creation of



time series is activated for the entire file and is not restricted to block boundaries. You can choose between several data types and input channels.

The type of time series can be selected from a list at the right side of the screen; for most types thereafter a data channel has to be selected that contains the signal from which the time series will be derived. Using the manual calibration option a scale factor and a zero level can be defined; with this option it is possible to rescale the derived time series. For instance, blood pressure sample values can be recalculated to 'mmHg' if the calibration values are known to the user. If variability measures become too large a scale factor can be set to, for instance, 0.1 to reduce values. The following types of time series are available (see chapter 2 and 3 for background information and algorithms):

- Interbeat interval: This data type will be computed from the R-peak data. Available choices are IBI, IBI Inverse ($1/IBI$), and dIBI ($\Delta IBI = IBI - \text{previous IBI}$, also referred to as 'successive differences'). Channel names can be changed manually.

- Electrocardiogram:

- *Manual Calibration:*

- *Scale Factor:* ECG values will be multiplied by this factor before use. In case the ECG is uncalibrated a value 1 will suffice.
- *Zero Level:* The zero level for the ECG channel in the sampled data file. This level will be added to the ECG values after multiplying with the scale factor.

- *ECG T-wave:* a T-wave amplitude value is computed for each beat.

- Blood pressure (BP):

- *Manual Calibration:*

- *Scale Factor:* Blood pressure values will be multiplied by this factor before use. This scale factor must be specified in mmHg.
- *Zero Level:* The zero level for the blood pressure channel in the sampled data file. This level (specified in mmHg) will be added to the blood pressure values after multiplying with the scale factor.

Note: In order to support the user, a tool will become available in the next CARSPAN version to calculate blood pressure zero and scale values from sampled calibration data.

You can compute one or more blood pressure time series:

- BPSys; systolic BP: This is the maximum of the BP signal between two Rpeaks.
- BPDia; diastolic BP: This is the minimum of the BP signal preceding the maximum systolic blood pressure in the considered R-R interval.
- Pulse: This is the difference between systolic and diastolic blood pressure for the considered R-R interval.
- Mean: This is the mean BP signal value between two R-peaks.

The time series are computed from the blood pressure data in the sampled data file. In principle data from more than one blood pressure channel can be inserted. Time series

names can be altered manually.

- Respiration:
 - *Manual Calibration:*
 - *Scale Factor:* Respiration values will be multiplied by this factor. In many cases, respiration is uncalibrated. The scale factor is then an arbitrary value that can be set, for instance, to 0.1 or 1.
 - *Zero Level:* This is the zero level for the respiration channel in the sampled data file. This level will be added to the respiratory values after multiplying with the scale factor.

Two respiratory time series can be chosen; these are computed from the respiration data in the sampled data file:

- *Resp Mean (mean respiration):* The mean respiration value of the time segment between two R-peaks is computed. The value is assigned to the time point of the second R-peak. The channel name can be manually modified.
- *Resp Sample (respiration sample):* A value around the time point of the R-peak is computed. A number of samples preceding and succeeding the R-peak is averaged and assigned to the Rpeak. The sample width has to be specified in milliseconds or number of samples. If one of the values is specified (i.e., time in milliseconds), then CARSPAN will automatically import the value for the other specification (i.e., number of samples). Time series names can be altered manually.

- *Precomputed:*

This can be any other time series on a beat-to-beat basis that is produced during data collection or during data preprocessing, outside the present CARSPAN session. These time series have to be separate data columns of the (extended) input event series files; such data columns could be created by former CARSPAN sessions or by other programs that give the same data format. If a data file has the correct format, it will be read automatically in the list below 'Create Timeseries', including the labels of these channels.

The option 'Precomputed' allows for selecting additional columns manually in case these are not presented automatically. The name of the time series ('Precomp') is set by default, but can be changed manually. Specify the data column in the event file that corresponds with the precomputed data. Event file column 1 is the first column after the 'time' column in the data file.

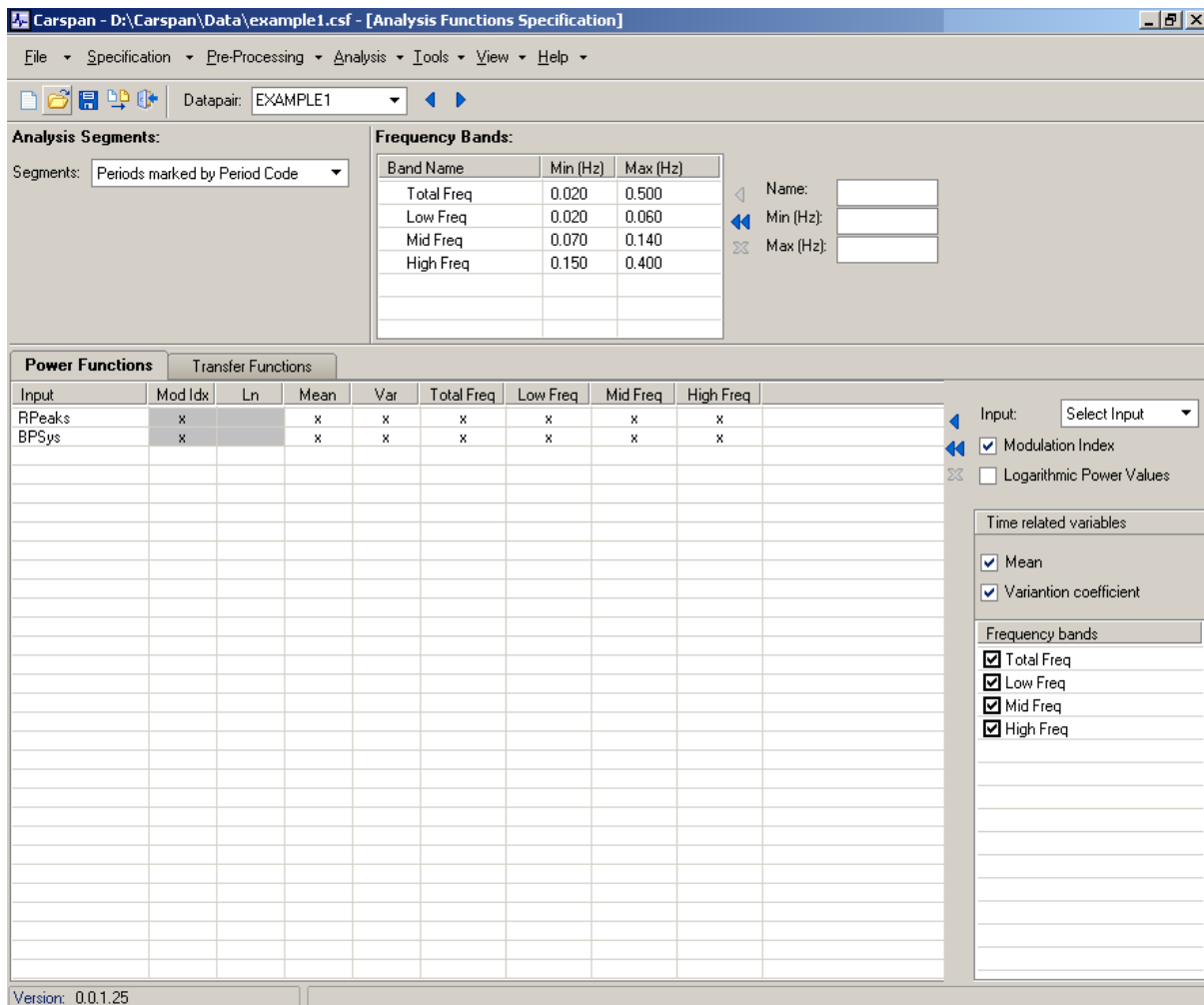
Note: Be careful to choose the correct data file channel that corresponds with the data type!

Note: Calibration values can be inputted manually when they are not already included in the header of the sampled data file. When no values are specified for 'scale factor', the default value will be 1. For 'zero level', the default value is 0.

Specify Analysis Functions

In the **Specification > Specify Analysis Functions** menu you can specify the spectral analysis options you want to perform on your data. The available data will be the (extended)

event files that were specified in the ‘specify data’ menu. Several options can be specified in order to compute spectral power density, coherence, and transfer functions from specified data segments:



- Analysis Segments

A data segment is a period across which a spectrum is calculated. Several options are available to specify the segments:

- *Periods marked by period codes:* The codes for the periods that are specified with ‘specify data’ (BOP, EOP) are used to define the segments. For each period a spectrum is calculated.
- *Period marked by Time:* The length of the segments can be specified manually in seconds. This means that a block is divided in a number of periods, according to the specified period length. It is possible, however, that not all periods match the block size, for example the last period may be shorter. Therefore, you have to decide what to do in such case: increase or decrease the period time that is defined manually with Time (sec), or skip the non-fitting last period. Make sure that the length of the segments is long enough for the specific frequency band limits. In this case, a check will appear in front of the specified frequency bands. The length of a period has to be at least the length of the inverse of the lowest frequency band boundary. For instance,

if the low frequency band boundaries are set at 0.02 and 0.06 Hz, the period length has to be at least 50 seconds. If the length of the segment is too short for a specific frequency band, this band will not be included in the analysis.

Note: If 'period marked by Time' is selected, then the period codes will become visible in the time series. However, this requires a specific procedure. After specifying the analysis functions, go to pre-processing Rpeaks. After that, click a second time on pre-processing Rpeaks. At this moment the period codes in the graphs will become visible. When you go to pre-processing Time series, the codes are visible as well. If you choose to export the event files at this time, the period codes will be recorded in the event file (.EVT). With this option CARSPAN will not use the previously specified period codes, but select a new set of BOP, EOP codes automatically. In order to prevent confusion, it is advised in this situation to remove the BOP, EOP specification from the 'Standard event codes' list in the 'Specify data' form.

- **Blocks:** Use the block codes that are specified with 'specify data' (BOB, EOB) to separate the segments. As a matter of fact, in this case block codes are used as period codes.

Note: the choice for type of segmentation depends on the preference of the user and is of course related to the data set to be analyzed.

- Frequency Bands

By clicking on the double arrow () the default frequency bands from the settings-ini file will be selected. Otherwise, specify the frequency band(s) with a minimum and maximum in Hz. The number of frequency bands you can specify is unlimited. However, not all blank rows have to be defined.

Also, a variable frequency band around the main respiration frequency can be specified. It is recommended to use a bandwidth of 0.03 Hz. Fill in the value of the bandwidth of choice both at the minimum and maximum Hz to use this function. By pressing the red arrow () the variable frequency band is selected.

- Power functions/Transfer functions

Now a table can be created for the spectral density functions and the band (power) values to be computed, as well as the transfer functions and accompanying band values. Moreover, some options are available. Both the time series and the options can be specified at the right side of the interface.

Power functions

Specify the time series you want to compute power densities for. If you press , power functions for all available time series will be selected for all specified frequency bands. If you want to select the power functions yourself, you have several options:

- **Input channel:** Here you can select the channel for which you want to compute a power density spectrum. You can only choose those time series you have specified with 'specify data'.

For the specified input channel you can check the following options:

- **Modulation Index:** squared modulation index, see box 5.5.
- **Use logarithmic values** (natural log).

The following time related values can be selected:

- Mean: mean value of the time series in the time segment.
- Variation coefficient of the time series in the time segment.

You can also specify from which frequency bands the power has to be computed. You can only select those frequency bands which are specified under the header 'frequency bands'.

Bands	
☒	Total Freq
☐	Low Freq
☒	Mid Freq
☒	High Freq

After filling in the specifications, press . The selected time series is now defined in a row in the window. The selected variables for each time series are indicated with an 'x'. On each row you can enter a time series, although not all rows need to be specified. You can remove one or several defined rows from the list; select the row and press DEL or . In a similar way options from a selected row can be edited by checking or un-checking the options at the right side.

Box 5.5 Modulation Index

In chapter 2 and 3 it was already explained that the mean heart rate values are computed from the number of beats in the measuring period, while spectral variability measures are handled and expressed in heart rate terms. In order to avoid large differences in spectral measures based on IBI rather than HR all spectral power values are computed relative to the mean of the time series in the concerning measuring period. In this way the spectral measures can be seen as variations around the mean value. When the modulation index-option is checked, the power density spectrum will be calculated with normalised scores.

In case the mean value of the time series is not a relevant measure, this option should not be chosen. This holds in most situations for respiration; respiration values are often uncalibrated which leads to a mean value close to zero. Normalisation will then result in incorrect indices.

Note that in the interface the box for modulation index (as well as the one for logarithmic values) is depicted in gray. This is done for reasons of distinction; the modulation index and the logarithmic values are options applied on the computed spectral values.

Transfer Functions

Specify the time series input/output pairs you want to compute transfer functions for. Pressing  will select all the possible transfer function combinations. Usually, a more adequate way is to select a specific transfer functions manually:

- Input and output channel: Here you can select the input and output channels for which you want to calculate the transfer function. As an example; if you want to study the effects of respiration on heart rate, respiration time series is the input channel and heart rate (Rpeaks) is the output channel. You can only choose from those time series you have specified with 'specify data'.

For the transfer function between the input and output channel you can specify if you want to calculate the following options:

- Coherence: coherence between the selected input and output channels (see Box 5.6)

- Modulus: gain-part of the transfer function between the selected input and output channels (see Box 5.6)
- Phase: phase-part of the transfer function between the selected input and output channels (see Box 5.6)
- Minimum coherence: lower limit of the coherence at each frequency point in the specified frequency band for which modulus and phase values are computed; specify a number between 0 and 1. The default value, specified in the original CarspanDefaultSettings.ini file, is set at 0.3.

Furthermore, you can specify for which frequency bands the analysis needs to be performed. You can only select those frequency bands which are selected under the heading 'frequency bands'.

Each row in the window-pane corresponds to one pair of time series (input and output channel). You can remove one or several defined pairs from the list; select the data pair and press DEL or . In a similar way selected options for a specific row can be edited.

Box 5.6 Coherence, modulus, and phase

*The **coherence** is the amount of linear coupling between the input and output channel as a function of frequency, and can be a value between 0 (no linear relationship at all) and 1 (relationship is completely linear). This measure corresponds to 'explained variance' in the time domain. It is possible to specify a minimum coherence. Frequency points with coherence lower than this minimum are excluded from the calculation for the transfer function. A minimum coherence value of 0.30 proves to be a useful boundary.*

*An additional way of comparing two signals is by means of the transfer function, which consists of a modulus (gain) and a phase part. The **modulus** is the ratio between output and input signal. As an example, if the effectiveness of the baroreflex (baroreflex sensitivity) has to be computed, systolic pressure is the input channel and interbeat interval time (IBI) the output channel. The modulus for this channel combination is expressed in ms/mmHg. The modulus function represents the multiplication factor for the amplitude of one signal to obtain the amplitude of the other signal. This factor can be different at each frequency, so the modulus is a function of frequency.*

*The **phase function** is a measure for the time delay of one signal in relation to the other signal at each frequency point. It corresponds to the phase-difference between two signals as a function of frequency, expressed in radians. In CARSPAN both the wrapped and the unwrapped phase function are computed; see chapter 2 and 3 for more explanation.*

Both modulus and phase functions are not reliable if the coherence is low, for instance lower than 0.3. For this purpose the 'minimum coherence' limit is useful.

Specify Analysis Profiles

With spectral profiles, spectral computations are performed on (partly) overlapping data-windows which result in, for instance, spectral power in a certain frequency band as a function of time. In a similar way, mean values (running mean) of a time series, variation coefficient as well as coherence, modulus, and phase in a specified frequency band can be computed. For this, go to **Specification > Specify Analysis Profiles**.

Spectral profiles are computed within a block; a block is a consecutive data segment which begins and ends at the block-start and stop code in the data file, respectively. When no codes are defined, the beginning and end of a file are used to define a block. The first window starts at the beginning of the block, whereas the beginnings of the following windows are determined by the amount of overlap (step size). In this way a time series of output band values is created which represents the time course of, for instance, spectral power in a specified frequency band or of the running mean of a time series. Each output value in a specific window is assigned to the time point corresponding to the middle of the window.

Several options can be specified in order to compute spectral profiles:

- Window Length

The profiles are calculated using a window that is shifting in time in order to get consecutive output (band) values. Here you can specify the length of this window. The default setting of the window length is specified in the CarspanDefaultSettings.ini file; in many cases 30 seconds is a low but useful value.

Note: The window length should at least be 15 seconds to fit the mid and high frequency band limits. Furthermore, a longer window length will increase the reliability of the spectral values.

- Step Size

The window is shifting in time by a variable called step size. For each step the specified band values are computed in that window. The step size (in seconds) should be smaller than the window length to create an overlap between segments. The minimum step size is 1; the default setting is set in the CarspanDefaultSettings.ini file; useful values could be 1, 5 or 10 seconds.

- Frequency Bands

By clicking on the double arrow () the default frequency bands from the CarspanDefaultSettings.ini file will be selected. Otherwise, specify the frequency band(s) with a minimum and maximum in Hz. In this case, make sure that the window length is long enough for the specific frequency band limits. The length of the window has to be at least the length of the inverse of the lowest frequency bands. For example, if the mid frequency band boundaries are set at 0.07 and 0.14 Hz, the length of the window should be at least 15 seconds. If not, a warning will appear:

 : the windowlength is too short for this bandlimit

The number of frequency bands you can specify is unlimited. Not all blank rows have to be defined, however.

- Power Functions/Transfer Functions

In the same way as with specifying analysis functions, a table can be created for the band (power) values to be computed, as well as the band values for the transfer functions

(mean (weighted) coherence, mean modulus and mean phase values). Moreover, some options are available. Both the time series and the options can be specified at the right side of the interface.

Power functions

Specify the time series you want to compute power densities for. If you press , power functions for all available time series will be selected for all specified frequency bands. If you want to select the power functions yourself, you have several options:

- Input channel: Here you can select the channel for which you want to compute a power density spectrum. You can only choose those time series you have specified with 'specify data'.

For the specified input channel you can check the following options:

- Modulation Index; squared modulation index, see box 5.5.
- Use logarithmic values (natural log)

The following time related values can be selected:

- Mean: mean value of the time series in the time segment.
- Variation coefficient of the time series in the time segment.

You can also specify from which frequency bands the power has to be computed. You can only select those frequency bands which are specified under the header 'frequency bands'.

Bands	
☒	Total Freq
☐	Low Freq
☒	Mid Freq
☒	High Freq

After filling in the specifications, press . The selected time series is now defined in a row in the window. The selected variables for each time series are indicated with an 'x'. On each row you can enter a time series, although not all rows need to be specified. You can remove one or several defined rows from the list; select the row and press DEL or . In a similar way options from a selected row can be edited by checking or un-checking the options at the right side.

Transfer Functions

Specify the time series input/output pairs you want to compute transfer functions for. Pressing  will select all the possible transfer function combinations. Usually, a more adequate way is to select specific transfer functions manually:

- Input and output channel: Here you can select the input and output channels for which you want to calculate the transfer function. As an example; if you want to study the effects of respiration on heart rate, respiration time series is the input channel and heart rate (Rpeaks) is the output channel. You can only choose from those time series you have specified with 'specify data'.

For the transfer function between the input and output channel you can specify whether you want to calculate the following options:

- Coherence: coherence between the selected input and output channels (see Box 5.6)
- Modulus: gain-part of the transfer function between the selected input and output channels (see Box 5.6)
- Phase: phase-part of the transfer function between the selected input and output

channels (see Box 5.6)

- Minimum coherence: lower limit of the coherence at each frequency point in the specified frequency band for which modulus and phase values are computed; specify a number between 0 and 1. The default value, specified in the original CarspanDefaultSettings.ini file, is set at 0.3.

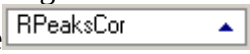
Furthermore, you can specify for which frequency bands the analysis needs to be performed. You can only select those frequency bands which are selected under the heading 'frequency bands'. Each row in the window-pane corresponds to one pair of time series (input and output channel). You can remove one or several pairs from the list;

select the data pair and press DEL or . In a similar way selected options for a specific row can be edited.

5.3.3. Pre-Processing

Process Rpeaks

After the RPeak code is specified with Specification > Specify Data Set, you can go to **Pre-Processing > Process RPeaks**. After specifying the options and settings a new graphical window opens with two graphs; both are representing the RPeak series. The Y-axis represents the IBI-durations in seconds; the X-axis the time in seconds. Each of the graphs has navigation options, see therefore Box 5.7. If these navigation options are not visible, press the down arrow (▼) in the left corner of the graph.

The graph above will always represent the original RPeak data; the window below contains corrected data after applying data correction. A graph can be folded in by pressing ▼ in the right corner of the graph. A beam will then appear in the bottom of the screen, for example . A click on the arrow in this beam will unfold the graph.

Box 5.7 Navigation options in graphs

-  - Increase/decrease vertical scale
-  - Increase/decrease horizontal scale (time axis)
-  - Interpolate selection
-  - Remove selection
-  - Edit chart layout
-  - Go to previous page
-  - Go to next page
-  - Go to the first data points
-  - Go to the last data points

When the mouse button is moved over the upper graph, the X- and Y-coordinates are shown next to the navigation options. Navigation occurs separately for each window; this means that time axes of different windows may become desynchronized. This can be restored by pressing the right mouse button and shifting over one of the graphs. In this case the time windows will be synchronized, while the time scale of each window will not change. By shifting the mouse with the right mouse button pressed over the graph, this graph is shifted over the screen, enabling a more continuous navigation than by means of the left/right arrows. Another probably useful function is that part of the original Rpeak time series can be selected in the upper screen and displayed in the screen below. Try this by pressing the left mouse button in the upper screen, keep depressed, and select a piece of data. Furthermore, several options become available by a click on the right mouse button, i.e., go to the raw data, and add or remove markers (see Box 5.8).

Box 5.8 Useful options

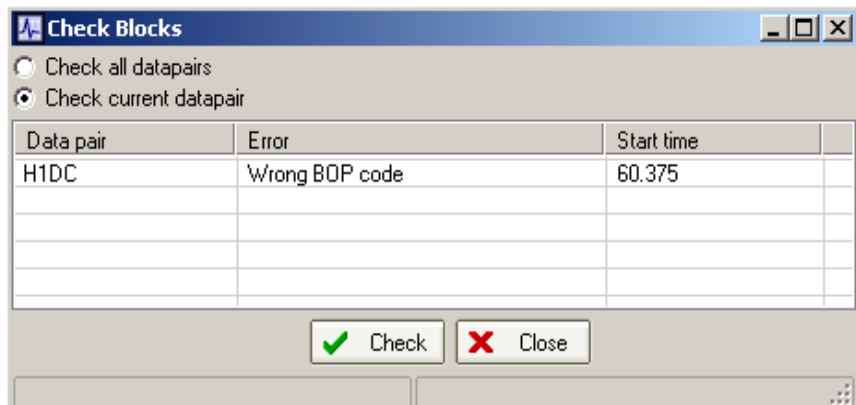
By a click with the right mouse button in the upper or lower graph, you will see the option **'jump to data'**. This option will direct you to the same location in the original (sampled) data. This is useful in case you want to check if the artifacts in the heart rate data could be clarified from the original signal.

Another option that becomes available if you click the right mouse button in the lower graph is **'add marker'**. This is useful in case codes were not specified in the 'specify data' menu. When you add markers manually make sure that you place a marker for 'begin of period' (BOP) after a marker for 'begin of block' (BOB). Periods fall within a block, so you have to define the beginning of a block first. Vice versa, after a marker for 'end of period' (EOP) follows a marker for 'end of block' (EOB). The codes will be recorded in the exported event file.

To remove a marker, click with the right mouse button on the place of the marker, and press **'remove marker'**. To remove all markers in the graph, use the option **'remove all markers'**. In this latter case, be aware that it is not easy to undo this.

Check Blocks

Before artifact correction can be performed, it will be required to check whether the necessary codes for begin and end of blocks and periods are indeed available in the correct order. Therefore, go to **Pre-processing > Check Blocks**. You can specify if you want to check all data pairs or only the current one. Press 'Check' to accomplish. In general, this check-option will be automatically activated when entering the Process Rpeaks option and the Process Time series option. When the codes are not filled in correctly in the 'specify data' section or when one or more block or period codes are not available, the program will give a message. A window will appear that indicates which data pair is affected by the possible wrong code(s), the type of error, and the start time of the error.



Autocorrect Rpeaks

With artifact detection it is possible to find deviating IBI values (i.e., irregularities in the event series) in order to prevent highly inflated values of HRV. Detection and correction of the R-peak event series will only take place within blocks. Artifact detection is performed using a moving window shifted along the event series. IBI-values are considered as too long or too short if a certain interval is deviating too much from the mean IBI-value in the current window.

You can detect and correct artifacts in the heart rate data by selecting **Pre-Processing > Auto Correct RPeaks**. A window will appear in which you can specify several options (see figure 5.3). Preset values in the CarspanDefaultSettings.ini file are set as standard but can be changed manually.

Note: See chapter 3 for the algorithms used with artifact correction.

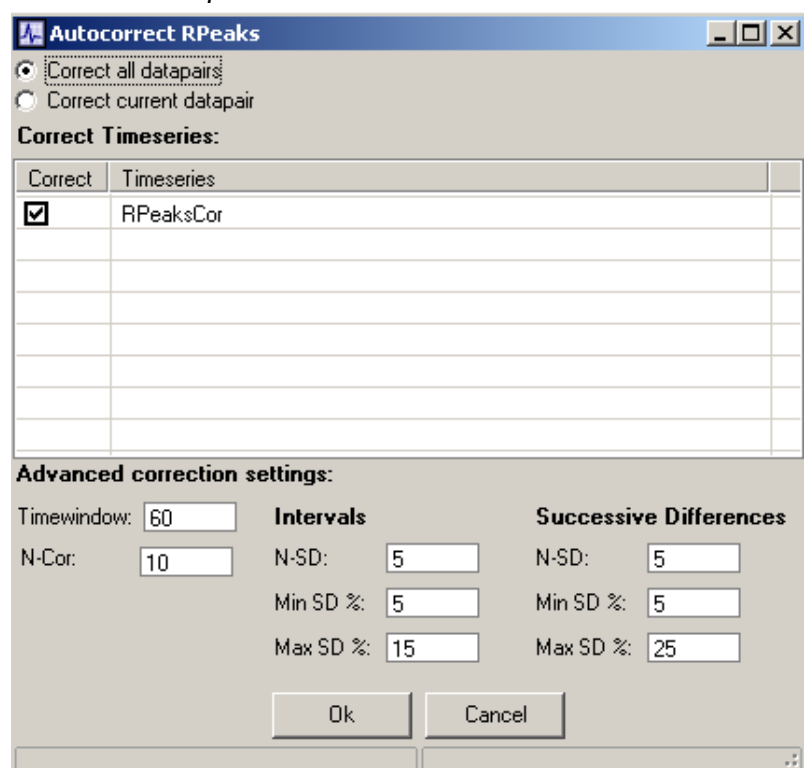
- Correct all datapairs or correct current datapair

Here, you can choose between correcting only one data pair and correcting all data pairs in the data set.

- Correct Timeseries

The time series that will be corrected are displayed here. In this phase of the pre-processing cycle only Rpeaks are available.

- Timewindow



A moving window that is shifted along the R-peak event series is used to detect and correct artifacts. The running mean and SD from the time window preceding the current time window is used to calculate the mean and SD of the current window. Here you can specify the length of the window in seconds.

- N-Cor
This is the maximum number of successive automatic corrections (interpolations) that is acceptable.
- *Intervals:*
 - N-SD: Maximum allowed number of standard deviations from the running mean of consecutive IBI-differences. Enter a positive number; in many cases a value of 5 or 4 will be acceptable.
 - Min SD %: The minimum allowed deviation boundary from the running mean in percentages. If the value of N-SD times the current SD in the actual time window deviates less than Min_SD from the current mean IBI values, then the 'Min SD line' will be taken as the lowest boundary to detect artifacts (instead of the N-SD criterion).
 - Max SD%: The maximum allowed deviation boundary from the running mean in percentages. If the value of N-SD times the current SD in the actual time window deviates more than Max_SD from the current mean IBI values, then the 'Max SD line' will be taken as the upper boundary to detect artifacts.
- *Successive Differences:*
 - N-SD: Maximum allowed difference (in SD) from the running mean of consecutive IBI-differences. If the successive difference value at a certain beat is larger than this specified number of SD of successive differences, then the difference between two consecutive IBI's is seen as an artifact.
Note: The SD of successive differences is computed from the same time window as used for the running mean and SD of IBI-values.
 - Min SD %: The minimum allowed deviation boundary from the running mean of consecutive IBI-differences in percentages. If the value of N-SD times the current SD in the actual time window deviates less than Min_SD from the current mean of consecutive IBI differences, then the 'Min SD line' will be taken as the lowest boundary to detect artifacts (instead of the N-SD criterion for successive differences).
 - Max SD%: The maximum allowed deviation boundary from the running mean of consecutive IBI-differences in percentages. If the value of N-SD times the current SD in the actual time window deviates more than Max_SD from the current mean of successive IBI differences, then the 'Max SD line' will be taken as the upper boundary to detect artifacts.

Box 5.9 Artifacts

Artifacts in the R-Peak event series may consist of either too short or too long intervals. For correction of prolonged IBI's, 'linear interpolation' between two preceding and two succeeding IBI's is used. If there are no preceding or succeeding values (i.e., at the beginning or end of an analysis segment) the running mean is assigned as the new value. Correction of a short interval is accomplished by leaving out an Rpeak. These procedures will also be used to correct other time series, i.e., blood pressure or respiratory time series.

In those cases that the number of succeeding artifacts exceeds the acceptable maximum number of successive automatic corrections (N-Cor), no correction is made. You have to inspect the data carefully, and probably divide the analysis segment into two segments, excluding the part with too many artifacts.

After specifying the options, a list with artifacts shows up next to the graphs. Indicated are type of artifact (short, long, complex, or manual), and begin- and end times of the artifact. Furthermore, at the bottom of the list the total number of artifacts in the channel as well as the total artifact time is shown. The artifacts are visible in the lower graph with the proposed corrections in red. A correction is calculated for an artifact using linear interpolation between two preceding and two succeeding correct values. A correction can be deselected by removing the check. Double-clicking an artifact shows the location in the graph. On the contrary, a double click on an artifact in the lower or upper graph will highlight the artifact in the list.

Include - Accept	Begin	End
☐ RPeaksCor		
☒ COMPLEX	311.734	316.083
Number, Time	1.000	4.349

You can also use *manual correction* (see also Box X, chapter 4): select a small time window around the corrupted beat by pressing the left mouse button, hold, move the mouse to the right, and release. By pressing the hammer () an interpolation will take place. The same can be reached by using the right mouse button instead of the 'hammer'. The manual correction is visible in the list with artifacts, and can be undone by deselecting the artifact.

By clicking on the left and right arrows, you can scroll through the different data pairs to see all the artifacts in the data files.

Datapair:  

*Note: The corrections will be used during further processing; however, the original data has not changed and no corrected data file has been saved up to now! To save all the specifications and data corrections in the session file at this point, go to **File > Save Session**. It is also recommended to export the event files after artifact correction; only through this action the corrections will be recorded in the event files.*

Process Time Series

After specifying the time series with 'Specify Data', the time series will be created and

Each of the graphs has navigation options in the action bar, which were explained in the paragraph on Rpeak processing (see Box 5.7). Navigation occurs separately for each window, so the time axis of the different windows may become desynchronized. This can be restored by pressing the right mouse button and shifting over one of the graphs. In this case the time windows will be synchronized with the middle of the window, while the time scale of each window will not change. When a piece of data in one of the graphs is selected by means of pressing the left mouse button and shifting over the data, the same time segment becomes highlighted in the other graphs as well (except for the original Rpeak data).

Auto Correct Timeseries

Note: See chapter 3 for the algorithms used.

- Here you can choose between correcting only the presently selected data pair and correcting all data pairs in the data set.

- The time series that can be corrected are displayed here. By selecting one or more time series they will be used for artifact correction. Only those time series that were specified with ‘Specify Data’ are available.

- **Timewindow**: A moving window that is shifted along the event series is used to detect and correct artifacts. The running mean and SD from the time window preceding the



current time window is used to calculate the mean and SD of the current window. Here you can specify the length of the window in seconds.

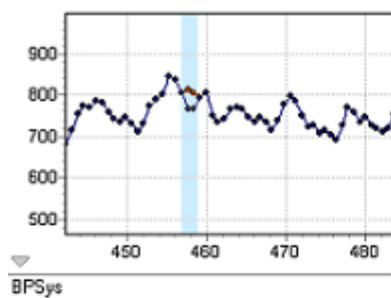
- N-SD: Maximum allowed number of standard deviations from the running mean of consecutive IBI-differences. Enter a positive number; in many cases a value of 5 or 4 will be acceptable.
- N-Cor: This is the maximum number of successive automatic corrections (interpolations) that is acceptable.
- Min SD %: The minimum allowed deviation boundary from the running mean in percentages. If the value of N-SD times the current SD in the actual time window deviates less than Min_SD from the current mean time series values, then the 'Min SD line' will be taken as the lowest boundary to detect artifacts (instead of the N-SD criterion)
- Max SD%: The maximum allowed deviation boundary from the running mean in percentages. If the value of N-SD times the current SD in the actual time window deviates more than Max_SD from the current mean IBI values, then the 'Max SD line' will be taken as the upper boundary to detect artifacts.

After specifying the options, a list with artifacts (long, complex, manual) appears next to the graphs, with begin- and end times. At the bottom of each list you can find the total number of artifacts as well as the total artifact time for that channel. The artifacts are shown for the different time series, and it is possible to (de)select one or all corrections of one or more channels by removing the check(s).

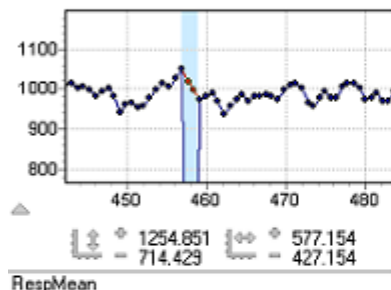
As with correcting Rpeaks, the artifacts are visible in the graph with the proposed corrections in red. A correction is calculated for an artifact using an interpolation between two preceding and two succeeding correct values ('linear interpolation'). Double-clicking an artifact shows the location in the graph of the time series. Vice versa, double-clicking an artifact in one of the graphs will highlight the artifact in the list.

If you correct an artifact manually, the correction will take place in all time series. However, this can be prevented by deselecting channels in which correction is not required. For instance, during manual blood pressure correction it may not be useful to correct IBI values and/or respiratory channels. An example is given in the figure below.

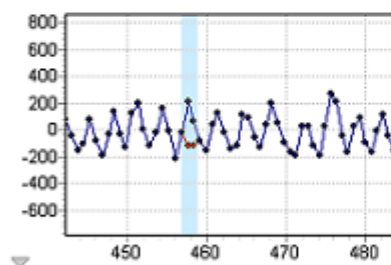
IBI



BPSys

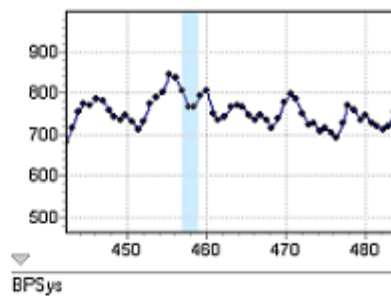


RespMean

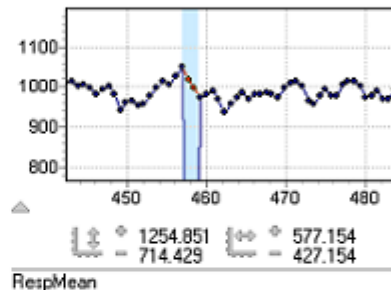


Include - Accept	Begin	End
☒ IBI		
☒ MANUAL	456.954	459.279
☒ BPSys		
☒ MANUAL	456.954	459.279
☒ RespMean		
☒ MANUAL	456.954	459.279

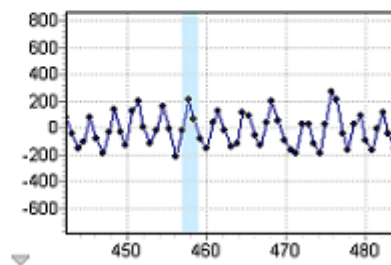
IBI



BPSys



RespMean



Include - Accept	Begin	End
☐ IBI		
☒ BPSys		
☒ MANUAL	456.954	459.279
☐ RespMean		

Two examples of artifact correction are shown above. The left figure shows artifact correction in all time series (IBI, BPSys, and RespMean). In the right figure, artifact correction is only carried through for the systolic blood pressure time series.

5.3.4. Analysis

Calculate Spectral Functions

After filling in all the specifications with **Specification > Specify Analysis Functions**, the spectral functions can be computed; go to **Analysis > Calculate Spectral Functions**. With this option the computation of spectral density, coherence, and transfer function from specified data segments is activated.

Note: After specifying the analysis functions, you can also press F7 at any time to (re)calculate spectral functions.

During calculation of the spectral profiles a window appears in which you can follow the progression. When something goes wrong during the calculation it will be visible in this window. After completing all computations, a browse tree appears with graphical and numerical output. By clicking on the '+' it can be unfolded. When you want to go back to the spectral results later on in the same session, go to '**View > Spectral results**'. The option 'Analysis > Calculate Spectral Functions' will lead to an unnecessary recalculation of the spectra.

The tree is organized according to the following sequence: filename, channel or channel combination (in case of transfer functions), block-period combination.

- Functions (Plots)

The plots of the computed spectra are organized in a list, starting with the power density spectrum of channel 1, followed by the spectra of all other specified channels; subsequently the channel combinations of the transfer functions are shown.

For the spectral power, both the time series and the spectral power density distribution are displayed per selected segment (block, period combination).

- *Time series:*

The names of the selected channels (selected in 'Specifications > Specify Analysis Functions' menu) are depicted per segment.

- Horizontal scale: time in seconds

- Vertical scale: heart rate in Hz; IBI in ms; for other channels it depends on the scale values that are specified in the 'specify data' menu, for instance, blood pressure in mmHg and respiration in uncalibrated units.

Note: see Appendix A.1 for useful navigation functions in the graphical data.

- *Spectra:*

This plot shows the distribution of signal variance over the frequency axis.

- Horizontal scale: frequency (Hz)

- Vertical scale: see explanation in Box 5.10.

Box 5.10 Explanation of scale factors

Spectra: The X-axis represents the frequency in Hz. The Y-axis displays two time series; for example IBI (units: msec^2) and SBP (units: mmHg^2). When the modulation index is marked in the 'specify data' menu, the power density spectrum will display normalised scores. Normalisation has no consequences for the units of the transfer function.

Coherence: The X-axis represents the frequency in Hz. The Y-axis displays a value between 0 and 1; 0 means no linear relationship at all, whereas 1 means a completely linear relationship between the two time series.

Modulus: The X-axis represents the frequency in Hz. The unit of the modulus is dependent of the chosen time series. When, for example, the transfer function between blood pressure (x) and interbeat interval (y) is calculated, the modulus has the unit ms/mmHg . Be aware that the modulus is not reliable for those frequencies where the coherence is lower than a specified value, i.e. 0.30.

Phase: There are two graphs; one for the wrapped, one for the unwrapped phase. The X-axis represents the frequency in Hz. The vertical units are radians (π). See chapter 3.3.4 for more explanation.

After the list of spectral power functions for separate channels, the computed (transfer) functions for channel combinations are listed. The corresponding plots contain subsequently the time series itself (both channels, combined), the power density spectra (both channels, combined), the coherence, the modulus and the phase (wrapped and unwrapped phase, combined).

The plots of the time series and the (power density) spectra are build up in the same way as already discussed above, now, however, with two graphs in the same plot; the scale at the left side corresponds with the first channel; the scale at the right side with the second channel of the combination.

- *Coherence:*

This displays the amount of linear coupling between two signals in the frequency domain, as a matter of fact this corresponds with explained variance in the time domain.

- Horizontal scale: frequency (Hz).
- Vertical scale: between 0 and 1.

- *Modulus:*

The modulus specifies the ratio between changes in one signal (output) and changes in the other signal (input) as a function of frequency.

- Horizontal scale: frequency (Hz).
- Vertical scale: unit of times series Y per unit of times series X. For instance, if the input channel is systolic blood pressure in mmHg, and the output channel is IBI in ms, the vertical scale of the modulus is in ms/mmHg.

- *Phase*

This indicates the phase difference between two signals as a function of frequency. In CARSPAN two phase functions are computed: the common phase function, specified between $-\pi$ and $+\pi$, and the unwrapped phase.

- Horizontal scale: frequency (Hz).
- Vertical scale: radians (π)

- Functions (Numerical)

This contains a summary file (man readable) and a numerical (listed) file with all results of the selected analysis, suitable for further statistical processing.

- *Summary output file:*

In the first part of this file a summary is given of:

- The selected input and output files
- The selected parameters, i.e., block and period codes, event code for the Rpeak, and selected frequency bands.
- The selected channels
- The selected analyses.

The second part contains start and stop time points, for each analysis segment. Mean values and band information for all selected analysis of each channel are also given (see Appendix C.2).

- *Numerical output file:*

The first part of this file is the same as the summary output file. The second part contains

the numerical output with the following information:

- Type of analysis, experiment -and subject number, block- and period number, length of the time segment, number of Rpeaks, and mean value and coefficient of variation.
 - Band information and the number of spectral points within these bands.
 - Spectral values of all frequency points with frequency resolution equal to 0.01 Hz.
- The last frequency point is half the mean heart rate.

(See Appendix C.1 for an example.)

Box 5.11 Useful navigation functions

When the transfer functions are shown, several graphs arise in the screen. To fold one of the graphs in, make use of the grey beam (arrows pointing up):



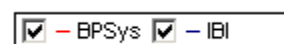
To unfold (arrows pointing down):



To enlarge or decrease the vertical scale of one of the graphs, move the cursor to the grey area below the graph in question, press the left mouse button, and keep depressed while moving the cursor up or down.

Furthermore, you can zoom in a piece of data by pressing the left mouse button and moving the cursor from the upper left corner downwards to the lower right corner. Moving the cursor upwards from right to left (while keeping the left mouse button depressed) will lead to a zoom out of the data. With the right mouse button depressed you can scroll through your data. Navigation occurs separately for each graph.

When two time series are represented in one plot, the legend shows which line corresponds to which channel. You can (de)select one or both channels.



Calculate Spectral Profiles

After specifying the analysis profiles, the spectral profiles can be computed; go to **Analysis > Calculate Spectral Profiles**. With this option, the same spectral computations are performed as with 'calculate spectral functions' (i.e., spectral density, coherence, and transfer function), but now on (partly) overlapping data-windows. This results in spectral profiles, e.g. spectral power in a certain frequency band as a function of time.

Note: After specifying the analysis profiles, you can also press F8 at any time to (re)calculate spectral profiles.

During calculation of the spectral profiles a window appears in which you can follow the progression. When something goes wrong during the calculation it will be visible in this window. After completing all computations, a browse tree appears with graphical and numerical output. By clicking on the '+' it can be unfolded.

- ⊕ Functions (Plots)
- ⊕ Functions (Numerical)
- ⊕ Profiles (plots)
- ⊕ Profiles (Numerical)

When you want to go back to the spectral results later on in the same session, go to **'View > Spectral results'**. The option 'Analysis > Calculate Spectral Profiles' will lead to an unnecessary recalculation of the spectra.

The tree is organized according to the following sequence: filename, channel or channel combination (in case of transfer functions), block, frequency band.

When the spectral profiles analysis is completed,

- Profiles (Plots)

This function shows plots of the spectral profiles analysis results. If selected in ‘specify analysis profiles’ the power spectral values, coherence, modulus, and phase values are shown per block for the specified frequency bands.

Navigations options are the same as for spectral functions plots. See Box 5.11 for further information.

- Profiles (Numerical)

The numerical output contains the time course (profile) of the results per selected frequency band of the selected analyses. Also the profiles of the selected mean values and variation coefficients are included if selected. The format of the output is about the same as the format of the original (extended) event file. The codes for Rpeak (not really Rpeaks, but used here for compatibility reasons), beginning and end of block and/or periods are visible in the header of the event file. Below the event codes you can find the labels of the specified analysis profiles.

The data part of the numerical profile file contains a first column with the codes that are copied from the original (extended) event file and additionally a fixed code (99), indicating the time points of the middle of each of the analysis periods. The second data column forms the time axis, while each of the further columns corresponds to one of the specified (spectral) profiles (See Appendix C.3).

This numerical file can be used for further statistical processing after selecting adequate pieces of data, for instance, using SPSS. Additionally, for viewing purposes, the numerical profiles output file can be read in a separate CARSPAN session using the ‘time series processing’ option.

5.3.5. View

View Spectral Results

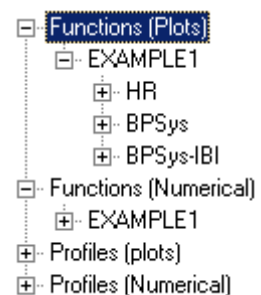
With **View > Spectral Results** you can see the results of the spectral analysis. The computed spectral results of the current session are displayed, exactly in the same format as obtained directly after the spectral computations.

View Raw Data

Information about the .NFF and the .EVT files is available with **View > View Raw Data**. First, you can select the file of interest in the browse tree. Next, you can see the text info (left tab, ASCII format) or sampled data (graphic info) by choosing the right tab. A double-click with the mouse button on the file gives you the option to see the ASCII file.

➤ View Text Info

The right side of this window shows the .EVT file. On the left side of this window the .NFF file information is displayed, which contains the following information (see figure 5.5):



- *Origin*: file name
- *Conversion*: not relevant
- *Number of Channels*: number of channels in the .NFF file.
- *Creation Date*: date when .NFF file was created
- *Block Time*: not relevant
- *AD Resolution*: resolution in bits
- *Start Time*: begin time of the .NFF file
- *Number of Trials*: value has to be 1 (no segmentation in trials); file is not trial-oriented.
- *Trial Length*: length of the trial in ms.
- *Channel number*: number of the channel in the .NFF file.
- *Label*: label of the channel
- *Interval*: sample interval in μ s
- *Delay*: not relevant
- *Number of Samples*: number of samples in this channel across the complete file.
- *Calibration*: not relevant
- *AdChan*: not relevant

View Text Info		View Graphic Info
EXAMPLE1.nff		
NFF File information		
Origin	2CROEMF2.nff	
Conversion		
Number of Channels	7	
Creation Date	20-OCT-2006 13:37:15.00	
Block Time	5120	
AD Resolution	12	
Start Time	286745	
Number of Trials	1	
Trial Length	1331200	
Channel information ◀ ▶		
Channel Number	1	
Label	ECG	
Interval	10000	
Delay	0	
Number of Samples	133120	
Calibration	0	
AdChan		

By clicking on the left and right arrows you can scroll through the different channels in the .NFF file.

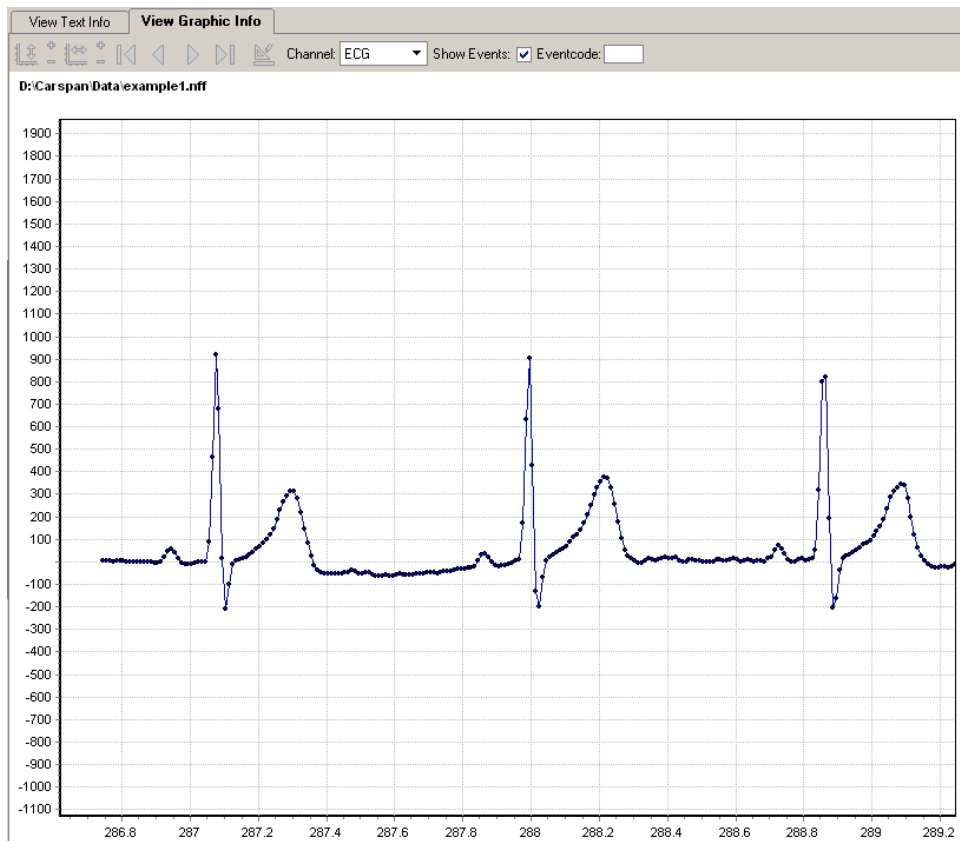
➤ View Graphic Info

This will show you the time plots of the sampled data (see figure below). The icons in this window are explained in Appendix A.1. Other options that are visible in this window are:

- Channel
Choose the channel for which you want to see the raw sample data.
- Event Code
Specify the Rpeak event code that corresponds to sampled cardiovascular signals.
- Show Events
After specifying the event code for Rpeak, mark 'show events' to see the Rpeak time points as vertical markers in the plot.

Note: It might be necessary to click more than once on 'Show Events' to see the Rpeak markers; in particular when the Rpeak event code has changed, the show event indication has to be switched off and on again to get the desirable result.

Note: Notice that this window is the same as you will get with 'jump to raw data' in the pre-processing menu.



6. Technical aspects

CARSPAN has been developed and tested under Windows XP. The requirements for the program are not higher than for this operating system itself, so it is expected that it will run under all versions of Windows XP. Probably it will run also on computers using Windows 2000, but this is not tested systematically.

Some problems are known related to the 'View raw data'-option, more specifically to the plotting of Rpeak events in the sampled data and on displaying the ASCII event files in this same option. Probably these problems are related to settings of the Windows XP-environment.

All CARSPAN interface components as well as computational routines are written in the programming environment Delphi.

7. Installing CARSPAN

The CARSPAN program can be installed by starting the CARSPAN Set-up file. The program is installed at C:\Program Files\IDP Software\Carspan, together with the necessary tools and components. Also the CarspanDefaultSettings.ini file with the configuration settings is installed in the same directory. Additionally, a number of subdirectories are created with the following goals:

- Sessions: for storing session files; the user can freely choose using another directory, for instance, connected to his data area;
- Help: this directory will contain the help file(s) in one of the next versions of the program;
- Tools: contains the sub-directories with tools as used in the Tool-menu; will be one of the main topics in the next version;
- Examples: contains all the data files, as well as the session files, as used in the examples worked out in the manual;
- Data: data area that can be used for storing the users' data files (.evt and .nff), although the user can freely choose using other directories for his data files.

8. Elaborated example

In chapter 2.5 it was indicated that the example which is worked out in this users guide is related to data of an experiment on mental task load. Of course, it is impossible to include a whole experiment, or even the complete data set of one of the subjects in this users guide. Therefore, only a part of the data of one of the subjects is worked out here. The following selection is made. In this chapter of the Users Guide a five minute period of baseline-rest and a task period of 5 minutes are taken for the analysis of spectral functions. All data preparation steps are included. For the example of the spectral profiles option the whole file consisting of one data block, including about five minutes of baseline-rest and five minutes of task performance of the same subject is chosen. Analysis is restricted to HR and SBP.

In order to get an idea what has to be done to analyze more data periods from the same file, an example is included with the CARSPAN distributed package. All input, output and intermediate files as well as a session description are included. The example files will be copied to the subdirectory 'examples' of the main CARSPAN directory (e.g. C:\Carspan\Examples) when CARSPAN is installed.

The same analysis options are specified as in the described example in the present chapter. Additionally, also respiratory data are analyzed in the example files; these are not described in this chapter.

One of the main characteristics of the CARSPAN package is that different phases of processing are available: session specification, actual processing, and viewing of results. At hand of the present example all these phases will now be reviewed.

In the specification part data files and its characteristics, (pre-)processing options, and analysis options are specified. In the pre-processing part artifact correction is done, while the real computations are performed in the analysis part.

The analysis specification part contains two options: spectral functions and spectral profiles. The first option specifies the computation of spectral density, coherence, and transfer function from specified data segments, while in the second option the same spectral computations are performed on (partly) overlapping data-windows which results in spectral profiles (e.g. spectral power in a certain frequency band as a function of time).

The example which is worked out, therefore, can be split in two parts: one which contains the analysis in terms of spectral functions, the other in terms of spectral profiles. In both parts the same data file is used, while only small differences are necessary in the specification of the analysis options. Artifact processing and other pre-processing steps only have to be executed once.

In the following sections each of these data specifications, (pre-)processing, and analysis functions are described in respect of the choices to be made for the described example.

8.1 Data-specification and pre-processing

Before the first analysis session can be started the configuration file (.csf) has to be specified. A new data session file will be created.

8.1.1 Description

Entering data

The data-set which is used in this example consists of the files Example1.evt and Example1.nff. The first is the event file in which the R-peak events and the begin/end events are stored; this is an ASCII-type file which can be looked at with the 'view text info' option from the View menu after specification of the file. The second contains the sampled values of, among others, the blood pressure and respiratory signal; this file is of a binary type. The contents of this file can be (partly) viewed and plotted using the 'view graphic info' option from the View menu.

The specification of the data set is performed in the specification menu with the 'specify data' option, which enables the specification of a number of standard labels used in the program, such as R-peak code, begin/end codes of blocks and periods as well as a number of user defined code labels, e.g. stimulus and response codes which are not directly coupled at the analyses to be performed (see fig. 8.1). In the same window the time series can be specified. Channel numbers have to be given and calibration data can be specified. Calibration data (scale, zero) have to be inputted manually. If the event file already contains pre-computed time series, either previously computed by CARSPAN or obtained by other programs, these can also be specified here. (Note: this option is applied in a separate example on file (Example 3).)

Check blocks

One of the first steps to be taken in the analysis sequence is a check whether all block and period begin/end codes are present and whether there are no hindering additional events with the same code. This can be done with the Check Blocks option in the pre-processing menu. When codes are not filled in correctly in the 'specify data' section or when one or more block or period codes are not available, it will be shown here. The window that appears displays which data pair is affected by the possible wrong code(s), the type of error, and the start time of the error.

R-R interval correction

The next option to be performed is to detect whether too short or too long intervals occur and to correct these in the R-peak event series itself. The results of the correction procedure are discussed in section 8.1.3.

Correct time series

With the option Correct time series the derived blood pressure and respiratory time series will be corrected by means of linear interpolation between two correct values preceding and two correct succeeding values in each of the specified time series. Correction will be performed for the specified channels.

8.1.2 Specification

Create a new session

With the option 'new session' in the file menu, a new CARSPAN session file will be created. Browse to the data directory C:\Carspan\Data, and type 'example1' as session

name. By specifying this name the complete description of the session status can be obtained when the session has to be continued at a later moment in time.

Dataset specification

Select 'example1.evt' as a dataset.

- Check the sampled data (.nff) check-box in order to make ECG, blood pressure and respiratory signals available.

Specify data

In this window, specify the standard event codes: Rpeak: 0; BOB (begin-of-block): 11; EOB (end-of-block): 12; BOP (begin-of-period): 21; EOP (end-of-period): 22.

Furthermore, select time series type:

Type: IBI, deselect IBI inverse, press 

Type: Blood pressure;

Data file channel: BP;

Select BP systolic (deselect others),

Manual calibration: Scale Factor: 0.125

Zero Level: 0

Press 

Now, all the relevant aspects with respect to your data are specified. You will have the same specifications as are displayed in figure 8.1:

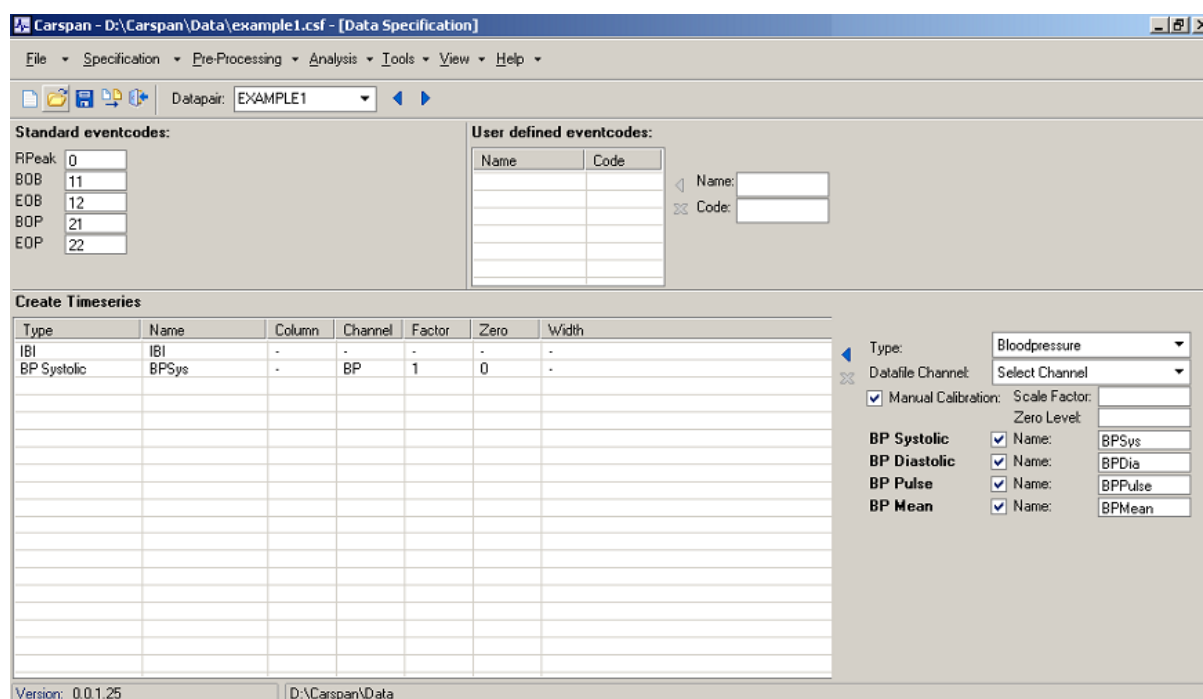


Figure 8.1: Data specification menu. Standard code labels for Rpeak, begin/end of block, and begin/end of period, user defined codes, and the selection of time series can be created here.

File > Save session; all the specifications with respect to your data is now saved.

8.1.3 Pre-processing, viewing, and interpretation of results

Pre-processing; Process Rpeaks;

- Two graphs are displayed, both representing the Rpeak series; the Y-axis represents the IBI-durations in seconds; the X-axis the time axis in seconds. Block- and period codes are shown in the graphs.

Pre-processing; Check blocks; check current data pair; press check;

- ‘No block or period errors found’. This means that all begin and end codes of the blocks and periods are available in the right order.

Pre-processing; Auto Correct Rpeaks;

Select ‘Correct current datapair’;

You will have the same specifications as are displayed in figure 8.2:

Figure 8.2 Specifications for Rpeaks corrections

Press ‘Ok’;

- One artifact was found at the beginning of the block, with a length of 4.4 seconds:

Include - Accept	Begin	End
☐ RPeaksCor		
☒ COMPLEX	311.734	316.083
Number, Time	1.000	4.349

This artifact is automatically corrected by CARSPAN.

Pre-processing; Process Time series;

- The data for Rpeaks, IBI, and BPSys are displayed.

Pre-processing; AutoCorrect Time series;

Choose 'Correct current datapair';

Select only 'BPSys' under Correct Timeseries;

You will have the same specifications as are displayed in figure 8.3:

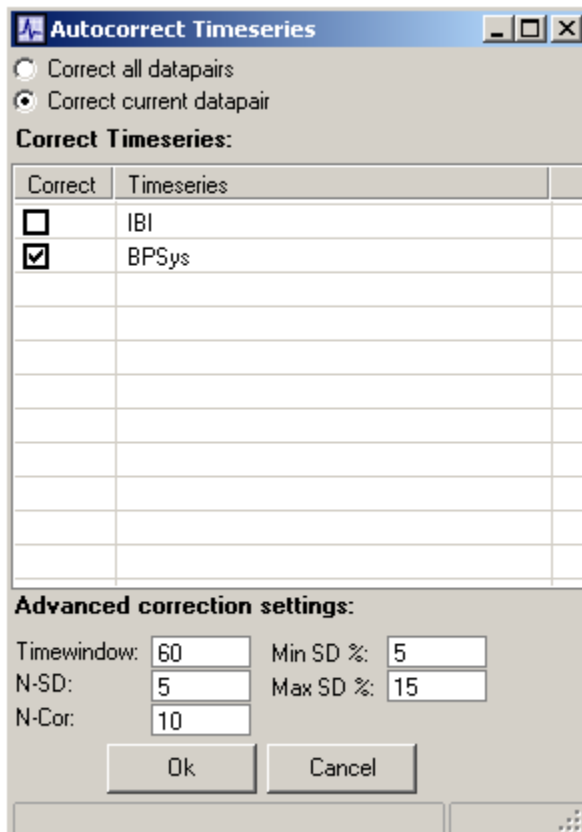


Figure 8.3 Auto Correct Time series window

Press 'Ok'

- The list with artifact corrections contains the series of detected artifacts indicated with begin- and end times of the artifacts. In the time series of the systolic blood pressure 13 artifacts were found, with a total artifact time of 31 seconds (figure 8.4). The artifacts are automatically corrected by CARSPAN.

Include - Accept	Begin	End
☒ IBI		
☒ BPSys		
☒ COMPLEX	316.821	319.104
☒ COMPLEX	355.506	357.617
☒ LOW	402.780	404.578
☒ COMPLEX	456.954	459.279
☒ COMPLEX	509.437	512.225
☒ COMPLEX	562.756	565.518
☒ COMPLEX	613.882	616.601
☒ COMPLEX	666.965	669.303
☒ COMPLEX	720.815	723.161
☒ COMPLEX	776.433	778.653
☒ COMPLEX	829.597	831.936
☒ COMPLEX	882.230	884.459
☒ COMPLEX	934.498	937.285
Number, Time	13.000	31.045

Figure 8.4 List with detected artifacts indicated with begin- and end times of the artifacts. At the bottom the total number of artifacts and total artifact time are shown.

Press 'File > Save session' to save all the specifications and corrections.

8.2 Spectral Functions

8.2.1 Description

Data-segmentation

First the blocks and periods have to be defined from which the spectral functions have to be computed. As already indicated, a data-block is specified between begin/end codes for a block (events 11 and 12 respectively), while a further period (analysis segment) specification is done with begin/end codes for periods (events 21 and 22 respectively).

Selection of functions

According to the specification in the first part of the example, power density spectra of the following time series can be computed: HR, IBI and SBP. However, in order to keep the example within useful limits, the power spectral analysis is restricted to HR and SBP. In the extended example 'Resp' is added. Coherence and modulus will be specified for the IBI/SBP combination; no phase spectra will be computed.

Furthermore, the default frequency bands will be selected. In addition, the squared modulation index will be used for HR and SBP. For the minimum coherence option, which is used in the mean-modulus computation to determine which frequency points have coherence higher than the specified value, a value of 0.3 is specified.

Output

With the 'calculate spectral functions', means and band information will be written to the summary file, additional to all information that is written in the numerical file, including raw

spectral data and summarized values per frequency band. The directory of these files is 'Output'. These raw spectral values allow the preview/plotting of spectral functions afterwards.

8.2.2 Specifications

Specify Analysis Functions;

Segments: select 'periods marked by period code';

Frequency bands: press double-arrow-left (frequency bands are selected, according to preset values in the CarspanDefaultSettings.ini file);

Choose tab: Power functions (at left side);

Select input: *Rpeaks* (at right side);

Set the parameters for Modulation Index, deselect Logarithmic Power Values;

Select Mean, deselect Variation Coefficient;

Select all frequency bands (all are set by default)

Press .

Select input: *BPSys* (at right side);

Set the parameters for Modulation Index, deselect Logarithmic Power Values;

Select Mean, deselect Variation Coefficient;

Select all frequency bands (all are set by default);

Press .

Choose tab: Transfer functions (at left side);

Select input: *BPSys*

Select output: *IBI*

Set the parameters for Coherence and Modulus (deselect phase), minimum coherence 0.30,

Select all frequency bands (all are set by default);

Press .

You will have the same specifications as are displayed in figure 8.5 (only the power functions display is shown here):

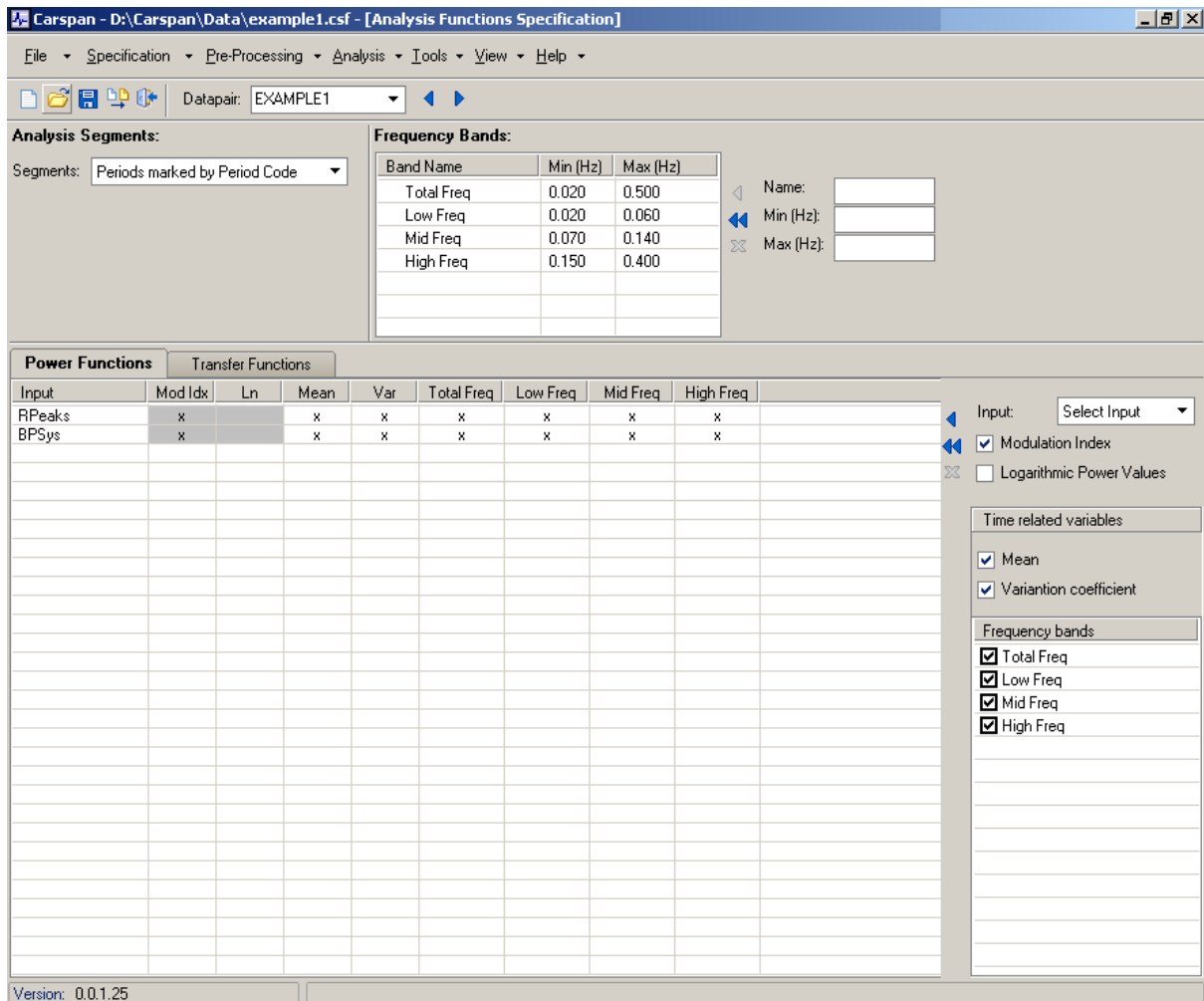


Figure 8.5: Specifications for the Analysis Functions.

File: Save session (information thus far specified is stored in the session file).

8.2.3 Processing and viewing results

Analysis; Calculate spectral functions;

- During calculation of the spectral functions a window appears in which you can follow the progression. When something goes wrong during the calculation it will be visible in this window. After completing all computations, a browse tree appears with graphical and numerical output.

Note: In case you want to have a look at the original data or at the specifications by moving back to the pre-processing or specification window, you can come back to the spectral results by selecting 'View > Spectral results'. Moving back to the 'Analysis > Calculate spectral functions' option would force the program to re-calculate the spectra.

8.2.4 Interpretation of results

Spectral functions

Results of the computed spectral functions are given in the summary output file and the numerical output file. The summary output file starts with a specification of all specified options (segmentation, spectral bands, selected channels and functions) and contains furthermore the start and stop time of the specified period as well as mean values of both channels (HR and SBP) and the values of the specified bands: total power in each band, weighted coherence and the mean modulus value. In these latter values it can be seen that not all frequency points in the low and high frequency band have a coherence which is higher than the specified boundary (0.30).

The numerical output contains the data in a format which can be used for plotting of the spectral functions. This file is also suitable for statistical analysis. Each spectral computation delivers a data line (record), which contains experiment and subject number, mean values of the time series, band values and finally, each of the frequency points of the computed spectral function (fixed resolution: 0.01 Hz). In the example the very low frequencies (below 0.04 Hz) can be recognized, as well as the mid frequency range, with a peak around 0.09 Hz, and an area around 0.28 Hz corresponding with the breathing rate of the subject. In this example, the spectral power of systolic blood pressure can be found at the same frequencies region as in IBI.

Be aware that the transfer function of SBP and *IBI* are displayed in the graphs, while for the computation of the power density spectrum, however, SBP and *HR* are used.

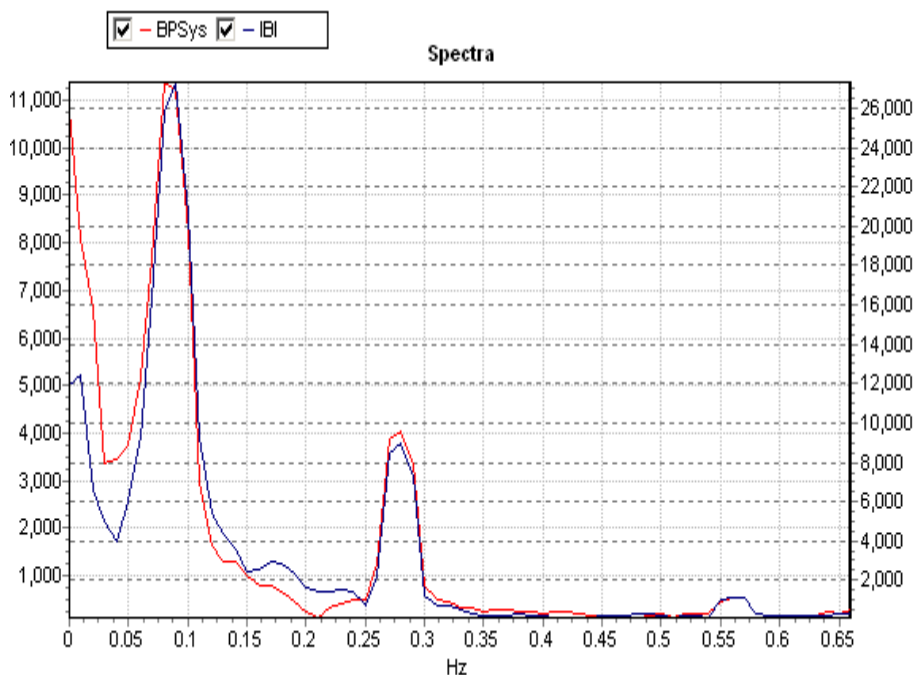


Figure 8.6a: Spectral results: Power density spectra of IBI and systolic blood pressure as analyzed in this example during task performance. Left axis: BPSys. Right Axis: IBI.

Note: In this figure the spectral results are taken that correspond to computation of the transfer function.

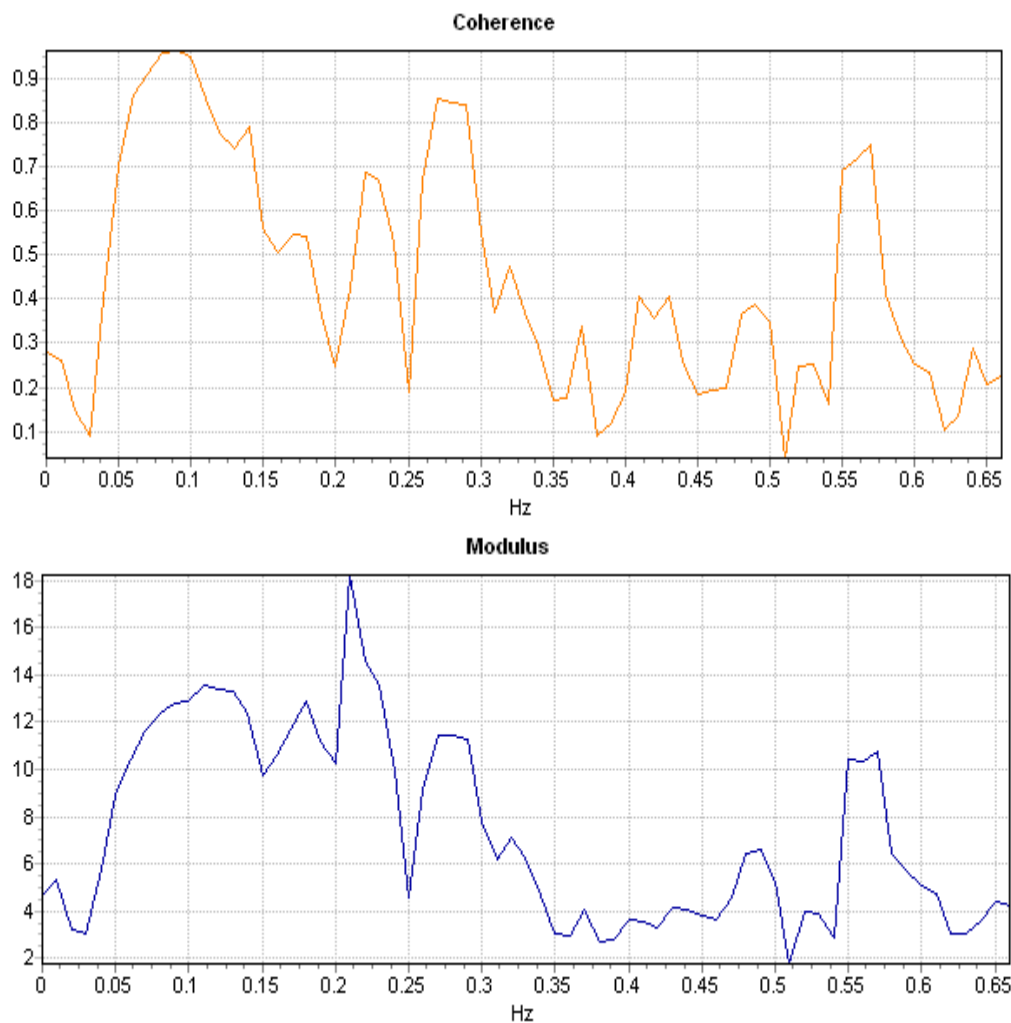


Figure 8.6b: Spectral results. Coherence and modulus function between IBI and systolic blood pressure during task performance. The scale values for the modulus are in ms/mmHg.

8.3 Spectral profiles

8.3.1 Description

Data-segmentation

First the blocks have to be defined from which the spectral profiles have to be computed. As already indicated, a data-block is specified between begin/end codes for a block (events 11 and 12 respectively). Additionally, in this option so-called 'windows' have to be specified. This means that the block to be analyzed is subdivided in (partly) overlapping segments (windows). From each window spectral variability in a number of specified frequency bands is computed. In this example the window size is chosen equal to 60 s, while the step size is 5 s. This means that there is an overlap of about 90% between succeeding windows. In this example the same file is used as within the computation of spectral functions; spectral profiles are computed from the whole block, including the whole baseline period and the

succeeding task period.

Selection of functions

According to the specification in the first part of the example, profiles of mean IBI and mean SBP, spectral variability profiles from the mid frequency band of HR and SBP as well as profiles of the modulus function between SBP and IBI will be computed. In the extended example, on file, spectral power in the high frequency band of HR and SBP will be added.

Analysis options

In the analysis options menu no additional frequency bands will be specified. The squared modulation index will be used for HR and SBP. In the minimum coherence option, which is used in the mean-modulus computation to determine which frequency points have coherence higher than the specified value, a value of 0.30 is specified.

8.3.2 Specifications

Specify Analysis Profiles;

Window length: 60 seconds, Step size: 5 seconds;

Frequency bands: press double-arrow-left (frequency bands are selected, according to preset values in the CarspanDefaultSettings.ini file);

Choose tab: Power functions (at left side);

Select input: *Rpeaks* (at right side);

Set the parameters for Modulation Index, deselect Logarithmic Power Values;

Select mid frequency bands (deselect high frequency band);

Press .

Select input: *BPSys* (at right side);

Set the parameters for Modulation Index, deselect Logarithmic Power Values;

Select mid frequency bands (deselect high frequency band);

Press .

Choose tab: Transfer functions (at left side);

Select input: *BPSys*

Select output: *IBI*

Set the parameters for Coherence and Modulus (deselect phase), minimum coherence 0.30;

Select mid frequency band (deselect high frequency band);

Press .

You will have the same specifications for the power functions as are displayed in figure 8.7:

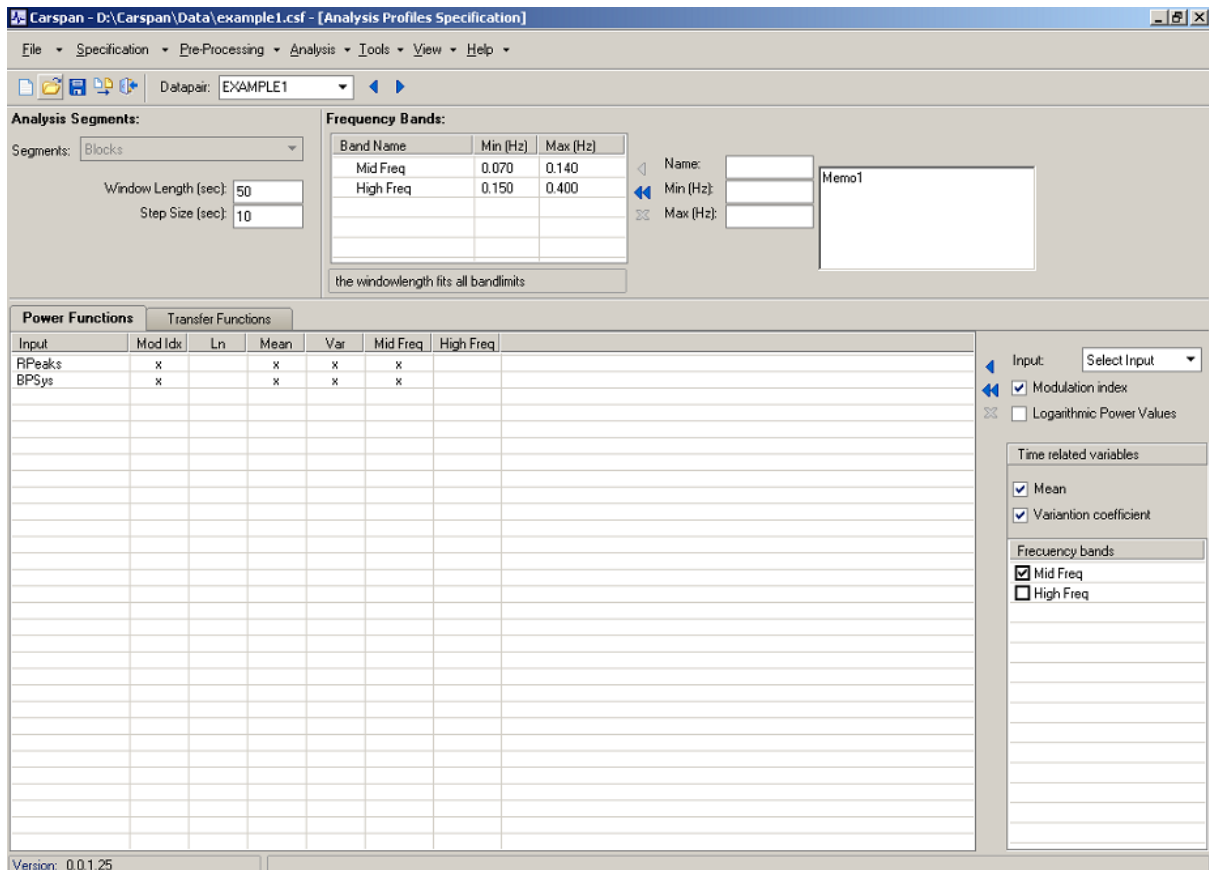


Figure 8.7: Specifications for the analysis profiles.

8.3.3 Processing and viewing of results

Analysis; Calculate spectral profiles;

- During calculation of the spectral profiles a window appears in which you can follow the progression. When something goes wrong during the calculation it will be visible in this window. After completing all computations, a browse tree appears with graphical and numerical output.

Note: In case you want to have a look at the original data or at the specifications by moving back to the pre-processing or specification window, you can come back to the spectral results by selecting 'View > Spectral results'. Moving back to the 'Analysis > Calculate spectral profiles' option would force the program to re-calculate the spectra.

8.3.4 Interpretation of results

Spectral profiles

Results of the computed spectral profiles can be found in the spectral profiles file.

The numerical output contains the time course (profile) of the results per selected frequency band of the selected analyses. Also the profiles of the selected mean values are included. The format of the output is about the same as the format of the original (extended) event file. The codes for Rpeak (not really Rpeaks, but used here for compatibility reasons), beginning and

end of block and/or periods are visible in the header of the event file. Below the event codes you can find the labels of the specified analysis profiles.

The data part of the numerical profile file contains a first column with the codes that are copied from the original (extended) event file and additionally a fixed code (99), indicating the time points of the middle of each of the analysis periods (windows). The second data column forms the time axis, while each of the further columns corresponds to one of the specified (spectral) profiles.

Note: Periods (rest and task period) are not visible in the graphs shown here. In this example there are (almost) no differences in values between rest and task period.

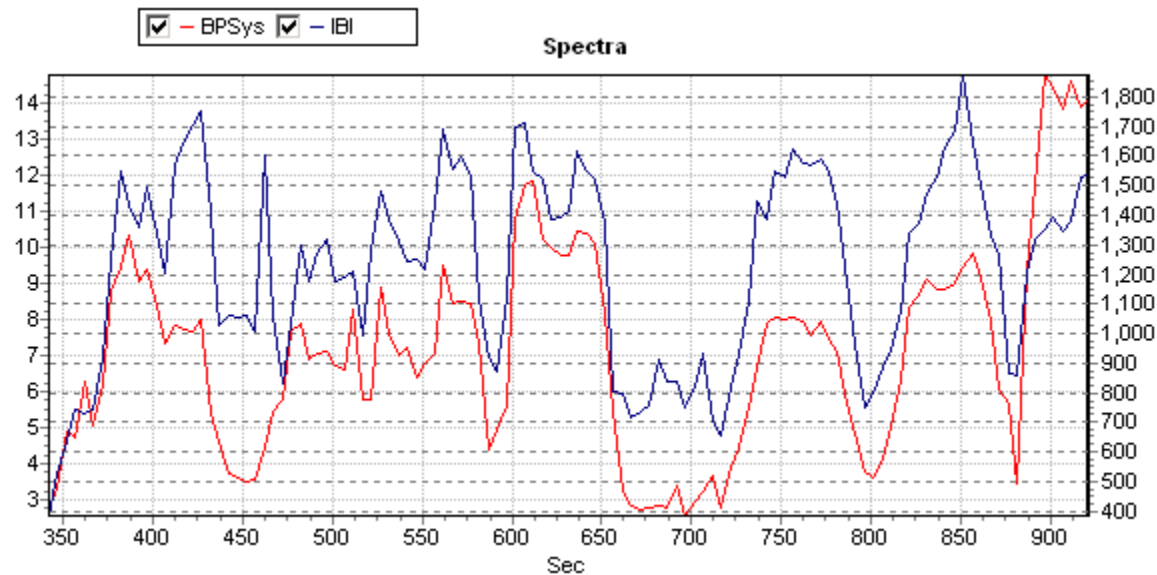


Figure 8.8a: Spectral power in the mid frequency band of IBI and systolic blood pressure as analysed in this example during task performance. Left axis: BPSys. Right Axis: IBI.

Note: In this figure the spectral results are taken that correspond to computation of the transfer function.

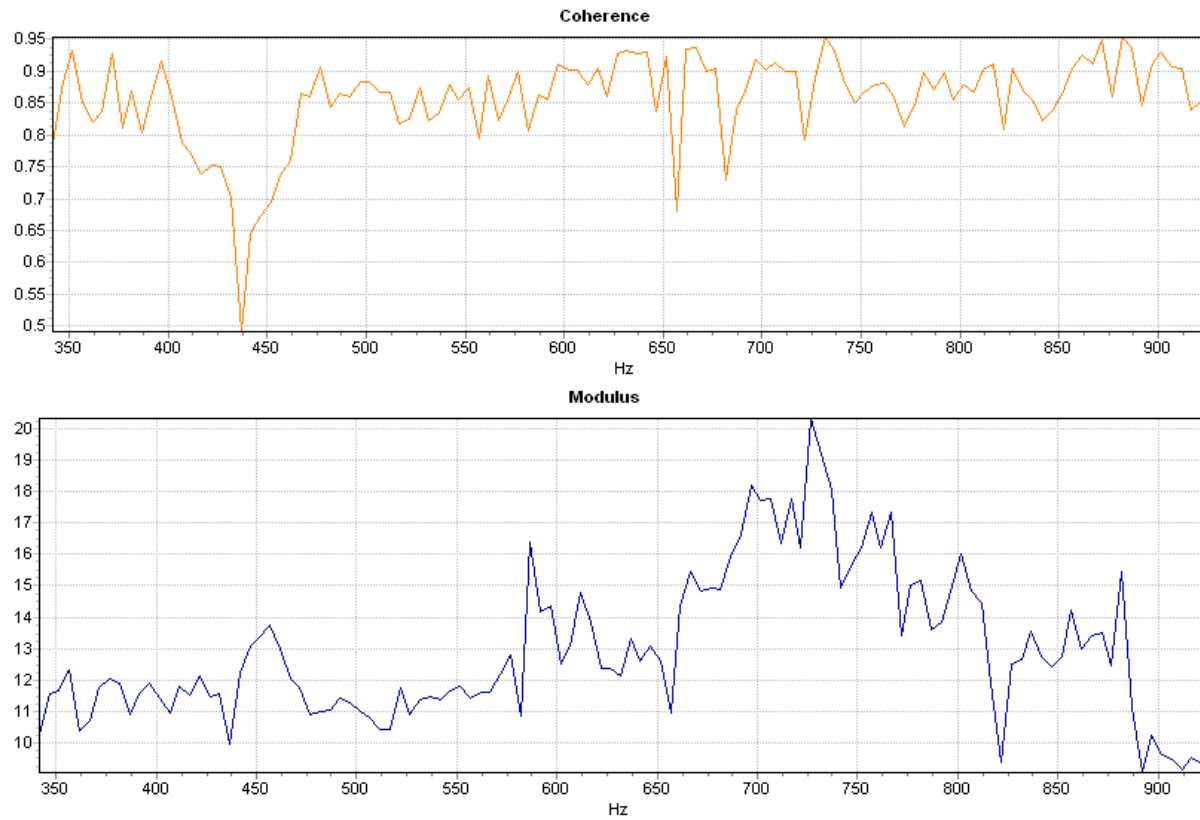


Figure 8.8b: Results of the spectral profiles option applied on the example data file. Shown here are the coherence and modulus functions between IBI and systolic blood pressure in the mid frequency band. The scale values for the modulus are in ms/mmHg.

Appendix

Appendix A: Buttons that are used in CARSPAN

Always available on top in the window

	Create a new session (Ctrl + X)
	Open an existing session (Ctrl + O)
	Save a session to file (Ctrl + S)
	Export event file (Ctrl + E)
	Exit Carspan (Ctrl + X)
	One data file back
	One data file further

Buttons with different meanings in different submenus

	Specify Data Set	Specify Data	Specify Analysis Functions	Specify Analysis Profiles
	Renew file list by reading all data files in the directory once again	x	x	x
	Remove data pair(s) from the file list	Remove selected specifications from the list	Remove selected specifications from the list	Remove selected specifications from the list
	Put the selection on top	x	Select (automatically) all the possible specifications and/or combinations	Select (automatically) all the possible specifications and/or combinations
	Move selection one file down	Select the specifications that are created	Select the specifications that are created	Select the specifications that are created
	Move selection one file up	x	x	x
	Place selection at the bottom	x	x	x

Buttons for navigation in graphs

	Increase/decrease vertical scale
	Increase/decrease horizontal scale (time axis)
	Interpolate selection
	Remove selection
	Edit chart layout
	Go to previous page
	Go to next page
	Go to the first data point
	Go to the last data point

Appendix B: Input files

B.1 CarspanDefaultSettings.ini file

```
[Profiles]
WindowLength=30
StepSize=5

[FrequencyBands Functions]
NrOfBands=4
Band1_Name=Total Freq
Band1_Min=0.020
Band1_Max=0.500
Band2_Name=Low Freq
Band2_Min=0.020
Band2_Max=0.060
Band3_Name=Mid Freq
Band3_Min=0.070
Band3_Max=0.140
Band4_Name=High Freq
Band4_Min=0.150
Band4_Max=0.400

[FrequencyBands Profiles]
NrOfBands=2
Band1_Name=Mid Freq
Band1_Min=0.070
Band1_Max=0.140
Band2_Name=High Freq
Band2_Min=0.150
Band2_Max=0.400

[Coherence]
Min_Coherence=0.3

[Corrections]
TWL=60
NCor=10
NSD=5
MinSD=0.05
MaxSD=0.15
NSucD=5
MinSucD=0.05
MaxSucD=0.25
```

B.2 Example of an event file (.EVT)

```
[Event file]
Textline=EXAMPLE1
Created=03-07-2007 10:05:06
Origin=Carspan

[Events]
RPeak=0
BeginBlock=11
EndBlock=12
```

BeginPeriod=21
EndPeriod=22

```
[Data]
  0   303.747 (R-peak event code)
  0   304.583
  0   305.388
  0   306.142
  0   306.852
  0   307.589
  0   308.392
  0   309.225
  0   310.028
  0   310.788
11   311.514 (Begin of block 1)
  0   311.734
  0   312.265
  0   313.096
21   313.900 (Begin of period 1)
  0   314.649
  0   315.369
  0   316.083
.....
  0   607.844
  0   608.562
  0   609.320
  0   610.101
  0   610.858
  0   611.594
22   611.740 (End of period 1)
  0   612.340
  0   613.120
  0   613.882
  0   614.596
  0   615.268
  0   615.937
.....
  0   629.391
  0   630.208
  0   631.029
21   631.747 (Begin of period 2)
  0   631.750
  0   631.844
  0   632.622
  0   633.371
  0   634.110
  0   634.823
  0   635.251
  0   635.526
  0   636.236
  0   636.927
  0   637.608
  0   638.281
  0   638.752
  0   638.955
  0   639.712
  0   640.450
.....
  0   944.953
  0   945.687
22   946.342 (End of period 2)
```



```

0  946.366
0  947.036
0  947.762
0  948.537
0  949.320
0  950.035
0  950.702
0  951.335
12 951.965 (End of block 1)
0  952.593
0  953.217
0  953.841
0  954.469
0  955.088
0  955.710
0  956.331
0  956.964
0  957.612
0  958.280
0  958.955
0  959.617
0  960.254
0  960.879

```

B.3 Example of an extended event file

```

[Event file]
Textline=EXAMPLE1
Created=03-07-2007 10:05:06
Origin=Carspan

```

```

[Events]
RPeak=0
BeginBlock=11
EndBlock=12
BeginPeriod=21
EndPeriod=22

```

```

[Timeseries]
Timeserie1=IBI
Timeserie2=BPSys

```

```

[Data]
0  292.211  765.000  128.250
0  292.976  765.000  128.250
0  293.734  758.000  123.875
0  294.492  758.000  120.500
0  295.304  812.000  124.000
0  296.223  919.000  127.500
0  297.097  874.000  130.875
0  297.969  872.000  132.000
0  298.878  909.000  126.750
0  299.767  889.000  126.500
0  300.624  857.000  126.625
0  301.421  797.000  123.875
0  302.189  768.000  118.750
0  302.936  747.000  116.500
0  303.747  811.000  121.750

```

0	304.583	836.000	122.750
0	305.388	805.000	125.375
0	306.142	754.000	124.750
0	306.852	710.000	119.000
0	307.589	737.000	118.625
0	308.392	803.000	122.000
0	309.225	833.000	127.000
0	310.028	803.000	129.125
0	310.788	760.000	127.250
11	311.514		
0	311.734	946.000	120.625
0	312.454	720.339	120.625
0	313.177	722.713	123.375
21	313.900		
0	313.906	728.604	126.000
0	314.639	732.892	129.000
0	315.369	730.452	121.625

B.4 Example of parts of a session file (.csf)

```
[DataSetSpec]
UseDataFile=1
DataDir=D:\CARSPAN\DATA\

[DataPair1]
EvtFileName=EXAMPLE2.EVT
DataFileName=EXAMPLE2.NFF
ShortName=EXAMPLE2
EvtFileDate=39245.3356712963
DataFileDate=39245.3362962963
EvtFileSize=36
DataFileSize=1866
Kind=8
IsReference=1

[DataPair2]
EvtFileName=EXAMPLE1.EVT
DataFileName=EXAMPLE1.NFF
ShortName=EXAMPLE1
EvtFileDate=39261.4216203704
DataFileDate=39245.3362962963
EvtFileSize=18
DataFileSize=1866
Kind=8
IsReference=0

[AnaFuncSegments]
Kind=0
Time=0
WindowLength=30
StepSize=5
AdjustPeriodTime=0

[AnaFuncBand1]
Name=Total Freq
Accepted=1
Used=1
Min=0.02
```

```

Max=0.5

...

[AnaFuncSpecPower1]
Name=RPeaks
Calcs=0 1
MinCoh=0
TSIn=0 0 -1
TSOut=12 -1 -1
Bands=1 1 1 1

[AnaFuncSpecPower2]
Name=BPSys
Calcs=0 1
MinCoh=0
TSIn=5 -1 2
TSOut=12 -1 -1
Bands=1 1 1 1

...

[EventCodesSettings]
FileAsBlock=0

[EventCodes]
RPeak=0
BeginBlock=11
EndBlock=12
BeginPeriod=21
EndPeriod=22

[TimeSerie1]
Accepted=1
Name=IBI
Kind=2
EvtColumn=-1
Channel=-1
ScaleFactor=1
ZeroLevel=0
ResSamples=10

[TimeSerie2]
Accepted=1
Name=BPSys
Kind=5
EvtColumn=-1
Channel=2
ScaleFactor=0.125
ZeroLevel=0
ResSamples=10

...

[EventFileEXAMPLE1RPeakCor1]
TimeB=311.734
TimeE=316.083
Kind=5
Accepted=1
NewData=720339 722713 728604 732892 730452

[EventFileEXAMPLE1DataCol5-12Correction1]

```

TimeB=316.821
TimeE=319.104
Kind=5
Accepted=1
NewData=123979167 124895833

[EventFileEXAMPLE1DataCol5-12Correction2]
TimeB=355.506
TimeE=357.617
Kind=5
Accepted=1
NewData=122416667 121395833

...

[Included Corrections]
RPeaks=1
TimeSeries=1

[EventFileEXAMPLE1Marker1]
Name=BOB
Code=11
Time=311.514
Kind=1

[EventFileEXAMPLE1Marker2]
Name=BOP
Code=21
Time=313.9
Kind=3

...

[EventFileEXAMPLE2Marker96]
Name=EOP
Code=22
Time=1268.875
Kind=4

[EventFileEXAMPLE2Marker97]
Name=BOP
Code=21
Time=1288.882
Kind=3

[EventFileEXAMPLE2Marker98]
Name=Event91
Code=17
Time=1288.885
Kind=6

...

Appendix C: Output files

C.1 Example of a numerical output file

```
-----

***   SPECTRAL FUNCTIONS - CARSPAN   ***

***           NUMERICAL OUTPUT FILE           ***

+++           11-03-2009    13:34:51           +++

-----

*   SELECTED FILES   *

+ EVENTFILE: C:\PROGRAM FILES\IDP SOFTWARE\CARSPAN\DATA\EXAMPLE1.EVT
  DATAFILE : C:\PROGRAM FILES\IDP SOFTWARE\CARSPAN\DATA\EXAMPLE1.NFF

*   SELECTED EVENTS   *

BLOCK(S)  BEGIN AT EVENT_CODE      11
BLOCK(S)  END    AT EVENT_CODE      12
PERIOD(S) BEGIN AT EVENT_CODE      21
PERIOD(S) END    AT EVENT_CODE      22
RPEAKS                    0

*   SELECTED BANDS   *

BAND      1      Total Freq
BAND      2      Low Freq
BAND      3      Mid Freq
BAND      4      High Freq

*   SELECTED CHANNELS   *

CHANNEL  1      HR
CHANNEL  3      IBI
CHANNEL  4      BPSys
CHANNEL  5      BPDia

*   SELECTED ANALYSES   *

POWER DENSITY SPECTRUM: raw values
IBI                      POWER DENSITY SPECTRUM (Mod. index)
BPSys                    POWER DENSITY SPECTRUM (Mod. index)
BPSys      -> IBI        TRANSFER FUNCTION (Coherence)
BPSys      -> IBI        TRANSFER FUNCTION (Modulus; min. coh. 0.30)
BPSys      -> IBI        TRANSFER FUNCTION (Phase shift; min. coh.
0.30)

-----
```

FILE: D:\CARSPAN\DATA\EXAMPLE1.EVT

FILE NAME: EXAMPLE1

Line number *)	Type of analyses**)	Exp. nr.	Subject nr.	Block nr.	Period nr.	Period start time (s)	Period end time (s)	Period length (s)	Number of IBI's	1st chan. nr.	2nd chan. nr.	1st channel mean	var. coeff (%)	2nd channel mean
1	1	0	0	1	1	313.917	611.594	298	403	3	0	738.638	6.700	0.000
2	1	0	0	1	2	631.844	945.687	314	422	3	0	743.872	6.800	0.000
3	1	0	0	1	1	313.917	611.594	298	403	4	0	121.511	4.300	0.000
4	1	0	0	1	2	631.844	945.687	314	422	4	0	119.160	4.000	0.000
5	2	0	0	1	1	313.917	611.594	298	403	4	3	121.511	5.256	738.638
6	2	0	0	1	2	631.844	945.687	314	422	4	3	119.160	4.807	743.872
7	3	0	0	1	1	313.917	611.594	298	403	4	3	121.511	5.256	738.638
8	3	0	0	1	2	631.844	945.687	314	422	4	3	119.160	4.807	743.872
9	4	0	0	1	1	313.917	611.594	298	403	4	3	121.511	5.256	738.638
10	4	0	0	1	2	631.844	945.687	314	422	4	3	119.160	4.807	743.872
11	5	0	0	1	1	313.917	611.594	298	403	4	3	121.511	5.256	738.638
12	5	0	0	1	2	631.844	945.687	314	422	4	3	119.160	4.807	743.872

Line number	Value band 1	#pts 1	Band 1 start [Hz]	Band 1 end [Hz]	Value band 2	#pts 2	Band 2 start [Hz]	Band 2 end [Hz]	Repeated for all bands
1	3684.692	49	0.02	0.50	813.449	5	0.02	0.06	
2	3735.550	49	0.02	0.50	511.805	5	0.02	0.06	
3	1446.451	49	0.02	0.50	291.295	5	0.02	0.06	
4	1356.732	49	0.02	0.50	279.139	5	0.02	0.06	
5	0.375	49	0.02	0.50	0.158	5	0.02	0.06	
6	0.517	49	0.02	0.50	0.215	5	0.02	0.06	
7	6.791	16	0.02	0.50	0.000	0	0.02	0.06	
8	9.030	14	0.02	0.50	7.118	1	0.02	0.06	
9	0.867	16	0.02	0.50	0.000	0	0.02	0.06	
10	0.296	14	0.02	0.50	-0.629	1	0.02	0.06	
11	0.082	16	0.02	0.50	0.000	0	0.02	0.06	
12	0.296	14	0.02	0.50	-0.629	1	0.02	0.06	

Line number	PDS or Value at 0.01 Hz	PDS or Value at 0.02 Hz	PDS or Value at 0.03 Hz	PDS or Value at 0.04 Hz	PDS or Value at 0.05 Hz	PDS or Value at 0.06 Hz	Repeated for all freq. points*8*)
1	39748.965	25268.344	12394.357	34265.159	932.091	39748.965	
2	38200.442	13256.059	16087.268	6542.653	4453.244	38200.442	
3	20202.430	6564.730	5158.154	7265.967	5694.219	20202.430	
4	10222.095	9147.745	5447.172	4551.223	2594.587	10222.095	
5	0.192	0.177	0.288	0.160	0.037	0.192	
6	0.214	0.229	0.072	0.032	0.228	0.214	
7	4.313	3.979	6.347	3.935	1.844	4.313	
8	4.445	4.928	2.295	1.629	3.817	4.445	
9	-2.506	-2.482	-2.802	-2.786	-2.822	-2.506	
10	-2.257	-2.227	-1.766	-0.863	-0.566	-2.257	
11	-2.506	-2.482	-2.802	-2.786	-2.822	-2.506	
12	-2.257	-2.227	-1.766	-0.863	-0.566	-2.257	

Notes:

*)Line numbers are just reference purposes only, not included in the file

**) Analysis type legend:

1=Power spectral density

2=Coherence

3=Modulus

4=Unwrapped phase

5=Wrapped phase

***) Frequency resolution equals 0.01 Hz, last frequency point is half the mean heart rate.

The numerical output file is also available as a tab delimited text file with extension .xls and this file can easily opened with Microsoft Excel. If changes are made and need to be saved, change the file type to standard Excel format before saving.

C.2 Example of a summary output file

```
-----

***   SPECTRAL FUNCTIONS - CARSPAN   ***

***           SUMMARY OUTPUT FILE           ***

+++           24-03-2009   16:59:26           +++

-----

*   SELECTED FILES   *

+ EVENTFILE: C:\PROGRAM FILES\IDP SOFTWARE\CARSPAN\DATA\EXAMPLE1.EVT
  DATAFILE : C:\PROGRAM FILES\IDP SOFTWARE\CARSPAN\DATA\EXAMPLE1.NFF

*   SELECTED EVENTS   *

BLOCK(S)   BEGIN AT EVENT_CODE           11
BLOCK(S)   END     AT EVENT_CODE           12
PERIOD(S)   BEGIN AT EVENT_CODE           21
PERIOD(S)   END     AT EVENT_CODE           22
RPEAKS                                0

*   SELECTED BANDS   *

BAND      1      Total Freq
BAND      2      Low Freq
BAND      3      Mid Freq
BAND      4      High Freq

*   SELECTED CHANNELS   *

CHANNEL  1      HR
CHANNEL  3      IBI
CHANNEL  4      BPSys
CHANNEL  5      BPDia

*   SELECTED ANALYSES   *

POWER DENSITY SPECTRUM: raw values
IBI                                POWER DENSITY SPECTRUM (Mod. index)
BPSys                             POWER DENSITY SPECTRUM (Mod. index)
BPSys      -> IBI                TRANSFER FUNCTION (Coherence)
BPSys      -> IBI                TRANSFER FUNCTION (Modulus; min. coh. 0.30)
BPSys      -> IBI                TRANSFER FUNCTION (Phase shift; min. coh.
0.30)
```

FILE: C:\PROGRAM FILES\IDP SOFTWARE\CARSPAN\DATA\EXAMPLE1.EVT

FILE HEADER: EXAMPLE1

```
---- BEGIN OF BLOCK      1 AT      311.514      ----
---- END OF BLOCK        1 AT      951.965      ----

-- BEGIN OF PERIOD      1 AT      313.900      --
* PERIOD LENGTH          297.840 SECONDS        *
* NUMBER OF SAMPLES      404                    *
-- END OF PERIOD        1 AT      611.740      --

>> MEAN OF CHANNEL  3 (IBI          )          738.638  <<
>> MEAN OF CHANNEL  4 (BPSys        )          121.511  <<

>> BAND INFORMATION          ENERGY          FREQUENCY          <<
>>                               IN BAND    POINTS    RANGE          <<
>> CHAN 3    POWER SP.    Total Freq    3684.692    49    0.02-0.50 Hz <<
>> (IBI          )    Low Freq    813.449    5    0.02-0.06 Hz <<
>>                               Mid Freq    2128.153    8    0.07-0.14 Hz <<
>>                               High Freq    716.373    26    0.15-0.40 Hz <<

>> BAND INFORMATION          ENERGY          FREQUENCY          <<
>>                               IN BAND    POINTS    RANGE          <<
>> CHAN 4    POWER SP.    Total Freq    1446.451    49    0.02-0.50 Hz <<
>> (BPSys        )    Low Freq    291.295    5    0.02-0.06 Hz <<
>>                               Mid Freq    610.904    8    0.07-0.14 Hz <<
>>                               High Freq    480.288    26    0.15-0.40 Hz <<

>> BAND INFORMATION          ENERGY          FREQUENCY          <<
>>                               IN BAND    POINTS    RANGE          <<
>> CHAN 4- 3 COHERENCE    Total Freq    0.375    49    0.02-0.50 Hz <<
>> (BPSys        -    Low Freq    0.158    5    0.02-0.06 Hz <<
>> IBI          )    Mid Freq    0.650    8    0.07-0.14 Hz <<
>>                               High Freq    0.216    26    0.15-0.40 Hz <<

>> BAND INFORMATION          ENERGY          FREQUENCY          <<
>>                               IN BAND    POINTS    RANGE          <<
>> CHAN 4- 3 MODULUS    Total Freq    6.791    16    0.02-0.50 Hz <<
>> (BPSys        -    Low Freq    0.000    0    0.02-0.06 Hz <<
>> IBI          )    Mid Freq    9.244    7    0.07-0.14 Hz <<
>>                               High Freq    5.565    7    0.15-0.40 Hz <<

-- BEGIN OF PERIOD      2 AT      631.747      --
* PERIOD LENGTH          314.595 SECONDS        *
* NUMBER OF SAMPLES      423                    *
-- END OF PERIOD        2 AT      946.342      --

>> MEAN OF CHANNEL  3 (IBI          )          743.872  <<
>> MEAN OF CHANNEL  4 (BPSys        )          119.160  <<

>> BAND INFORMATION          ENERGY          FREQUENCY          <<
>>                               IN BAND    POINTS    RANGE          <<
>> CHAN 3    POWER SP.    Total Freq    3735.550    49    0.02-0.50 Hz <<
>> (IBI          )    Low Freq    511.805    5    0.02-0.06 Hz <<
```

```

>>                                Mid Freq      2173.427      8      0.07-0.14 Hz <<
>>                                High Freq     1015.713     26      0.15-0.40 Hz <<

>> BAND INFORMATION                                ENERGY      FREQUENCY      <<
>>                                IN BAND    POINTS    RANGE      <<
>> CHAN 4      POWER SP.      Total Freq    1356.732     49      0.02-0.50 Hz <<
>> (BPSys      )      Low Freq      279.139      5      0.02-0.06 Hz <<
>>                                Mid Freq      602.227      8      0.07-0.14 Hz <<
>>                                High Freq     426.398     26      0.15-0.40 Hz <<

>> BAND INFORMATION                                ENERGY      FREQUENCY      <<
>>                                IN BAND    POINTS    RANGE      <<
>> CHAN 4- 3 COHERENCE Total Freq      0.517      49      0.02-0.50 Hz <<
>> (BPSys      -      Low Freq      0.215      5      0.02-0.06 Hz <<
>> IBI          )      Mid Freq      0.809      8      0.07-0.14 Hz <<
>>                                High Freq     0.341     26      0.15-0.40 Hz <<

>> BAND INFORMATION                                ENERGY      FREQUENCY      <<
>>                                IN BAND    POINTS    RANGE      <<
>> CHAN 4- 3 MODULUS      Total Freq      9.030      14      0.02-0.50 Hz <<
>> (BPSys      -      Low Freq      7.118      1      0.02-0.06 Hz <<
>> IBI          )      Mid Freq      9.668      8      0.07-0.14 Hz <<
>>                                High Freq     8.392      5      0.15-0.40 Hz <<

```

C.3 Example of a profiles output file

```

[EventFile]
Textline=
Origin=EXAMPLE1
Created=03-07-2007 10:25:01
[Events]
RPeak=99
BeginBlock=11
EndBlock=12
BeginPeriod=21
EndPeriod=22
[Timeseries]
Timeserie1=Mean_HR
Timeserie2=Mean_BPSys
Timeserie3=PDS_Mid Freq_HR
Timeserie4=PDS_Mid Freq_BPSys
Timeserie5=MOD_Mid Freq_BPSys-IBI
Timeserie6=COH_Mid Freq_BPSys-IBI

;Code      Time      Tserie1  Tserie2  Tserie3  Tserie4  Tserie5  Tserie6

[Data]
11      311.514
21      313.900
99      341.734      1.352    124.078  1324.030      2.705    10.251    0.787
99      346.734      1.345    124.207  2183.741      3.303    11.540    0.883
99      351.734      1.346    123.847  2033.294      4.874    11.637    0.933
99      356.734      1.341    123.630  2503.791      4.728    12.299    0.851
99      361.734      1.335    122.998  2165.080      6.279    10.359    0.821
99      366.734      1.331    122.591  2339.793      5.051    10.725    0.836
99      371.734      1.335    121.975  3112.897      6.229    11.728    0.927
99      376.734      1.322    121.938  3852.637      8.785    12.012    0.810
99      381.734      1.330    121.557  4506.955      9.465    11.870    0.869
99      386.734      1.334    121.736  4046.726     10.312    10.881    0.803

```

99	391.734	1.332	121.631	3765.555	9.033	11.577	0.868
99	396.734	1.324	121.677	4416.457	9.371	11.883	0.916
99	401.734	1.318	121.926	3575.816	8.414	11.430	0.861
99	406.734	1.317	121.722	3689.835	7.353	10.931	0.788
99	411.734	1.324	121.658	4991.898	7.840	11.786	0.771
99	416.734	1.323	121.783	4904.335	7.727	11.499	0.737
99	421.734	1.326	121.869	5196.965	7.647	12.143	0.752
99	426.734	1.318	122.142	5109.423	7.995	11.470	0.750
99	431.734	1.316	122.466	4149.173	5.366	11.577	0.701
99	436.734	1.325	122.465	3203.991	4.570	9.966	0.491
99	441.734	1.323	122.829	3371.012	3.678	12.208	0.646
99	446.734	1.325	122.399	3377.499	3.613	13.081	0.672
99	451.734	1.322	122.459	3357.340	3.475	13.358	0.694
99	456.734	1.325	122.707	3244.834	3.560	13.754	0.737
99	461.734	1.351	122.522	7455.554	4.449	12.966	0.761
99	466.734	1.365	122.681	3906.531	5.423	12.012	0.864
99	471.734	1.368	122.937	3629.206	5.770	11.738	0.860
99	476.734	1.368	122.934	4478.116	7.666	10.903	0.907
99	481.734	1.372	122.762	5454.576	7.899	10.966	0.844
99	486.734	1.377	122.799	4934.397	6.884	11.034	0.864
99	491.734	1.387	122.540	5333.248	7.064	11.433	0.859
99	496.734	1.394	122.588	5406.563	7.132	11.267	0.884
99	501.734	1.399	122.376	4837.519	6.741	10.982	0.883
99	506.734	1.390	122.363	5055.413	6.576	10.788	0.867
99	511.734	1.389	121.800	4719.641	8.264	10.420	0.868
99	516.734	1.391	121.321	4000.138	5.748	10.423	0.818
99	521.734	1.366	121.497	5549.989	5.748	11.767	0.824
99	526.734	1.362	120.980	5180.731	8.902	10.911	0.875
99	531.734	1.356	120.559	5028.318	7.626	11.354	0.822
99	536.734	1.358	119.893	4430.968	6.988	11.462	0.834
99	541.734	1.364	119.749	4299.577	7.215	11.348	0.880
99	546.734	1.374	119.574	4233.429	6.405	11.655	0.855
99	551.734	1.381	119.488	4252.675	6.797	11.792	0.873
99	556.734	1.387	119.732	5342.567	7.047	11.410	0.794
99	561.734	1.391	119.413	6406.374	9.530	11.619	0.893
99	566.734	1.395	119.140	5711.363	8.433	11.602	0.822
99	571.734	1.401	119.288	5943.110	8.496	12.178	0.860
99	576.734	1.398	119.562	5984.660	8.468	12.784	0.900
99	581.734	1.408	119.403	4404.368	7.316	10.852	0.806
99	586.734	1.407	119.737	4543.874	4.367	16.417	0.863
99	591.734	1.413	119.819	3997.344	4.904	14.166	0.856
99	596.734	1.422	119.904	5478.623	5.573	14.361	0.911
99	601.734	1.410	120.833	8321.881	10.775	12.508	0.901
99	606.734	1.402	120.892	6836.845	11.706	13.081	0.902
99	611.734	1.399	120.815	6307.181	11.839	14.772	0.879
22	611.740						
99	616.734	1.382	120.701	5752.800	10.254	13.851	0.904
99	621.734	1.376	120.766	5250.712	10.008	12.353	0.861
99	626.734	1.372	120.888	5430.966	9.799	12.378	0.928
99	631.734	1.367	120.941	5284.999	9.766	12.108	0.932
21	631.747						
17	631.750						
16	635.251						
99	636.734	1.362	121.067	5989.020	10.434	13.336	0.929
17	638.752						
99	641.734	1.358	121.200	5738.937	10.395	12.614	0.930
17	642.253						
16	645.754						
99	646.734	1.347	121.424	5258.210	10.132	13.085	0.836
16	649.255						

Appendix D: List of abbreviations and symbols

Unit=unit of the time series used

Abbreviation	Meaning	Unit
ASCII	American Standard Code for Information Interchange	
$B(f_1, f_2)$	Power in the frequency range $f_1..f_2$	(unit) ²
$B'(f_1, f_2)$	Normalized power in the frequency range $f_1..f_2$	mMI ²
BOB	Begin of block	s
BOP	Begin of period	s
BP	Blood pressure	mmHg
BRS	Baroreflex sensitivity	ms/mmHg
CARSPAN	CARDiovascular time series SPectral ANalysis	
$CW(f_1, f_2)$	Weighted coherence in the frequency range $f_1..f_2$	-
DBP	Diastolic blood pressure	mmHg
DFT	Discrete Fourier Transform	
ECG	Electrocardiogram	V
EEG	Electroencephalogram	V
EOB	End of block	s
EOP	End of period	s
f	Frequency	Hz
FFT	Fast Fourier Transform	
H(f)	Transfer function	unit _{out} /unit _{in}
HF	High frequency band 0.15-0.40 Hz	(unit) ²
HIGH	High frequency band 0.15-0.40 Hz	(unit) ²
HR	Heart Rate	Hz, beats/min
HRV	Heart rate variability	Hz ² , ms ²
i	Beat index	-
IBI	Interbeat interval	Ms
Im{}	Imaginary part of a complex number	Unit
IPFM	Integrated Pulse frequency Model	
j	Unit complex number, $j^2=-1$	-
LF	Low frequency band 0.04-0.14 Hz	(unit) ²
LOW	Low frequency band 0.02-0.06 Hz	(unit) ²
MBP	Mean blood pressure	mmHg
MI	Modulation index	
MIBI	Mean interbeat interval	ms
MID	Mid frequency band 0.07-0.14 Hz	(unit) ²
N	Number of	
PBP	Pulse blood pressure (SBP-DBP)	mmHg
Re{}	Reel part of a complex number	Unit
RESP	Respiration frequency band centered around the respiratory frequency	(unit) ²
S(f)	Spectral density function	(unit) ² /Hz
S'(f)	Normalized spectral density function	mMI ² /Hz
SBP	Systolic blood pressure	mmHg
SD, sd	Standard deviation	unit
SuD	Successive differences	unit
SVES	Supra-ventricular extra systolic beat	
t	Time	s
T	Time segment of the analysis	s
TASK	Task frequency band centered around the stimulus repetition frequency	(unit) ²
TOTAL	Total frequency band 0.02-0.50 Hz	(unit) ²
T _{refr}	Refractory period	ms
vc	Variation coefficient	%
VLF	Very low frequency band 0.004-0.03 Hz	(unit) ²
VES	Ventricular extra systolic beat	
X(f)	Fourier transform of x(t)	unit/Hz

Abbreviation	Meaning	Unit
$x(t)$	Time series	unit
$\gamma(f)$	Coherence function	-
$\delta(t)$	Dirac function	-
π	Pi, $\approx 3.14159..$	-
$\varphi(f)$	Phase spectrum	rad/Hz